

Forensic Medicine AIIMS 2017

Question 1:

A 35-year-old male came to the OPD with complaints of an increase in his heart rate and palpitations. On examination, there was episodic tachycardia and occasional extrasystole. Ocular examination showed visual field defects. Which of the following is the likely cause?

- a) Cocaine
- b) Nicotine
- c) Cannabis
- d) Atropine

Question 2:

Which of the following is true regarding corporobasal index of the sacrum?

- a) Breadth of fifth lumbar vertebra X 100 / breadth of all sacral vertebra
- b) Breadth of first sacral vertebra X 100 / breadth of fifth lumbar vertebra
- c) Breadth of fifth lumbar vertebra X 100 / breadth of base of sacrum
- d) Breadth of first sacral vertebra X 100 / breadth of base of sacrum

Question 3:

A person was brought dead and an autopsy was performed to find the cause of death. The following findings were noted on examination. What is the probable cause of death?



- a) Throttling
- b) Strangulation
- c) Hanging
- d) Mugging

Question 4:

First carpal bone to ossify:

- a) Lunate
- b) Capitate
- c) Hamate
- d) Scaphoid

Question 5:

A 14-year-old rape victim presents with 26 weeks of pregnancy. What should not be done?

- a) A female doctor should examine the patient
- b) Vaginal swab examination is not necessary
- c) No need to confirm pregnancy by UPT
- d) Termination of pregnancy with her consent by a Gynaecologist

Question 6:

Match the following:

- a) 1=a, 2=b, 3=c, 4=d
- b) 1=b, 2=c, 3=d, 4=a
- c) 1=d, 2=a, 3=b, 4=c
- d) 1=c, 2=d, 3=a, 4=b

Question 7:

In a case of a gun shot injury to skull with entry wound at left temporal region surrounded by abrasion collar & tattooing and a exit wound at right temporal region along with metal fragments of bullet inside the bullet track. Which of the following should be preserved to determine the range of firing ?

- a) Hair
- b) Bullet pieces
- c) Blood
- d) Clothes

Question 8:

Order of appearance of rigor mortis:

- a) Orbicularis oculi, facial muscles, thorax, upper limb
- b) Orbicularis oculi, neck, facial muscles, upper limb, thorax
- c) Neck, muscles of eyelid, upper limb, thorax
- d) Neck, muscles of eyelid, facial muscles, thorax, upper limb

Question 9:

Black foot is seen in which poisoning

- a) Arsenic
- b) Lead
- c) Mercury
- d) Phosphorus

Question 10:

Criminal responsibility of Insane is defined in:

- a) McNaughten's Rule
- b) Currens Rule
- c) Durham's Rule
- d) American Law Institute Test

Question 11:

An autopsy was performed to find the cause of death of a person brought dead to the casualty and it showed a fracture of the thyroid cartilage and bruising around the neck. The cause of death is?

- a) Hanging
- b) Throttling
- c) Strangulation
- d) Mugging

Question 12:

A 30-year-old drowsy male was brought to the casualty. His pulse was 130/min, respiratory rate -30/min. His lab reports are given below. He showed improvement on nasogastric aspiration and administration of sodium bicarbonate, calcium gluconate, and 4-methyl pyrazole. What substance did he most likely consume?

- a) Formaldehyde
- b) Ethylene glycol
- c) Paraldehyde
- d) Methyl alcohol

Question 13:

Which of the following is false regarding teeth features and ethnicity?

- a) In Africanoids the cusps of molars are wide and deep and shovel shaped cusps in incisors

- b) Caucasians have carabelli cusps
- c) Upper third molar is most commonly absent in Mongolians
- d) Prominent lingual ridge and labial ridge in mongols

Question 14:

Jack knife position causes death due to:

- a) Wedging
- b) Burking
- c) Positional asphyxia
- d) Traumatic asphyxia

Answer Key

Question No.	Correct Option
1	b
2	d
3	a
4	b
5	d
6	b
7	a
8	a
9	a
10	a
11	b
12	b
13	a
14	c

Detailed Explanations

Solution to Question 1:

The given clinical scenario is suggestive of nicotine poisoning.

Chronic poisoning of nicotine also known as nicotine addiction leads to a condition called tobacco heart. It is characterized by irregularities of the heart with occasional extrasystoles and angina. It also causes amblyopia and narrowing of the field of vision.

Effects on CVS and eyes by different poisons are given below:

Poison	Effect on CVS	Effect on ocular system
Cocaine	Tachycardia Systemic arterial hypertension MI (due to coronary vasoconstriction) Tachyarrhythmias Dilated cardiomyopathy Aortic dissection & rupture	Madarosis (Loss of eyebrow/eyelashes) Corneal abrasions/ulcerations CRAO Bilateral blindness
Nicotine	Irregularities of heart with occasional extrasystole Episodes of chest pain suggestive of angina	Amblyopia Narrowing of field of vision Blindness
Cannabis	Tachycardia Palpitation Hypertension	Bloodshot eyes (conjunctival congestions), Occasional mydriasis Nystagmus
Atropine	Tachycardia Arrhythmias	Dilated pupil (Mydriasis) Diplopia

Solution to Question 2:

Corporobasal index is given by the formula:

Corporobasal index = Breadth of first sacral vertebra x 100 / breadth of the base of the sacrum.

Corporobasal index of the sacrum is one of the indices used to determine the sex of the individual.

In males, it is ≥ 42 and in females, it is ≤ 42 .

Other indices to determine the sex of the individual are:

Index	Formula	Male	Female
Washburnischiopubic index	Pubic length in mm x 100 / Ischial length in mm	73-94	91-115
Sciatic notch index	Width of the sciatic notch x 100 / Depth of sciatic notch	4-5	5-6
Sacral index	Transverse diameter of base of sacrum X 100 / Anterior length of sacrum	<114	>114

Index	Formula	Male	Female
Sternal index	$\frac{\text{Length of manubrium} \times 100}{\text{Length of body}}$	46.2	54.3
Medullary index of long bones	$\frac{\text{Diameter of medulla} \times 100}{\text{Diameter of whole bone}}$		

Solution to Question 3:

The given autopsy images are suggestive of throttling.

Thumbnail impression over the neck



Petechial haemorrhages in the eye



Extravasated blood clots surrounding larynx



In throttling, the following mechanisms may lead to death:

- Occlusion of blood supply to the brain - carotid arteries
- Occlusion of the airway - minor role
- Carotid sinus pressure may result in vagus nerve stimulation resulting in cardiac arrest (vasovagal shock)

Other signs in deaths due to asphyxia:

- Petechial hemorrhages in the face, eyebrows, eyelids, forehead, ears, conjunctiva, sclera
- When they occur in the visceral pleura, they are called Tardieu's spots
- Congested larynx with slight oozing
- Marked venous congestion of internal organs
- Emphysematous, congested lungs with serosanguinous fluid exudates in alveoli
- Dark-coloured blood due to accumulation of CO₂

Option B: Strangulation is a very generalised term used to describe deaths due to violent asphyxia. Eg- ligature strangulation, throttling etc

Option C: In the case of hanging, there will be a ligature mark, injury to the cervical spine etc.

Option D: In case of mugging the neck is compressed or squeezed by holding it between the crook of the elbow or knee. The attack is usually made from behind. Though a victim killed by this method shows signs of asphyxia and other struggle marks, the examination neck will not show any evidence of violence, as the compression was affected by the soft muscles of the arm or leg.

Solution to Question 4:

Capitate is the first carpal bone and pisiform is the last carpal bone to ossify.

Age of ossification centres of carpal bones:

Carpal bone	Age of ossification
Capitate	1-3 months
Hamate	2-4 months
Triquetrum	2-3 years
Lunate	2-4 years
Scaphoid	4-6 years
Trapezium	4-6 years
Trapezoid	4-6 years
Pisiform	8-12 years

Solution to Question 5:

According to the MTP Act 2021 amendments, pregnancy beyond 24 weeks of gestation cannot be terminated. Also, the guardian's written consent is required in the case of a minor girl.

- Up to 20 weeks of pregnancy, any qualified practitioner can perform an abortion. However, he must have undergone compulsory training.
- Between 20-24 weeks of pregnancy, the opinion of a second practitioner must be obtained to justify performing the abortion.
- The woman to be wanting an abortion should be above 18 years of age to consent. If not, consent can be obtained from the parents or guardian when she is a minor or mentally unsound.

In the cases of sexual assault of a female child, a female doctor is appointed for the examination (Option A). In case, such a doctor is unavailable, then the head of the institution can nominate a woman for that purpose. These examinations are carried out in the presence of a parent or guardian.

Vaginal swab samples are typically collected within 96 hours (ideally up to 72 hours), as the chances of detecting evidence decrease significantly after this period. (Option B)

Solution to Question 6:

- Common popular street names for cocaine are—Snuff, Rock, Crack, Coke, Snow, Cadillac, White lady, etc
- Common popular street names for LSD are –acid, blotter acid, blue caps, blue dots, brown caps, crackers, Deeda, green caps, orange wedges, Paisley caps, pink dots, pink wedges, purple Owsleys, purple wedges, The Beast, The Chief, The Ghost, The Hawk, white lightning, window panes, yellow caps, yellow dots, 25.

- Common names of *Abrus precatorius* are Jequirity bean, rosary pea, Buddhist rosary bead, rosary bead, Indian bead, Indian liquorice, Seminole bead, prayer head, crab's eye, weather plant, lucky bean, ojo de pajarito, gulagunchi, rati, gunja
- Capsaicin, active principle of red chilly or capsicum cause Dermatitis and burning of hands and fingers known as 'Hunan hand' which is common among the pickle industry workers, who use their hands for handling chilly paste or powder for prolonged period.

Solution to Question 7:

The main objective of the study of a firearm wound (Ballistics) is to establish the range and direction of the firing. The ultimate purpose is to determine the weapon used for firing.

Firearm wounds are of two fundamental types: wounds of entry and wound of exit.

The wound of entry is the wound through which the bullet or pellets enter the body, and the wound of exit is the wound through which the bullet emerges out of the body.

To determine the Range of firing, the following parameters are used:

- Entry wound size and shape
- Blast effect – Muzzle end imprint and edge eversion
- Blackening, Tattooing, Singeing of hair, Abrasion collar, and Grease collar.

In the above options, Hair should be preserved in order to determine the range of firing which can be examined for carbonization/desiccation in close range, also from lead deposition around the wound to assess the range of firing.

Characteristics	Contact shot	Close Shot	Near Shot	Distant Shot
Range	In-touch with skin	2.5-7.5 cm	30-60 cm	>60 cm
Size	Largest size	Bullet size	Smaller	Smallest
Shape	Stellate	Circular	Circular	Circular
Blast effect				
Muzzle end imprint	+ve	-ve	-ve	-ve
Edge eversion	+ve	-ve	-ve	-ve
Other findings				
Blackening	-ve	+ve	-ve	-ve
Tattooing	-ve	+ve	+ve	-ve
Singeing	-ve	+ve	-ve	-ve
Abrasion collar	-ve	+ve	+ve	+ve
Grease collar	-ve	+ve	+ve	+ve

Solution to Question 8:

The correct order is orbicularis oculi, facial muscles, thorax, and upper limb.

Nysten's rule describes the sequential onset of rigor mortis in various muscle groups.

Rigor mortis begins from the head and progresses down the body:

- The order of appearance of rigor mortis is heart (left chamber) → eyelids → muscles of the neck → lower jaw → muscles of the face → muscles of the chest(thorax) → upper limb → abdomen → lower limb → fingers and toes (proximo-distal progression)

It disappears in the same order of appearance.

Note that in option B, the upper limb is mentioned before thorax and hence is not the answer.

Solution to Question 9:

Blackfoot's disease is a form of arsenic induced peripheral vascular disease resulting in gangrene.

- Arsenic is a heavy metallic inorganic irritant poison. Metallic arsenic is not poisonous as it is insoluble in water and cannot be absorbed from the gastrointestinal tract. However arsenious oxide or arsenic trioxide (sankhyal or somalkar) is poisonous.
- They are found in shell fish, cod, and haddock.
- Arsenic compounds act by inactivating the sulfhydryl enzymes, which in turn interfere with the cellular metabolism, in the liver, lungs, intestinal wall, and spleen. Arsenic can replace phosphorus in the bones where it may remain for years. It also gets deposited in the hairs.
- Chronic poisoning with arsenic presents with a sequence of five different set of manifestations.
- Gastrointestinal : The victim presents with gradual weight loss, malnutrition, fatigue, loss of appetite, cirrhosis of liver, nausea, vomiting, etc.
- Catarrhal changes : The victim presents with running nose, headache, conjunctivitis, bronchial catarrh, etc.
- Raindrop pigmentation : It is known to produce milk and roses complexion initially, followed by patchy brown pigmentation of the skin (especially face), which resembles the raindrops. It might also show hyperkeratosis of the skin of the palm and soles (Black Foot), which is prone to change into basal cell carcinoma at a later stage. The scalp may also show alopecia (baldness).
- Meese's lines : The victim's nail manifests with whitish lines 1 to 2 mm breadth across the nail of the finger and toes representing the deposition of the poison as a result of high sulphhydryl content of the keratin.
- Arsenical neuritis : The victim presents with polyneuritis, optic neuritis, anesthetics, paresthesias, atrophy of extensors resulting in wrist and foot drop, etc.

Solution to Question 10:

- McNaughten's Rule was formed by the Judges of McNaughten's case, to clarify the common law of insanity in England during 1843. It is similar to IPC 84/ BNS 22 which states "Nothing is an offence which is done by a person who at the time of doing it by reason of unsoundness of mind is incapable of knowing the nature of the act, or that he is doing what is either wrong or contrary to law".
- This forms the basis of the 'test' of insanity that the jury should be instructed to apply in each case of alleged insanity.
- The rules are:
 - Every man is presumed to be sane and to possess a sufficient degree of reason to be responsible for his crimes, until the contrary be proved to their satisfaction.
 - To establish an insanity defence, it must be clearly proved that, at the time of the committing of the act, the accused was laboring under 'such a defect of reason, from a disease of the mind, as not to know the nature and quality of the act he was doing; or if he did know it, that he did not know he was doing what was wrong'.
 - Curren's rule of 1964 states that an accused person is not criminally responsible, if at the time of committing the act, he did not have the capacity to regulate his conduct to the requirements of the law, as a result of mental disease or defect.
 - Durham 's rule of 1954 States that an accused person is not criminally responsible if his unlawful act is the product of mental disease or mental defect.

Solution to Question 11:

Both bruising around the neck and the fracture of the thyroid cartilage are findings seen in throttling. Throttling/manual strangulation is when bare hands are used to compress the neck and strangle.

Mechanism of death:

- Occlusion of blood vessels supplying blood to the brain (carotid arteries).
- Occlusion to airway probably plays a minor role if any, in causing death
- Carotid sinus pressure may result in vagus nerve stimulation, resulting in cardiac arrest (vagal inhibition)

Autopsy findings:

- Bruising of neck: due to the grasping of the neck by the assailant.
- Abrasions on the neck: fingernail marks from assailant or victim
- Fracture of thyroid cartilage
- Inward fracture of greater cornu of the hyoid bone.

Solution to Question 12:

The given scenario point towards ethylene glycol poisoning.

Ethylene glycol is converted to glyoxyaldehyde by alcohol dehydrogenase. Glyoxyaldehyde causes kidney damage.

Methanol is converted to formaldehyde by alcohol dehydrogenase. Formaldehyde causes retinal damage like blurring of vision/loss of vision.

The principles of treatment for both these conditions are the same. It predominantly involves using fomepizole or ethanol to inhibit the metabolic conversion of these relatively non-toxic alcohols to their toxic metabolites.

Solution to Question 13:

The cusps of molars are wide and deep and cusps of incisors are shovel-shaped in Mongoloid races.

Africanoid races tend to have large teeth and often have more cusps on their molars, even up to eight, with two lingual cusps on the mandibular first premolars as an additional common finding.

In Caucasian races, the lateral incisors in the upper jaw are usually smaller than the central. They also have long pointed canine roots. Small nodules on the lingual surface of maxillary molars, called Carabelli's cusp are most common in Caucasian races.

In mongoloid teeth, enamel pearls and small nodules of enamel on the tooth surface, are much more frequent. The condition of bull-tooth or taurodontism is most common in Mongoloid people. The pulp cavity of molars is wide and deep, and the roots are fused and bent. A congenital lack of the third upper molar is most commonly seen. The lingual and labial ridges are more prominent.

Solution to Question 14:

Jack-knife position leads to positional asphyxia.

It is a position where the victim's thighs and knees are driven against the chest causing indirect compression of the chest and restriction of breathing movements.

Forensic Medicine AIIMS 2018

Question 1:

Which of the following is incorrect regarding the image shown?



- a) Only the roots of this plant are poisonous
- b) Causes AV block
- c) Drug of choice for poisoning is Atropine
- d) Mitha zaher is one of the variety of this plant

Question 2:

A girl was allegedly kidnapped by a man. In court, she says that she is a major and is married to the man. X-ray images of her elbow, wrist and pelvis are given. What is the approximate age of the girl ?



- a) 14 to 15 years
- b) 16 to 17 years
- c) 18 to 19 years
- d) 21 to 22 years

Question 3:

First-hand knowledge rule is applicable to:

- a) Common witness
- b) Hand writing expert
- c) Doctor
- d) Hostile witness

Question 4:

Crown Rump Length is 21 cm, length of lower limb is 10 cm, gestational age of the fetus will be:

- a) 6 to 7 months
- b) 4 to 5 months
- c) 7 to 8 months
- d) Term

Question 5:

Identify the correct pair i- Mummification, ii- Adipocere, iii- Marbling, iv- Lividity of death



- a) i-4, ii-3, iii-2, iv-1
- b) i-3, ii-4, iii-2, iv-1
- c) i-3, ii-1, iii-4, iv-2
- d) i-3, ii-4, iii-1, iv-2

Question 6:

Match the following: i- Heroin ii- Eve iii-Joint iv-LSD 1-Ganja 2-Acid 3-Brown sugar 4-Amphetamine

- a) i-2, ii-1, iii-4, iv-3
- b) i-3, ii-2, iii-1, iv-4
- c) i-4, ii-2 iii-1, iv-3
- d) i-3, ii-4, iii- 1, iv-2

Question 7:

Which of the following trait is a feature of the female skull?

- a) Larger teeth
- b) Frontal and maxillary sinuses are larger

- c) Frontal and parietal eminences are larger
- d) Supratympanic crest is present

Question 8:

If a person starts to serve a second employer, the new employer is responsible for his actions by:

- a) Vicarious liability
- b) Doctrine of negligent choice
- c) Contributory negligence
- d) Borrowed servant doctrine

Question 9:

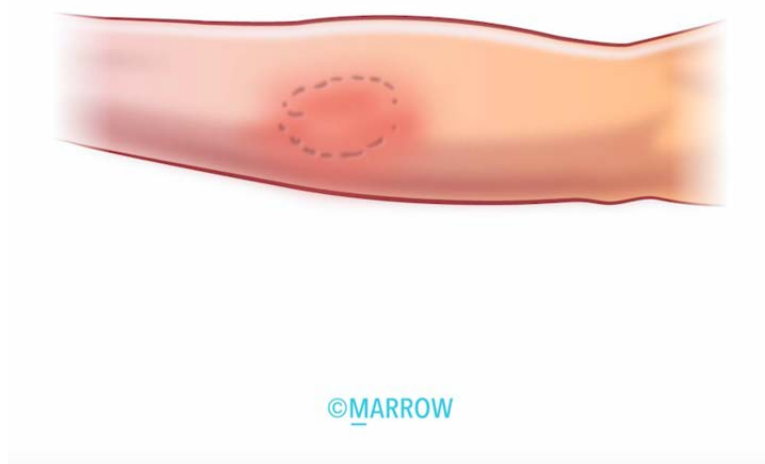
Identify the following post-mortem change.



- a) Tattooing
- b) Suggilation
- c) Putrefaction
- d) Decomposition

Question 10:

All of the following statements are true about the given condition except:



- a) Swabs should be taken immediately after moistening with sterile water
- b) It is most commonly caused by incisors and canines
- c) Photograph is taken with two scales at right angle to one another in the vertical plane
- d) 3D imaging is possible in the given condition

Question 11:

An 18-year-old girl was brought to the OPD. Genital examination revealed the findings as given below. These findings are indicative of:

- a) True virgin
- b) False virgin
- c) Premenstrual stage
- d) Molestation

Question 12:

In case of professional misconduct, the patient's records on demand should be provided within:

- a) 72 hours
- b) 42 hours
- c) 24 hours

d) 7 days

Question 13:

A registered medical practitioner is requested by a survivor of sexual violence to examine her. Within what time must the practitioner take a vaginal swab to look for the presence of spermatozoa?

- a) Within 72 hours**
- b) Within 36 hours**
- c) Within 5 days**
- d) Within 48 hours**

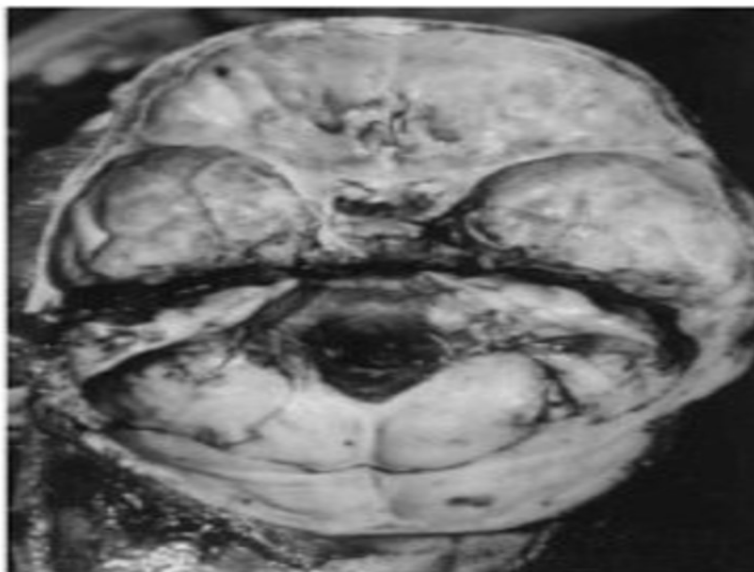
Question 14:

According to POCSO, aggravated sexual assault includes:

- a) Sexual assault done by police officer**
- b) Threatening**
- c) Gang penetrative sexual assault**
- d) All of the above**

Question 15:

Name the type of skull fracture.



- a) Depressed fracture
- b) Hinge fracture
- c) Pond fracture
- d) Comminuted fracture

Question 16:

Bullet fingerprinting involves:

- a) Primary markings
- b) Secondary markings
- c) Both
- d) None

Question 17:

Identify the injury in the image below.



- a) Abrasion
- b) Laceration
- c) Bruise
- d) Imprint

Question 18:

What is the cause of death in smoke inhalation?

- a) Histotoxic hypoxia
- b) Ischemic hypoxia
- c) Anemic hypoxia
- d) All of the above

Question 19:

Paradoxical undressing is seen in:

- a) Heat stress
- b) Voyeurism
- c) Hypothermia
- d) LSD overdose

Answer Key

Question No.	Correct Option
1	a
2	d
3	a
4	a
5	d
6	d
7	c
8	d
9	b
10	c
11	b
12	a
13	a
14	a
15	b

16	c
17	a
18	c
19	c

Detailed Explanations

Solution to Question 1:

The image is of aconite. All parts of this plant (especially the roots and seeds) are poisonous.

Varieties of this plant are monkhood, mitha zaher, bish, and bikh.

It is a cardiac poison and causes AV block, bradycardia. Atropine is used in the treatment of its toxicity.

Solution to Question 2:

The most approximate age of this girl is 21-22 years.

Since the ischial tuberosity is fused in this scenario the girl must be aged ≥ 21 years.

Bone/ joint	Maximum age at fusion	Interpretation of given x-ray
Elbow joint centres (lower end of humerus, upper end of radius, olecranon to ulna)	16 years	Fused, >16 years
Wrist joint centres (lower end of ulna and radius)	19 years	Fused, > 19 years
Hip bone centres (head of femur, greater and lesser trochanter, ischial tuberosity)	22 years	Fused, > 21 years

Solution to Question 3:

The first-hand knowledge rule is applicable to a common witness who gives evidence about the facts observed or perceived by him.

An expert witness is someone who has been trained or is skilled or has knowledge and experience and is able to draw conclusions from the facts observed by him. Doctors, handwriting/ fingerprints/ firearms experts come under this category.

A hostile witness is a person who is supposed to have some interest or motive for concealing a part of the truth or for giving completely false evidence.

Solution to Question 4:

Gestational age of this fetus is 6-7 months.

Crown rump length + length of lower limb = crown heel length

Rule of Haase enables the estimation of the age of the fetus in lunar months from its crown-heel length (in cm).

According to the rule,

- Age in months = Square root of crown-heel length (when the crown-heel length is 25 cm or less/ first 5 months).
- Age in months = crown-heel length divided by 5 (when the crown-heel length is more than 25 cm/ last 5 months).

Here, crown heel length = Crown rump + lower limb length = 31 cm.

Hence, gestational age = $31 / 5 = 6-7$ months.

Solution to Question 5:

Image 1. Marbling: It is the discoloration of superficial veins due to the formation of sulphmethemoglobin, which starts in 24 hours and becomes prominent in 36-48 hours.

Image 2. Lividity of death: Postmortem lividity (suggillation, postmortem hypostasis/postmortem staining/livor mortis/suggillation/vibices) develops in the dependent portions of the body due to engorged vessels showing through the skin.

Image 3. Mummification: It occurs in the absence of moisture in the air and continuous action of dry or warmed air.

Image 4. Adipocere: It forms under warm and humid weather due to hydrolysis of lipids.

Solution to Question 6:

Heroin	Brown sugar, Smack, Dope
Ganja	Joint , Reefer, Weed, Pot, Grass, Purple haze
Ethylene dioxymethamphetamine	Eve
LSD	Acid

Solution to Question 7:

Frontal and parietal eminences are larger in a female skull.

Differentiating features between male and female skull:

Note: The features given above develop after puberty and are modified by senility, so are applicable only between the 20th and the 55th year.

Trait	Male	Female
General size	Large	Small
Architecture	Rugged	Smooth
Supraorbital ridges	Prominent	Less prominent
Mastoid process	Prominent	Less prominent
Occipital region	Marked muscle markings	No marked muscle markings
Frontal eminences	Small	Large
Parietal eminences	Small	Large
Orbits	Square with round margins	Round with sharp margins
Mandible	Large, broad ramus	Small, small ramus
Chin shape	U-shaped	V-shaped

Solution to Question 8:

When an employee is being borrowed, the lending employer temporarily surrenders control over this worker and the borrowing employer temporarily assumes control. This is known as borrowed servant doctrine.

Vicarious liability is by which the employer is responsible for his employees' negligence if such activity occurs in the course of the employment and within its scope.

Solution to Question 9:

The image shows suggilation.

Postmortem lividity (suggilation/ postmortem hypostasis/ postmortem staining/ livor mortis/ vibices) develops in the dependent portions of the body due to engorged vessels showing through the skin.

Solution to Question 10:

The above image shows a human bite mark. The photograph is taken with two scales at the right angle to one another in the horizontal plane.

Human bites are usually semi-circular or crescentic caused by the front teeth (incisors, canine).

Swabs of bite marks should be taken immediately using a swab moistened with sterile water. A sample of the control area adjacent to the mark and of the victim's saliva should also be taken using swabs moistened with sterile water.

3D imaging is done in which the image is measurable with high precision.

A plastic substance such as rubber is laid over the bite mark which produces a permanent negative cast.

Solution to Question 11:

The given findings are suggestive of a false virgin.

The examination findings of a separated labia majora, a flabby labia minora, and a fourchette tear are indicative of forceful penetration.

Hymen usually gets ruptured with the first act of sexual intercourse. However, in some cases, hymen remains intact in spite of sexual intercourse. It is indicative of a false virgin.

Solution to Question 12:

In case of professional misconduct, the patient's records on demand should be provided within 72 hours.

If a doctor does not maintain the medical records of the patients for a period of three years as per the regulation, and refuses to provide the same within 72 hours, when the patient or his/her authorized representative makes a request for it, then it is professional misconduct.

Solution to Question 13:

A vaginal swab of a rape victim demonstrates spermatozoa if the vaginal swab is taken within 72 hours after an assault.

However, a vaginal swab can be taken within 96 hours to demonstrate semen.

Evidence on the outside of the body and on materials such as clothing can be collected even after 96 hours.

Solution to Question 14:

Sexual assault done by a police officer is an aggravated sexual assault according to POCSO (Protection of Children from Sexual Offences).

The POCSO act defines a child as any person below eighteen years of age. It defines various forms of sexual offenses which include:

- Sexual harassment - Threatening
- Sexual assault - Any act with sexual intent that involves physical contact without penetration.
- Aggravated sexual assault - Similar to sexual assault along with the criteria explained under the term aggravated
- Penetrative sexual assault - An act with sexual intent that involves physical penetration of the penis or any object into the vagina, mouth, urethra, or anus.
- Aggravated penetrative sexual assault - Similar to penetrative sexual assault along with the criteria explained under the term aggravated
- Pornography

The term aggravated is used for those assaults done:

- By a person in a position of trust or authority such as a police officer, security force, relative, doctor
- In the course of communal violence
- On mentally & physically disabled
- Gang sexual assault
- Using deadly weapons
- Causing grievous hurt

Depending on if there was penetration or not, it can be aggravated sexual assault (if there is no penetration) or aggravated penetrative sexual assault (if there was penetration). Both are different.

The Act prescribes stringent punishment graded as per the gravity of the offence, with a maximum term of rigorous imprisonment for life and fine.

Note- Gang penetrative sexual assault comes under aggravated penetrative sexual assault and not under aggravated sexual assault. Therefore, option C cannot be correct in this case.

Solution to Question 15:

The image shows hinge fracture/ motorcyclists fracture. Here, the base of the skull is divided into 2 halves, each moving independently of the other like a hinge.

Solution to Question 16:

Bullet fingerprinting involves both primary and secondary markings.

Primary markings are formed when a bullet passes through a barrel and the sides are marked by the rifling of the barrel (which is specific to the make and model of the gun).

Irregularities on the inner surface of barrel itself which are specific for that particular weapon are also seen on the bullet and are called secondary markings.

Solution to Question 17:

The given image shows an abrasion.

An abrasion is the most superficial of injuries. It does not penetrate the full thickness of the epidermis.

A pure abrasion does not bleed, as blood vessels are confined to the dermis. However, the dermal papillae are corrugated in nature. Thus, many abrasions enter the dermis and bleeding commonly occurs.

Solution to Question 18:

Victims of smoke inhalation are affected by carbon monoxide poisoning. In carbon monoxide poisoning, the cause of death is anemic hypoxia.

Carbon monoxide has a higher affinity to hemoglobin compared to oxygen and hence the amount of hemoglobin available for oxygen transport is markedly reduced. It displaces oxygen from its combination with hemoglobin and forms a relatively stable compound known as carboxyhemoglobin. It thus reduces the oxygen content of the blood, and hence, that of the tissues. It acts as a chemical asphyxiant and produces death due to anemic anoxia.

It is important to note that in the case of anemic hypoxia, O₂ content is low but dissolved O₂ is within normal limits so it does not stimulate peripheral chemoreceptors.

Note: Carbon monoxide poisoning causes both anemic (due to CO-Hb) and histotoxic (due to inhibition of cytochrome C oxidase) hypoxia. But the main cause of death is due to anemic hypoxia.

Option A - Histotoxic hypoxia- The most common example is cyanide poisoning. Cyanide inhibits important enzymes in the oxidative phosphorylation process, which is essential to the survival of our cells. Additional oxygen supplementation will not be able to overcome this inhibition, so it will have limited benefit. Hyperbaric oxygen may be helpful to some extent.

Option B - Stagnant hypoxia refers to ischemic hypoxia that results from inadequate circulation due to conditions such as heart failure and hypoxia in the setting of shock. In such conditions, administering intravenous fluids can improve the patient's condition because it is the circulation that is affected. If the shock is prolonged it can result in acute respiratory distress syndrome (ARDS), which can be treated with positive pressure ventilation and not simply by increasing oxygen administration.

Solution to Question 19:

The phenomenon of paradoxical undressing is seen in hypothermia victims.

Its occurrence is attributed to a paradoxical feeling of heat and may lead to an erroneous assumption of a sexual offense.

Fatal hypothermia-induced stupor, loss of powers of judgment, disorientation, and increasing incapacitation are associated with this phenomenon.

Forensic Medicine AIIMS 2019 (May & Nov)

Question 1:

Identify the following post-mortem change.



- a) Tattooing
- b) Suggilation
- c) Putrefaction
- d) Decomposition

Question 2:

Paradoxical undressing is seen in:

- a) Heat stress
- b) Voyeurism
- c) Hypothermia
- d) LSD overdose

Question 3:

Identify the plant which produces a deliriant toxin.



- a) *Erythroxylum cocca*
- b) *Datura*
- c) Hyoscine
- d) *Digitalis purpurea/cannabis*

Question 4:

An 11-year-old girl was found to be quiet and shy in her classroom with intermittent crying, for the past few days. It was discovered that her uncle was touching her genitalia inappropriately. The duty of the principal is to report this to?

- a) Magistrate
- b) Police
- c) Child Welfare Committee
- d) Parents

Question 5:

Which of the following postmortem changes is shown in the given image?



- a) Marbling
- b) Algor mortis
- c) Rigor mortis
- d) Postmortem lividity

Question 6:

What is bullet wipe?

- a) Residual from barrel of gun
- b) Blackening
- c) Gutter fracture of skull
- d) Tattooing

Question 7:

A 28-year-old man presented with dry mouth, urinary retention, and constipation after consuming an unknown substance. On examination, his PR - 110bpm, and his pupils were dilated. What did he most likely consume?

- a) Heroin
- b) Morphine
- c) Malathion
- d) Belladonna

Answer Key

Question No.	Correct Option
1	b
2	c
3	a
4	b
5	a
6	a
7	d

Detailed Explanations

Solution to Question 1:

The image shows suggilation.

Postmortem lividity (suggilation/ postmortem hypostasis/ postmortem staining/ livor mortis/ vibices) develops in the dependent portions of the body due to engorged vessels showing through the skin.

Solution to Question 2:

The phenomenon of paradoxical undressing is seen in hypothermia victims.

Its occurrence is attributed to a paradoxical feeling of heat and may lead to an erroneous assumption of a sexual offense.

Fatal hypothermia-induced stupor, loss of powers of judgment, disorientation, and increasing incapacitation are associated with this phenomenon.

Solution to Question 3:

The given plant is *Erythroxylum coca*.

Cocaine is the principal alkaloid found in the cultivated varieties of coca.

Solution to Question 4:

The principal has to report this case to the police.

The POCSO Act, 2012 is a comprehensive law for the protection of children from the offences of sexual assault, sexual harassment, and pornography. This is done while safeguarding the interests of the child at every stage of the judicial process by incorporating child-friendly mechanisms for reporting, recording of evidence, investigation, and speedy trial of the offence through designated special courts.

Section 21(1) of the POCSO Act, 2012 requires mandatory reporting of cases of child sexual abuse to the law enforcement authorities and applies to everyone including parents, doctors, and school personnel. Failure to report a suspicion of child abuse is an offence under the Act.

Solution to Question 5:

The postmortem change shown above is marbling.

The blood in the vessels is hemolyzed, which stains the vessel walls and the adjacent tissues, giving rise to a marbled appearance. The marbling of the skin becomes prominent about 36–48 hours after death in summer and is distinctly appreciable in the superficial veins of the abdomen, shoulders, chest, and inguinal region.

Solution to Question 6:

Bullet wipe is due to the grease, soot or debris from the barrel of the gun.

While examining the clothing of a victim of bullet injury, there is a deposition of various residues surrounding the entrance defect at various ranges known as Bullet wipe. It helps in determining the wounds of inlet and outlet upon the body.

Solution to Question 7:

The given clinical scenario with dilatation of the pupil indicates Atropa belladonna consumption.

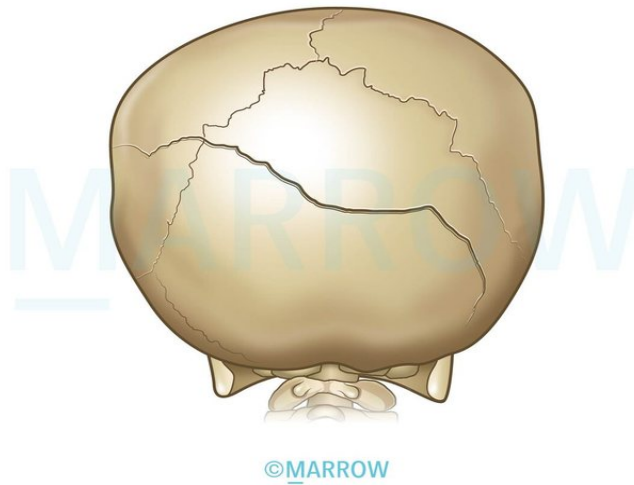
Atropa Belladonna acts by inhibiting the muscarinic effects of acetylcholine. All the parts of the plants are toxic. The active principle is 1-hyoscyamine.

Heroin, morphine, and malathion cause constriction of the pupil.

Forensic Medicine AIIMS 2020 (May and Nov INI-CET)

Question 1:

Identify the type of skull fracture in the given image?



- a) Ring
- b) Depressed
- c) Fissured
- d) Comminuted

Question 2:

A middle-aged man was brought to the ER after an assault with a sharp axe. Identify the type of wound?



- a) Incised wound
- b) Chop wound
- c) Lacerated wound
- d) Abrasion

Question 3:

A patient comes to the casualty with life-threatening injuries. There was no time for taking consent. The doctor started doing life-saving procedures with all precautions, and despite best efforts, the death of the patient occurred. The doctor is not responsible under which doctrine?

- a) Res ipsa loquitur
- b) Doctrine of anticipation
- c) Doctrine of extended consent
- d) Doctrine of conjugated consent

Question 4:

Which of the following is false regarding the management of snakebite in India?

- a) Atropine and neostigmine are very useful in the management of krait bite
- b) Neostigmine and ventilatory support are used in treatment along with ASV
- c) ASV is the mainstay of treatment

- d) Current ASV in India is not effective against humped pit viper

Question 5:

A 14-year-old girl comes to the OPD with her mother. History of penovaginal penetration by the neighbour is reported to the doctor. The girl admits to giving the neighbor the consent for the same and says she does not want to be examined. As a medical officer, which of the following should you do?

- a) 1,2,3 are true
- b) 2,3,4 are true
- c) 1,3,5 are true
- d) 3,4,5 are true

Question 6:

A surgeon performs a hysterectomy for uterine fibroids after obtaining relevant informed consent from the patient. Despite adequate precaution, the ureter was injured intraoperatively. Under which of the following principles will the doctor not be held responsible for this act?

- a) Novus actus interveniens
- b) Medical maloccurrence
- c) Res ipsa loquitur
- d) Physician error

Question 7:

The punishment for the death of a patient by spurious drug:

- a) 5 years
- b) 7 years
- c) Life imprisonment
- d) 1 year

Question 8:

Identify the seeds shown in the image below.



- a) Tropicana canna
- b) Nigella sativa
- c) Datura seeds
- d) Opium seeds

Question 9:

A 12-year-old boy presents with the following injury that he sustained on being tackled by an opponent during the course of a football match. As the Casualty medical officer, you would enter the wound description in the case sheet as _____



- a) Incised wound
- b) Abrasion
- c) Laceration
- d) Bruise

Question 10:

You are posted as an intern in casualty when an MLC case is brought by the police with the following appearance. She was declared brought dead. What would you mark as the likely cause of death before sending the patient for postmortem?



- a) Drowning
- b) Hanging
- c) Strangulation
- d) Traumatic asphyxia

Question 11:

What is the approximate age of a person in whom the medial end of the clavicle is fused and also the sternum is completely fused

- a) <18 years
- b) 18-22 years

- c) 22-25 years
- d) >25 years

Question 12:

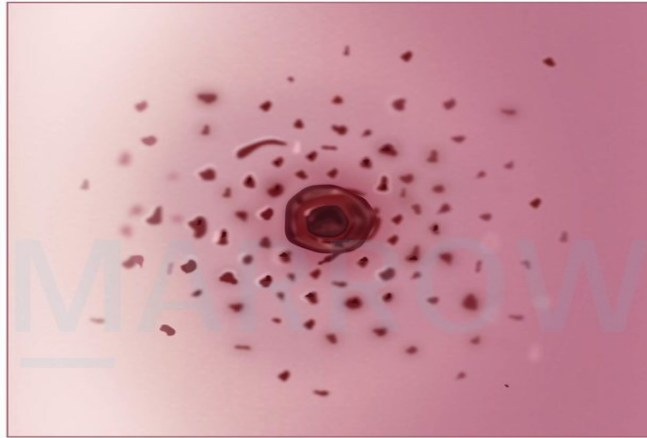
Identify the condition shown in the image.



- a) Postmortem lividity
- b) Marbling
- c) Filigree burns
- d) Subcutaneous emphysema

Question 13:

Identify the image shown below

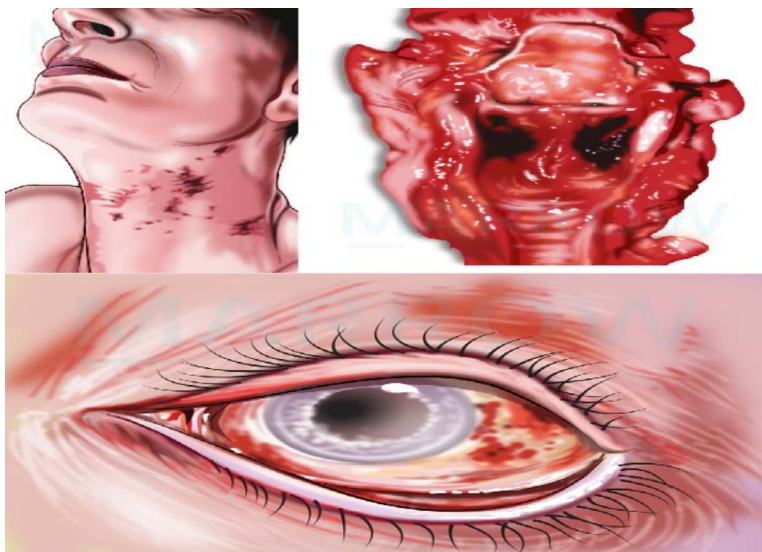


©MARROW

- a) Shot gun wound
- b) Tattooing
- c) Arsenic poisoning
- d) Graze wound

Question 14:

A 22-year-old woman was found dead at home. She was brought to the mortuary by the police and the following findings were noted on examination. Which of the following is the likely cause of her death?



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- a) Strangulation by ligature

- b) Hanging
- c) Throttling
- d) Drowning

Question 15:

Which of the following is seen in freshwater drowning?

- a) A,B,C,D
- b) A,B,C
- c) B,C,D
- d) A,C,D

Question 16:

Match the following weapons with the possible injuries caused by them:

- a) A-4, B-3, C-1, D-2
- b) A-1, B-3, C-4, D-2
- c) A-1, B-2, C-4, D-3
- d) A-4, B-1, C-3, D-2

Question 17:

You are working as a medical officer at a PHC. A 17-year-old girl presents to you after an alleged sexual assault, the girl refuses physical examination. Which of the following steps would you take to proceed in this case?

- a) 1, 2 and 3
- b) 2, 3 and 5
- c) 1, 2 and 5
- d) 2 and 3

Question 18:

Match the following poisons with their probable mechanism of action: A. Zinc chloride B. Chloral hydrate C. Quinine D. Potassium carbonate 1.Abortifacient 2. Irritant 3. Corrosive 4.

Stupefying

a) A-2, B-4, C-1, D-3

b) A-4,B-3, C-2,D-1

c) A-4, B-2, C-1,D-3

d) A-2, B-3, C-4,D-1

Answer Key

Question No.	Correct Option
1	c
2	b
3	b
4	a
5	c
6	b
7	c
8	b
9	c
10	a
11	d
12	b
13	a
14	c
15	d
16	a
17	a
18	a

Detailed Explanations

Solution to Question 1:

The given image shows a fissured fracture of skull.

Solution to Question 2:

The given wound caused by an axe, is likely to be a chop wound.

Chop wounds are so-called because of the action in which they are produced. Depending on the type of instrument, they may resemble an incised or a lacerated wound. The shape of the wound may give clues to the type of weapon. It would be a deep, well-margined wound, gaping with limited damage to the surrounding skin.

For example, in the given image, the wound resembles an injury produced by a sharp axe. Such wounds may be inflicted with homicidal intent on the skull/scalp.

Solution to Question 3:

Under IPC 92, the doctor can perform life-saving procedures in a patient with life-threatening injuries without obtaining consent. The guiding principle is whether the delay in treatment in order to obtain consent would result in harm to the patient. This is the doctrine of anticipation.

Under some circumstances like emergencies, informed consent may be neither possible nor necessary.

IPC 92 states that nothing is an offense by reason of any harm which it may cause to a person for whose benefit it is done in good faith, even without that person's consent, if the circumstances are such that it is impossible for that person to signify consent, or if that person is incapable of giving consent, and has no guardian or other person in lawful charge of him from whom it is possible to obtain consent in time for the thing to be done with benefit.

Other options:

'Res Ipsa Loquitor' means 'the thing or fact speaks for itself'. It is obvious evidence for the negligent act committed. For example, swabs left in the abdominal cavity post laparotomy.

The doctrine of extended consent: It is used in unanticipated conditions during surgery. This comes into play when a surgeon encounters an unanticipated abnormality within the field of surgery. It is called the "extension doctrine," and it assumes that the surgeon is using reasonable judgment. For example, a surgeon incurred no liability for draining some ovarian cysts during the course of an appendectomy. The condition must be one that was unforeseen, and the patient must not have expressly refused such an intervention.

Doctrine of conjugated consent - Dummy option (No such consent has been defined).

Solution to Question 4:

Atropine is not useful in the management of krait bite. The toxin in common krait's venom acts presynaptically, hence atropine is not used in the treatment of envenomation by krait.

As the cobra venom acts postsynaptically, atropine can be used in the management.

ASV(Anti Snake Venom) is the mainstay of snakebite treatment.

ASV used in India is effective against:

- Cobra
- Common krait
- Russel viper
- Saw scaled viper

Neostigmine and ventilatory support are used in the treatment of snakebite.

Solution to Question 5:

In the given scenario 1, 3, 5 are true.

The scenario describes a 14-year-old girl with a history of sexual intercourse with her neighbor albeit with consent. Consent for intercourse is valid only after 18 years in an unmarried girl. As she is less than 18 years, it is presumed to be a medicolegal case of sexual assault.

Examination of a victim of sexual assault should be done after the consent of the victim (if she is less than 12 years of age or unsound mind, consent from the guardian needs to be taken). If she denies examination, counseling has to be done regarding the need for examination and how it could be held up against her in the court of law. If she still refuses for part or whole of examination or sample collection, an informed refusal needs to be documented.

Information to the police is given in all medicolegal cases.

Solution to Question 6:

The doctor will not be responsible for the above act under medical maloccurrence.

Medical Maloccurrence/Inevitable Accident: Every human being responds differently to different drugs as well as to different treatment modalities. On certain occasions, despite all proper care given by the doctor during treatment, the patient fails to respond properly or may suffer some adverse reactions to the drug which may lead to severe injuries or permanent deformities. This is known as a medical maloccurrence/inevitable accident.

If the doctor can prove this before the court, it will be an absolute defence against malpractice.

'Res ipsa loquitur' means 'the thing or fact speaks for itself'. It is obvious evidence of the negligent act committed. For example, swabs left in the abdominal cavity post laparotomy.

'Novus actus interveniens': This essentially means that the negligence at the core of the lawsuit stems from the actions of a third party and that this negligence was unforeseeable. The viability of this defence heavily relies on the specific details of each case. For instance, if a doctor writes a prescription that is allegedly illegible, and the pharmacist misreads it, fails to verify it with the doctor, and then dispenses the wrong medication to the patient, resulting in an injury, the doctor might use this defence in such a scenario.

Solution to Question 7:

The punishment for the death of a patient by a spurious drug is imprisonment for life.

In case of an adulterated, spurious, or any drug that when used by any person for or in the diagnosis, treatment, mitigation, or prevention of any disease or disorder is likely to cause ill health or is likely to cause such harm on his body as would amount to grievous hurt within the meaning of Section 320 of IPC, shall be punishable with imprisonment for a term that shall not be less than 3 years but that may extend to a term of life.

Solution to Question 8:

The seeds shown in the given image are of the plant *Nigella sativa*. These are also known as black cumin or kalonji.

The seeds are the most commonly used parts of the plant in traditional medicine. Seeds have a pungent and bitter taste with a considerable amount of oil inside. The seeds are used as a spice in cooking. They are also used as a food preservative and the most often used therapeutic agent is its oil.

Solution to Question 9:

The given image shows a laceration.

A laceration is a tear or a split in the skin, mucous membrane, muscle, or any internal organ, with a depth spanning more than the covering epithelium of the skin or an organ, and is produced by blunt force.

Features of laceration:

- Margins are irregular with pieces of tissue attached in between called tissue tags or bridges
- Bruising is seen around the margin
- Deeper tissues are unevenly divided with tissue tags
- Hair bulbs are crushed
- Depth of the wound contains abundant foreign material such as dust particles, paint material of the vehicle involved, etc
- In skin lacerations, usually one of the margins overhangs the other, which could help in assessing the direction of the force
- A laceration, on healing, produces a permanent scar.

Classification of a laceration:

- Split laceration (incised - like laceration)
- This occurs when the skin is crushed between two hard objects
- For example, during a fall, the scalp gets crushed between the skull from inside and hard ground outside. This results in a wound, which usually mimics an incised one unless examined closely.
- Stretch laceration

- As the name implies, it occurs due to the over-stretching of the skin, beyond its elasticity.
- This typically occurs in the setting of a fractured femur, where the fracture fragment may stretch and pierce the skin to project out
- Avulsed laceration
- An avulsion is a lesion where the skin only separates by tearing from the underlying tissues or peels off and overhangs the wound
- This is common in road traffic accidents.

Solution to Question 10:

The given image shows frothing from nostrils and no clear signs of facial or conjunctival congestion. It is likely a case of drowning.

Frothing may also be seen in throttling or strangulation, but it might be associated with signs of conjunctival congestion, facial congestion, patterned bruise on neck etc. In the absence of these features, it is unlikely to be caused by strangulation.

Traumatic asphyxia is unlikely as the face of this patient is all whitened without signs of congestion or any obvious signs of trauma

Solution to Question 11:

Approximate age for the given person with fused medial end of clavicle and completely fused sternum is most likely >25 years.

The sternum is formed from the fusion of Manubrium, 4 sternbrae(body), and the xiphoid process. The fusion of sternbrae occurs from below upwards, followed by its fusion to the xiphoid process and then fusion of manubrium to the fused sternum.

Age	Age determination
<5 years	The appearance of ossification centers
5-16 years	Analysis of hand and wrist
>16 years(male)	Complete fusion of distal radius or ulna
>16 years(female)	Complete fusion of distal ulna
>18 years(male)	Complete fusion of distal femur or proximal tibia
>18 years(female)	Complete fusion of iliac crest
>21 years	Complete fusion of medial end of clavicle

Manubrio-sternal joint	60-70 years
1st and 2nd Sternebrae	25 years
2nd and 3rd sternebrae	20 years
3rd and 4th sternebrae	15 years
Xiphoid process and body	40-45 years

Solution to Question 12:

The image shows postmortem marbling. It is a visible postmortem change that is characterized by linear branching patterns on skin that correspond to the course of superficial veins.

Intestinal bacteria infest the venous system as blood acts as a nutrient medium for their proliferation. There is subsequent hemolysis which stains the vessel walls and the adjacent tissues, giving rise to marbling. This becomes prominent in about 36–48 hours after death in the summer and is distinctly appreciable in the superficial veins of the abdomen, shoulders, chest, and inguinal region.

Solution to Question 13:

The given image shows entry wound of a shot gun from a mid-range or long distance.

- There is a central round hole, with multiple pellet holes scattered around the central hole.
- This wound is most likely to have been caused by a shotgun from a range of 1-5 m.

Following are the appearances of shot gun entry wound from different distance:

Appearance of shotgun entry wounds at different ranges:

Contact shot

- Soft tissues - single circular wound about the diameter of the muzzle with muzzle imprint and no soot soiling, tattooing, singeing.
- Bone - same as with soft tissues, but wound may be ragged and split because of gas rebound.

<30 cm

- Circular wound with surrounding soot soiling.
- Burnt hairs (singeing).
- Punctate skin lesions from powder particles (tattooing).

30cm - 1m

- Circular rat-hole wound.

- Soot soiling.

- Powder tattooing.

1 - 5 meters

- Central rat-hole wound.

- No soot, tattooing +/-.

- Satellite pellet holes around the periphery.

>5 meters

- Diffuse pellet pattern

- No central hole, no soot soiling, tattooing, singeing.



Solution to Question 14:

In the given case scenario and autopsy images showing carotid intimal tear and muscle hematoma with nail marks on neck and absence of ligature mark point towards a diagnosis of throttling.

Other options

Option A- Strangulation by ligature - ligature mark with signs of a struggle would've made this diagnosis more likely.

Option B- Hanging - ligature mark with a suicide note would've made it more likely.

Option C- Drowning- Body recovered from a pond or other water body with other suggestive features like frothing etc would've made this more likely.

Solution to Question 15:

In cases of freshwater drowning, hyperkalemia is seen due to hemolysis of swollen RBCs and not hypokalemia.

Freshwater is hypotonic to plasma and is rapidly absorbed from the alveoli and into the RBCs. The RBCs finally swell up and lyse resulting in the release of intracellular potassium causing hyperkalemia. Increased intravascular blood volume combined with hyperkalemia causes arrhythmia.

Seawater is hypertonic to plasma and causing a shift of fluid from the blood to alveoli causing pulmonary edema. There is a net decrease in blood volume causing a fall in blood pressure.

The table below summarises the cause of death in different types of drowning.

Type of drowning	Cause of death
Freshwater drowningSeawater drowning	Disruption of surfactant
Dry drowning	Laryngeal spasm/Vagal inhibition
Post immersion Syndrome / Secondary drowning	Cerebral anoxiaPneumoniaPulmonary edemaChemical pneumonitisMetabolic acidosis

Solution to Question 16:

The correctly matched sets of the above-mentioned weapons and the possible injuries are as follows:

Mechanism of injury with the above mentioned weapons:

Axe - it is a heavy weapon that may or may not be sharpened. Injury is produced by chopping action of the weapon and it may be in a spectrum of an incised wound to laceration depending upon the sharpness of the blade.

Road traffic accident - rubbing of epidermis tangential to a rough or hard surface gives rise to graze abrasion also called brush burn. It can occur in a road traffic accident after a pedestrian is being hit by a vehicle and the victim is slid/grazed for some distance before he comes to a halt.

Blade - it gives rise to a clean-cut wound which is usually deepest at its starting. Tailing of the wound indicates the direction of injury. Length is the biggest parameter of this type of wound.

Lathi- It is a blunt force injury giving rise to a patterned abrasion with 2 parallel lines and a normal unbruised area on site of impact or stick/lathi. The lathi forces the blood from the site of impact to either side which gives rise to tramline appearance.

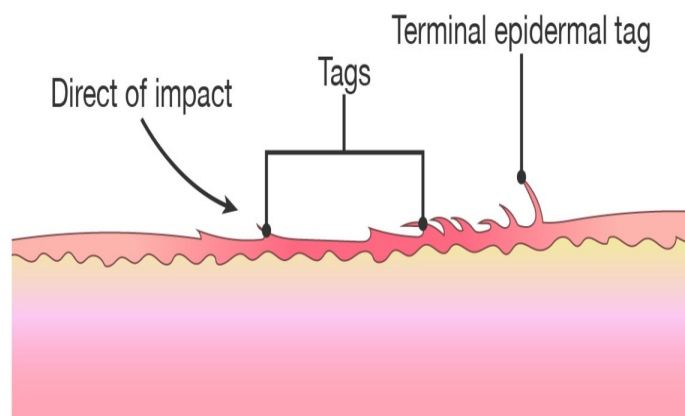
The image below shows a chop wound inflicted on the scalp:

A. Axe	4. Chop wounds
B. Road traffic accident	3. Graze abrasions
C. Blade	1. Incised wound

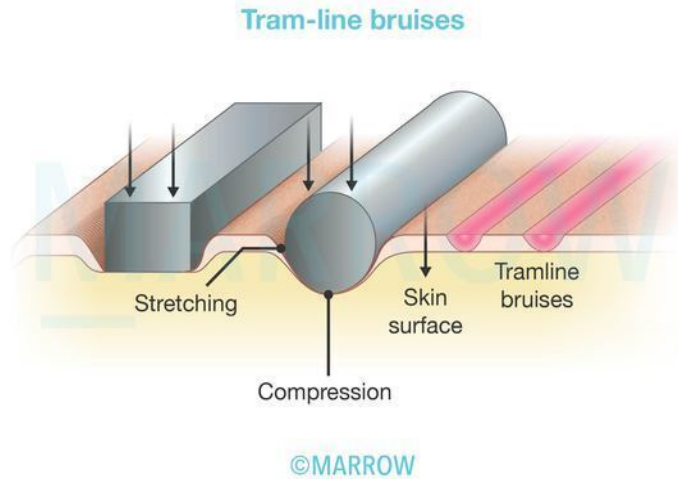
A. Axe	4. Chop wounds
D. Lathi	2. Tram track bruise



The image below shows a schematic illustrating heaping up of epidermis in a graze abrasion or brush burn:



The image below shows production of a tramline bruise associated with a lathi strike:



Solution to Question 17:

In the given scenario, you need to start treatment for STI, inform the police, and record refusal of consent for examination.

Examination of a sexual assault victim shouldn't be forced upon, but an informed refusal is to be documented after explaining to them the consequences of choice.

Medical treatment and testing for pregnancy have to be done for all victims of sexual assault free of cost and police should be informed as per Section 397 of BNSS (357-C of CrPc).

As the girl is under 18 years of age, as per POCSO (The Protection of Children from Sexual Offences) act 2012, the medical practitioner should initiate treatment for exposure to sexually transmitted diseases (STDs) including prophylaxis for identified STDs (including HIV if suspected) and possible pregnancy and emergency contraceptives should be discussed with the victim and her parent/guardian.

Solution to Question 18:

For the above-given options, the correct match is given as

Rationale for the classification

Zinc chloride also known as butter of zinc is highly irritant to the eyes on topical application. It can cause erosive esophagitis on ingestion. It has a fatal dose of 350mg. It could've also been classified as a corrosive, but since potassium carbonate is an alkali with stronger corrosive action, it is better classified as an irritant for this question

Chloral hydrate is a CNS depressant. It is also known as knockout drops because of its rapid action and used in robbery or rape to quickly render the victim helpless.

Quinine is a cardiotoxic drug with the additional action of being an abortifacient. It's overdose may lead to an echolic action (contract the pregnant uterus)

Potassium carbonate is a caustic alkali with a predominant action of corrosive due to its high pH.

Poison	Mechanism
A.Zinc Chloride	2. Irritant
B.Chloral hydrate	4. Stupefying
C. Quinine	1. Abortifacient
D.Potassiumcarbonate	3. Corrosive

Forensic Medicine NEET 2018

Question 1:

Which is the first organ to putrefy?

- a) Brain
- b) Heart
- c) Kidney
- d) Prostate

Question 2:

Which of the following constitutional article is not related to children?

- a) 21A
- b) 24
- c) 45
- d) 42

Question 3:

McNaughten's rule falls under which Section?

- a) BNSS Section 85 (CrPC 83)
- b) BNS Section 22 (IPC 84)
- c) BNS Section 219 (IPC 184)
- d) BNSS Section 204 (CrPC 184)

Question 4:

Luminescent stools are associated with which of the following?

- a) Mercury
- b) Cadmium
- c) Phosphorus

d) Oxalic acid

Question 5:

Which of the following poisons cause bluish discoloration of the stomach mucosa on post mortem examination?

- a) Oxalic acid**
- b) Sodium amytal**
- c) Soneryl**
- d) Arsenic**

Question 6:

Saturnine gout is seen in:

- a) Lead poisoning**
- b) Cadmium poisoning**
- c) Beryllium poisoning**
- d) Mercury poisoning**

Question 7:

At what age, basiocciput fuses with basisphenoid?

- a) 18-22 years**
- b) 22-25 years**
- c) 14-16 years**
- d) 12-14 years**

Question 8:

Edmond Locard is most famous for?

- a) Study of fingerprints**
- b) Theory of exchange**
- c) Stature estimation**

- d) Forensic ballistics

Question 9:

What is the odor of the mummified body?

- a) Odorless
- b) Pungent
- c) Putrid
- d) Offensive

Question 10:

Which metallic poisoning causes proximal convoluted tubule damage with proteinuria and muscle pain ?

- a) Arsenic
- b) Chromium
- c) Mercury
- d) Lead

Question 11:

Posthumous child is a

- a) Child delivered after the death of parents
- b) Child delivered after the death of biological father
- c) Child born out of wedlock
- d) Fictitious child

Answer Key

Question No.	Correct Option
1	a
2	d

3	b
4	c
5	b
6	a
7	a
8	b
9	a
10	a
11	b

Detailed Explanations

Solution to Question 1:

Among the given options, the first organ to putrefy is the brain.

The order of organ putrefaction (from first to last):

- Larynx and trachea
- Stomach, intestine, spleen
- Liver and lungs
- Brain
- Heart
- Kidney and the bladder
- Prostate and uterus
- Skin, muscle, tendons
- Bones

Solution to Question 2:

Article 42 is not related to children. Article 42 says the state shall make provision for securing just and humane conditions of work and maternity relief.

Constitutional Guarantees that are meant specifically for children include:

- Article 21 A - Right to free and compulsory elementary education in the 6 -14 year age group.
- Article 24 - Right to be protected from any hazardous employment till the age of 14 years.
- Article 39 E - Right to be protected from being abused.
- Article 39 F - Right to equal opportunities and facilities.

- Article 45 - Right to early childhood care and education for all children until they complete the age of 6 years.

Solution to Question 3:

McNaughten's rule falls under Section 22 of the Bharatiya Nyaya Sanhita (IPC 84).

It states, "A person of unsound mind is not liable for an act."

Defect: In Mc Naughten's rule, only intellectual factors are considered but emotions, delusion, and control of impulses are ignored.

Solution to Question 4:

Luminescent stools are seen in acute phosphorus poisoning. The vomitus and stools will be luminous in the dark.

The stools may give rise to faint fumes, constituting the smoky stool syndrome.

Solution to Question 5:

In sodium amytal poisoning, the capsules stain the stomach mucosa and its contents, turquoise blue. Bluish discolouration of the stomach mucosa is also seen in copper sulphate poisoning.

The appearance of the gastric mucosa in specific poisons:

Poison	Appearance of Gastric Mucosa
Arsenic	Red velvety mucosa
Sulphuric acid	Brown or black with erosion perforation and Carbonization
Carbolic acid	Grey leathery and thickened mucosa
HNO ₃	Yellow
HCN	Pink
KCN	Brick red to brown
Phosphorous	Yellow or grey white
Sodium amytal, Copper sulphate	Blue

Solution to Question 6:

Saturnine gout is seen in chronic lead intoxication or poisoning.

The features of saturnine gout are the same as primary gout but it affects the knee more than the first metatarsophalangeal joint and tophi rarely develop. Lead exposure should be suspected if a patient presents with the triad of:

- Saturnine gout
- Hypertension
- Renal insufficiency (renal failure and interstitial fibrosis)

Solution to Question 7:

Basiocciput fuses with basisphenoid around 18-22 years.

Solution to Question 8:

Edmond Locard is famous for his theory of exchange or Locard's exchange principle.

This theory states that every contact leaves a trace. When two objects come into contact with one another, each will take something from the other object or leave something behind.

Locard is also renowned for his contribution to the improvement of dactylography, an area of study that deals with fingerprints. He developed the science of poroscopy, the study of fingerprint pores, and the impressions produced by these pores.

Solution to Question 9:

The mummified body is odorless.

Solution to Question 10:

Arsenic poisoning causes PCT damage with proteinuria and muscle pain.

Nephropathy is caused by most of the heavy metals. However, muscle pain is seen with arsenic poisoning.

Salient features of Arsenic poisoning:

Acute poisoning: Resembles cholera; garlicky odor

Chronic poisoning: Following features are seen:

- Raindrop pigmentation seen on temples, eyelids and neck

- Aldrich-Mee's lines (transverse white striae on the fingernails)
- Hyperkeratosis of palms and soles
- Painful paraesthesia of hands and feet
- Red velvety appearance of stomach
- Blackfoot disease - a gangrenous condition affecting the extremities

Tests to detect Arsenic poisoning:

- Marsh test and Reinsch test

Treatment:

Freshly prepared hydrated ferric oxide.

BAL(British Anti Lewisite)

Penicillamine(If BAL not available)

Note: Mercury does not cause muscle pain though it commonly causes PCT damage.



Mee's lines- Transverse white striae of the fingernails



Hyperkeratosis of palms and soles



Rain drop pigmentation

Solution to Question 11:

The child delivered after the death of the biological father, or a child delivered after the death of the biological mother, usually when delivered by a cesarean section is a posthumous child.

Illegitimate child is a child born out of wedlock i.e. when the parents are not married.

Supposititious child or fictitious child is when a woman pretends to go through pregnancy and delivery. Later, she produces a living child as her own for obtaining money or for the purpose of claiming the property.

Forensic Medicine NEET 2019

Question 1:

Which Section of the IPC/BNS deals with the punishment of perjury?

- a) 191/ BNS 227
- b) 192/ BNS 228
- c) 193/ BNS 229
- d) 197/ BNS 234

Question 2:

Gastric lavage can be performed in case of poisoning by:

- a) Sulphuric acid
- b) Strychnine
- c) Carbolic acid
- d) Kerosene oil

Question 3:

Frotteurism is _____

- a) Obtaining sexual pleasure by wearing clothes of opposite sex
- b) Desire to seek surgery to become member of opposite sex
- c) Sexual gratification by rubbing his private parts against another person
- d) Exposure of one's genitals to an unsuspecting stranger

Question 4:

Identical twins have similarities in all of the following except:

- a) Blood group
- b) DNA fingerprinting
- c) Fingerprint pattern

- d) Iris colour**

Question 5:

Which of the following statements is true about battered baby syndrome?

- a) Skin lesions and bruises of various ages are seen on the body**
- b) It occurs due to the tendency of a child to fall repeatedly**
- c) Multiple stab wounds are seen**
- d) The syndrome is a result of firearm injuries**

Question 6:

Which of the following tests is based on the principle that prior knowledge of the events/activity will influence the suspect's reaction?

- a) Truth Serum**
- b) Polygraph**
- c) Brain Mapping**
- d) Hypnosis**

Question 7:

In alleged history of gunshot injury, there is burning, blackening, tattooing around the wound. The injury is:

- a) Close shot entry wound**
- b) Close shot exit wound**
- c) Distant shot entry wound**
- d) Distant shot exit wound**

Question 8:

Rigor mortis occurs first in:

- a) Eyelids**
- b) Intestines**

- c) Myocardium
- d) Neck

Question 9:

A patient died from bronchopneumonia following an intracerebral hemorrhage caused by cerebral metastases from a squamous cell carcinoma of the left main bronchus. What should be recorded in the line I (c) of the death certificate?

- a) Bronchopneumonia
- b) Cerebral metastasis
- c) Intracerebral hemorrhage
- d) Squamous cell carcinoma of the left main bronchus

Question 10:

Which of the following acts as a physiological antidote to scorpion venom?

- a) Physostigmine
- b) Norepinephrine
- c) Prazosin
- d) Atropine

Question 11:

Which of the following is commonly used for preservation of viscera in toxicology autopsy?

- a) Glycerin
- b) Rectified spirit
- c) Formalin
- d) Saturated salt solution

Answer Key

Question No.	Correct Option
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1	c
2	c
3	c
4	c
5	a
6	c
7	a
8	c
9	b
10	c
11	d

Detailed Explanations

Solution to Question 1:

The punishment of perjury is dealt with under IPC 193, has now been updated to BNS 229.

Perjury is defined as the offence of wilfully telling an untruth or making a misrepresentation under oath.

Solution to Question 2:

Gastric lavage can be performed in case of carbolic poisoning.

In carbolic acid poisoning, if the patient survives for 48 hrs, there will be carboluria followed by anuria. The victim will pass dark, smoky urine which soon turns olive green on standing due to oxidation of hydroquinone and pyrocatechol.

It is a corrosive acid, however, it hardens the GI mucosa producing a leather bottle stomach and anesthetizes the nerve endings. This allows for gastric lavage.

Contraindications of gastric lavage:

- Corrosives poisoning (except carbolic acid)
- Convulsants/CNS stimulants
- Comatose patients
- Volatile poison
- Upper GI pathology (eg. esophageal varices)
- Hypothermic patient
- Severe heart disease

- Advanced pregnancy

Sulphuric acid and kerosene are corrosives and strychnine is a convulsant. Hence gastric lavage is not done in these poisonings.

Solution to Question 3:

- Frotteurism is obtaining sexual gratification by contact with another person (rubbing his private parts against female buttocks, breasts, thighs etc. in crowds etc). It is punishable under Section 292 BNS (290 IPC, with a fine of up to 200 Rs). Under the BNS, it is permissible to be fined up to Rs. 1000.
- Eonism (also known as Transvestism) is a sexual perversion where a person's whole personality is dominated by a desire to be identified as a member of the opposite sex. Usually seen in males who obtain sexual pleasure by wearing female clothes. There is no hormonal/genital abnormality.
- Transsexualism: It is the desire to become members of the opposite sex and seek surgery for anatomical alterations.
- Exhibitionism: It involves behaviours over a period of 6 months or more, involving the exposure of one's genitals to an unsuspecting stranger. This is a punishable offence. Section 296 BNS (294 IPC with imprisonment up to 3 months or fine.)

Under the BNS, this is punishable with imprisonment up to 3 months, or a fine of Rs. 1000, or with both.

Solution to Question 4:

Fingerprint pattern differs in identical twins.

Fingerprints are formed during intrauterine life due to friction ridges. These ridges are created by pressure on the fingers from touching the surroundings. Hence, it is unique to every individual.

Solution to Question 5:

Skin lesions and bruises of various ages are a classical feature of a child with battered baby syndrome.

A battered child is one who has received repetitive physical injuries as a result of non-accidental violence, produced by a parent or guardian, there may also be deprivation of nutrition, care, and affection.

The syndrome must be considered in any child:

- In whom the degree and type of injury varies with the history given
- When injuries are of different ages and in different stages of healing
- When there is a purposeful delay in seeking medical attention despite serious injury

- Who exhibits evidence of fracture of any bone, subdural hematoma, failure to thrive, soft tissue swelling or skin bruising
- Who dies suddenly

Skeletal survey, which includes the following, should be done in a case of suspected child abuse:

- AP views of the entire skeleton
- Dedicated views of hands and feet
- Lateral views of appendicular skeleton, skull and spine

Solution to Question 6:

Brain Mapping is based on the principle that if the suspect has prior knowledge of the event/activity being described it will reflect in the suspect's brain wave responses.

It is a technique that measures recognition of familiar stimuli by measuring electrical brain wave responses(P300) to words, phrases, or pictures that are presented on a computer screen.

Truth serum : This is based on the principle, that at a point very close to unconsciousness, the subject will be mentally incapable of resisting questioning and will be incapable of lying.

Polygraph : It is based on the theory, that when a person lies in the answer to a question, if he/she experiences fear, the emotion of fear will result in certain physiological changes.

The physiological changes are commonly observed are :

- Relative rise in blood pressure
- Pulse rate increases
- Slowing down of breathing/erratic breathing
- Suppression of involuntary muscular movements
- Increased activity of sweat glands

Solution to Question 7:

Based on the ballistics, the gunshot injury is close shot entry wound.

- The main objective of study of a firearm wound (Ballistics) is to establish the range and direction of firing, the ultimate purpose is to determine the weapon used for firing.
- Firearm wounds are fundamentally of two main types: wounds of entry and wound of exit. The wound of entry is the wound through which the bullet or pellets enter the body, and the wound of exit is the wound through which the bullet emerges out of the body.
- To determine the Range of firing, following parameters are used:
- Entry wound size and shape
- Blast effect – Muzzle end imprint and edge eversion

- Blackening, Tattooing, Singeing of hair, Abrasion collar and Grease collar
- In the above options, Hair should be preserved in order to determine the range of firing which can be examined for carbonisation/dessication in close range, also from lead deposition around the wound to assess range of firing.

Characteristics	Contact shot	Close Shot	Near Shot	Distant Shot
Range	In-touch with skin	2.5-7.5 cm	30-60 cm	>60 cm
Size	Largest size	Bullet size	Smaller	Smallest
Shape	Stellate	Circular	Circular	Circular
Blast effect				
Muzzle end imprint	+ve	-ve	-ve	-ve
Edge eversion	+ve	-ve	-ve	-ve
Other findings				
Blackening	-ve	+ve	-ve	-ve
Tattooing	-ve	+ve	+ve	-ve
Singeing	-ve	+ve	-ve	-ve
Abrasion collar	-ve	+ve	+ve	+ve
Grease collar	-ve	+ve	+ve	+ve

Solution to Question 8:

Rigor mortis occurs first in the myocardium.

The order of appearance of rigor mortis: Left chamber of heart is affected first. Other muscles are affected in the given order:

- Involuntary muscles: Heart
- Eyelids
- Neck and lower jaw
- Face
- Chest
- Upper extremities
- Abdomen
- Legs
- Fingers and toes

Note- Rigor mortis first occurs in the myocardium but is first seen in eyelids. Here the question is first occurs in and the answer is myocardium.

Solution to Question 9:

Cerebral metastasis will be filled in the line I (c) of the death certificate, as one of the antecedent causes in the order of the events that took place.

The death certificate will be filled in the order of the events that took place, as follows:

Cause of Death		Approximate interval between onset and death
I		
Disease or condition directly leading to death*	(a) <u>Bronchopneumonia</u>	_____
	due to (or as a consequence of)	
<i>Antecedent causes</i>	(b) <u>Intracerebral hemorrhage</u>	_____
Morbid conditions, if any, giving rise to the above cause, stating the underlying condition last	due to (or as a consequence of)	
	(c) <u>Cerebral metastasis</u>	_____
	due to (or as a consequence of)	
	(d) <u>Squamous cell carcinoma of the left main bronchus</u>	_____
II		
Other significant conditions contributing to the death, but not related to the disease or condition causing it.	_____	_____
	_____	_____

*This does not mean the mode of dying e.g. heart failure, respiratory failure. It means the disease, injury, or complication that caused death.

Source : Reference No. 6
Fig. 1 : International form of medical certificate of cause of death.

Solution to Question 10:

Prazosin acts as a physiological antidote for scorpion venom.

The venom is a clear, colorless, proteinous toxalbumen, having a hemolytic and neurotoxic effect. Its toxicity is more than that of snakes, but only a small quantity is injected. The venom is a potent autonomic stimulator, resulting in the release of massive amounts of catecholamines from the adrenals.

If the scorpion has hemolytic venom, the reaction is mainly local and simulates the viper snake bite. The scorpion sting will have only one hole in the center of the reddened area. The extremity will have edema, pain, and reddening. This usually lasts for one to two hours.

The symptoms produced by a neurotoxic venom are similar to a cobra bite. There is usually no marked reaction in the local area. Nausea, vomiting, extreme restlessness, fever, various types of paralysis, cardiac arrhythmias, convulsions, coma, respiratory depression, and death may occur within hours from pulmonary edema or cardiac failure.

Prazosin can counteract the alpha-adrenergic effects of autonomic stimulation, such as hypertension.

Solution to Question 11:

A saturated salt solution is used commonly for the preservation of viscera in toxicology autopsy.

Rectified spirit is indicated for most of the poisons, exceptions being alcohol, kerosene, phosphorus, paraldehyde, carbolic acid, acetic acid, etc.

Forensic Medicine NEET 2020

Question 1:

A deceased male was brought to the morgue for a post-mortem examination. His relatives reported that he had a tattoo over his skin, which was not found on examination. Which of the following structures would you examine?

- a) Lymph nodes
- b) Liver
- c) Arteries
- d) Skin

Question 2:

Which Section of the IPC/BNS deals with the punishment of perjury?

- a) 191/ BNS 227
- b) 192/ BNS 228
- c) 193/ BNS 229
- d) 197/ BNS 234

Question 3:

The most common drug which causes physical dependence is?

- a) Ketamine
- b) Heroin
- c) LSD
- d) Phencyclidine

Question 4:

A 5-year-old child with a history of perianal pain is brought to the OPD. On subjecting the specimen from the perianal region to a test, it produces yellow rhombic crystals of spermine picrate. What is the test done?

- a) Barberio test
- b) Florence test
- c) Takayama test
- d) Teichmann test

Question 5:

In the case of rape of a young child, the hymen is usually:

- a) Ruptured since it is superficially situated
- b) Ruptured since it is very thin
- c) Unruptured since it is deeply situated
- d) Unruptured since it is highly elastic

Question 6:

In the civil negligence case against a doctor who bears the onus of proof?

- a) Patient
- b) Police not under rank of sub inspector
- c) Magistrate
- d) Doctor

Question 7:

Which of the following poison can be found even in skeletonised remains of the body?

- a) Arsenic
- b) Carbon monoxide
- c) Hydrogen sulphide
- d) Phosphorus

Question 8:

A child who is a victim under POCSO act is brought to the department of forensic medicine for age estimation. The X-ray image of the hand is shown below. What is the likely age of the child?



- a) 4 years**
- b) 7 years**
- c) 10 years**
- d) 13 years**

Answer Key

Question No.	Correct Option
1	a
2	c
3	b
4	a
5	c
6	a
7	a
8	d

Detailed Explanations

Solution to Question 1:

The lymph nodes should be examined in this case.

Some amount of material or dye used for creating tattoo marks is transported through lymph to the regional lymph nodes. Hence, in cases where a tattoo mark has disappeared naturally or has been removed, the lymph nodes must be examined.

Solution to Question 2:

The punishment of perjury is dealt with under IPC 193, has now been updated to BNS 229.

Perjury is defined as the offence of wilfully telling an untruth or making a misrepresentation under oath.

Solution to Question 3:

Heroin is the most common drug which can cause physical dependence.

Physical dependence is defined as a condition where withdrawal of a drug after its prolonged use causes physiological and emotional illness, often referred to as the withdrawal syndrome. Neither physical dependence nor withdrawal symptoms occur with hallucinogens. Although it is necessary to note that a user can develop a psychological dependence on the insight-inducing experiences of episodes of hallucinogen use. Hence, all 3 hallucinogens - ketamine, LSD, and Phencyclidine (PCP) does not cause physical dependence.

According to the world drug report 2019, released by the United Nations office on drugs and crime (UNODC), the most common drugs used across the world (in decreasing order) include:

- Cannabis
- Opioids
- Amphetamine and prescription stimulants
- Ecstasy
- Cocaine

Among the opioids, the commonly used group of drugs are opiates which include drugs like opium, morphine, and heroin.

Solution to Question 4:

The test done in the scenario is the Barberio test.

It is used to detect seminal fluid. When saturated aqueous or alcoholic solution of picric acid is added to the spermatic fluid, yellow needle-shaped rhombic crystals of spermine picrate are produced.

Florence test:

- Used to detect: semen
- Stain used: 10% hydrochloric acid and Florence solution (potassium iodide, iodine, and water)
- Finding: If semen is present, dark brown rhombic or needle-shaped crystals of choline iodide appear immediately

Takayama test (Haemochromogen crystal test):

- Used to detect: bloodstains
- Stain used: Takayama reagent
- Finding: If bloodstain is present, pink feathery crystals of haemochromogen or reduced alkaline haematin appear in a few minutes

Teichmann test (Haemin Crystal Test):

- Used to detect: bloodstains
- Stain used: glacial acetic acid and a small crystal of sodium chloride
- Finding: If bloodstain is present, faint yellowish-red to brownish-black rhombic crystals of haemin or haematin chloride appear. When hydrogen peroxide is added, the haematin crystals give bubbles of gas

Solution to Question 5:

In the case of a young child, the hymen is deeply situated and usually intact after intercourse. As the vagina is very small, it is impossible for the penetration of the adult organ to take place.

In children, there are few or no signs of general violence, as the child is unaware of the assault and incapable of resisting. A few indicators for inquiry are:

- Pain on urination and/or defecation
- Abdominal pain
- Inability to sleep
- Sudden withdrawal from peers/adults
- Feelings of anxiety
- Nervousness
- Helplessness
- A feeling of ending one's life

Solution to Question 6:

In the case of civil negligence, the burden of proof is on the patient.

The question of civil negligence arises when a suit is filed in a civil court:

- By a patient or his/her relatives against the doctor for getting compensation from the doctor if he/she has suffered an injury due to the doctor's negligence.
- By a doctor against the patient or his/her relatives to collect the fees, the payment of which has been refused due to alleged professional negligence by the doctor.

In either case, the patient has to prove that the doctor has committed professional negligence.

To prove negligence, all the following four elements must be present:

- Duty - A duty of care by the doctor must exist.
- Dereliction - The physician must maintain the standard of a 'prudent physician' under similar circumstances. If the physician fails to maintain the standard of care, it is called dereliction.
- Direct causation - The negligent act or omission must directly cause the injury without any intervening cause.
- Damage - The negligent act must cause damage of a type that would have been foreseen by a reasonable physician.

Solution to Question 7:

Arsenic can be found even in skeletonized remains of the body.

It becomes fixed in the cancellous tissues of the long bones by the conversion of phosphates to arsenates. The elimination of arsenic in this form is much slower and can be detected in the bones for several years after death.

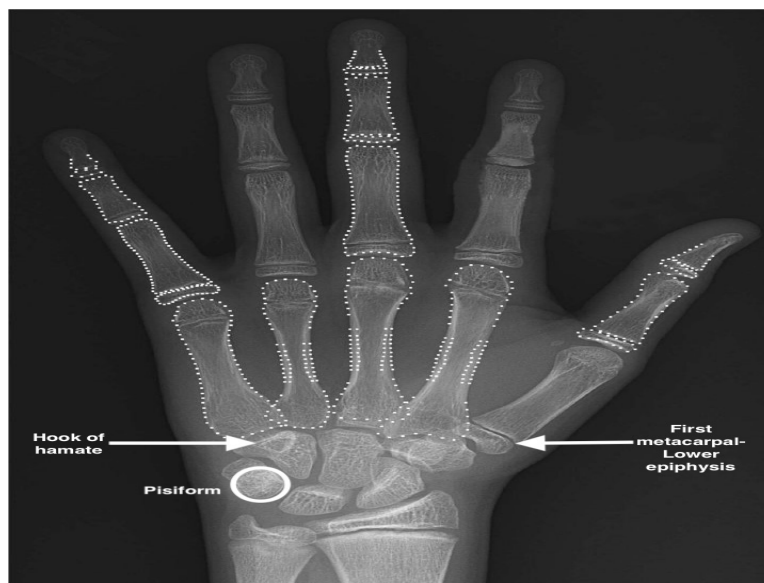
Heavy metals such as lead, copper, mercury, including metalloids like arsenic and antimony, may be found in bones for a prolonged period in cases of chronic poisoning. They are also deposited in hair, nails, etc. For these reasons, when heavy metal poisoning is suspected, these specimens are collected in addition to the routine viscera and body fluids.

Solution to Question 8:

The child's age is likely to be 13 years.

In the given X-ray image of the wrist and hand, all the carpal bones are seen. The lower epiphysis of the first metacarpal has not united with the shaft. This suggests that the age of the child is between 8 to 15 years.

Additionally, the presence of the hook of the hamate in the radiograph suggests that the age is between 13-14 years.



The age of ossification of the carpal bones:

The age of fusion of joints:

Carpal bone	Age of ossification
Capitate	1-3 months
Hamate	2-4 months
Triquetrum	2-3 years
Lunate	2-4 years
Scaphoid	4-6 years
Trapezium	4-6 years
Trapezoid	4-6 years
Pisiform	8-12 years
Hook of the Hamate	13-14 years

Bone/ joint	Maximum age at fusion
Elbow joint centers Metacarpals	16 years
Wrist joint centers	19 years
Hip bone centers	22 years

Forensic Medicine INI-CET 2021

Question 1:

Arrange the following in the correct order of symptomatology seen in iron toxicity:

- a) 2 -> 3 -> 1 -> 4
- b) 2 -> 3 -> 4 -> 1
- c) 3 -> 2 -> 4 -> 1
- d) 3 -> 4 -> 1 -> 2

Question 2:

Match the following weapons with their possible injury type

- a) A-4,B-3,C-1,D-2
- b) A-1,B-3,C-4,D-2
- c) A-1,B-2,C-4,D-3
- d) A-4,B-1,C-3,D-2

Question 3:

Match the following postmortem findings and their time of appearance:

- a) A-3, B-2, C-1, D-4
- b) A-3, B-1, C-2, D-4
- c) A-4, B-2, C-1, D-3
- d) A-1, B-2, C-3, D-4

Question 4:

Which of the following is true about the the cause of the given finding?



- a) Lightning strike
- b) Electrocution
- c) Marbling
- d) Trickling down of acid

Question 5:

A 24-year-old man who was accused of murder was taken into police custody. As a part of the interrogation, the prisoner was kept in a prone position with both his wrists and ankles bound behind his back and secured with a rope. What is this method known as?

- a) Choke/carotid hold
- b) Carotid sleeper hold
- c) Hog tie
- d) Bar arm

Question 6:

An 18-year-old female was sexually assaulted by a 20-year-old male. There was presence of white stains on the clothes of the victim. Barberio's test was performed on the stains. Which of the following is detected by this test?

- a) Spermine
- b) Acid phosphatase
- c) Choline

d) Fructose

Question 7:

Match the following with the respective ages of fusion:

- a) A-1, B-2, C-3, D-4**
- b) A-4, B-1, C-2, D-3**
- c) A-4, B-3, C-1, D-2**
- d) A-1, B-4, C-3, D-2**

Question 8:

A skull was recovered from a forest. According to inquest papers, a girl had gone missing 15 days back. The skull was sent to the forensic research lab. Which of the following would identify it as a female skull?

- a) 1 and 4 only**
- b) 2, 3 and 4**
- c) 1, 3 and 5**
- d) 3, 4 and 5**

Question 9:

The dead body of a 20-year-old man found in the sea was brought in for postmortem examination. Which of the following findings would you see in seawater drowning?

- a) 1 and 3**
- b) 1 and 4**
- c) 2, 3 and 5**
- d) 2, 4 and 5**

Question 10:

A pelvis is brought for forensic examination. Which of the following characteristics will identify the given pelvis as a male?

- a) 2,3,4,5**

- b) 4,5,6
- c) Only 5
- d) 1 and 3

Question 11:

Match the following in relation to fingerprint changes.

- a) 1-e, 2-c, 3-b, 4-a, 5-d
- b) 1-d, 2-b, 3-c, 4-e, 5-a
- c) 1-c, 2-a, 3-d, 4-b, 5-e
- d) 1-d, 2-c, 3-a, 4-e, 5-b

Question 12:

A patient came to the casualty after a road traffic accident. A lacerated wound of 4x1 cm, which is bone-deep, is noted over the scalp of the forehead. Which of the following is the correct mechanism of this lacerated wound?

- a) Shearing force
- b) Impact of a semi sharp-edged object
- c) Overstretching of skin and subcutaneous tissue
- d) Tissue crushed between weapon and underlying bone

Question 13:

The friend and victim had alcohol and got into an argument over a common girlfriend, following which the friend kills the victim by sitting on his chest and covering his nose and mouth. This is an example of?

- a) Burking
- b) Smothering
- c) Traumatic asphyxia
- d) Overlaying

Question 14:

A person comes to the OPD with a burn injury to the right thigh. On examination, a soddened injury is seen along with erythema, blisters, and a line of demarcation that extends down the limb. What is the most likely cause of the injury?

- a) Electric burn
- b) Chemical burn
- c) Dry heat
- d) Moist heat

Question 15:

Which of the following rules are used in case as tests of insanity?

- a) 1, 2, 3 and 4
- b) 2, 3, 4 only
- c) 1, 2 only
- d) 3, 4 only

Question 16:

A person working in a dye factory presented with nausea, vomiting, dark bloody stools, conjunctivitis, and a burning sensation in the throat and stomach. Which poisoning do you suspect in this case?

- a) Potassium permanganate
- b) Arsenic
- c) Thallium
- d) Lead

Question 17:

Following a blast, a man was brought from the scene to the emergency with multiple small puncture wounds to the right side of the chest, below the nipple, and multiple small abrasions and bruising over the left lower abdomen. Which of the following mechanisms is responsible for these injuries?

- a) Flying missiles
- b) Blast of air

- c) Explosion burns
- d) Building collapse

Question 18:

On conducting the autopsy on a victim of hanging, the ligature mark is seen at the lower 1/3rd of the neck. The victim is seen to have a protruded tongue. He was found with his head hanging to his left side with saliva dribbling from the left angle of his mouth. Both the pupils are dilated and the right eye is open. Compression of which of the following structures is the probable reason for an open right eye in this case?

- a) Right internal jugular vein
- b) Cervical sympathetic chain
- c) Left vagus nerve
- d) Right internal carotid artery

Answer Key

Question No.	Correct Option
1	c
2	a
3	a
4	a
5	c
6	a
7	b
8	c
9	b
10	c
11	a
12	d
13	a
14	d
15	c
16	b

17	a
18	b

Detailed Explanations

Solution to Question 1:

The correct order of symptomatology in iron toxicity is 3 > 2 > 4 > 1.

Symptoms of iron toxicity are divided into 4 stages:

- Stage 1 (few hours after ingestion): vomiting, abdominal pain, gastrointestinal bleeding, shock, acidosis, and coma
- Stage 2 (6-24 hours): Asymptomatic phase
- Stage 3 (24-48 hours): Metabolic acidosis, jaundice, hypoglycemia, shock, coma with hepatic and renal failure.
- Stage 4 (1-2 weeks): Late complications such as gastric stricture and pyloric stenosis.

Iron exerts its toxic effects by increasing capillary permeability, release of hydrogen ions, inhibition of mitochondrial function, and direct corrosive action on the gastric mucosa. Ingestion of 20 to 30 grams is considered to be fatal and the fatal period is around 24-30 hours.

Desferrioxamine is the mainstay of treatment with supportive gastric lavage, administration of egg and milk to form iron-protein complexes, oral MgOH solution, and hemodialysis in severe cases.

Postmortem changes in iron toxicity include:

- Hemorrhagic necrosis of gastric mucosa with perforation of gastric or jejunal wall
- Acute hepatic necrosis
- Degenerative renal tubular changes

Solution to Question 2:

For the above-given weapons, possible injury type is matched as under:

Mechanism of injury with the above weapons

Axe - it is a heavy weapon that may or may not be sharpened. Injury is produced by chopping action of the weapon and it may be in a spectrum of an incised wound to laceration depending upon the sharpness of the blade.

RTA - rubbing of epidermis tangential to a rough or hard surface gives rise to graze abrasion also called brush burn. It can occur in an RTA after a pedestrian is being hit by a vehicle and the victim is slid/grazed for some distance before he comes to a halt.

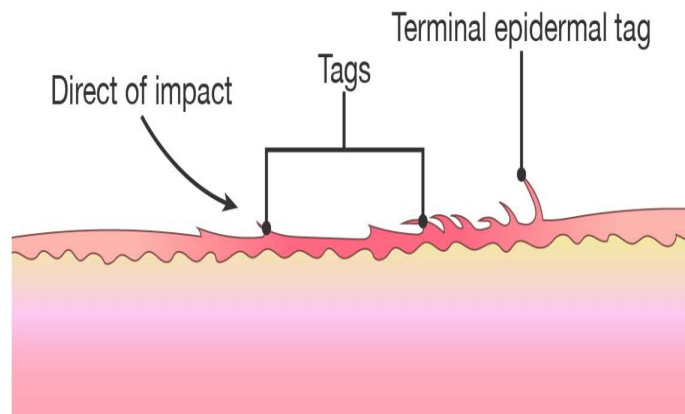
Blade - it gives rise to a clean-cut wound which is usually deepest at its starting. Tailing of the wound indicates the direction of injury. Length is the biggest parameter of this type of wound.

Lathi- It is a blunt force injury giving rise to a patterned abrasion with 2 parallel lines and a normal unbruised area on site of impact or stick/lathi. The lathi forces the blood from the site of impact to either side which gives rise to tramline appearance.

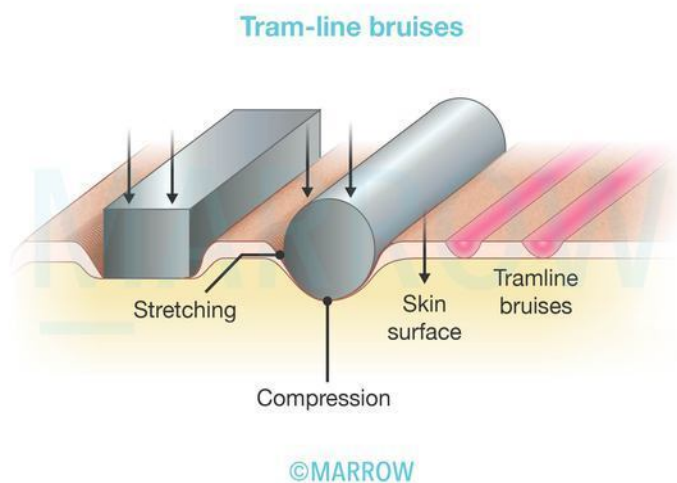
A. Axe	4. Chop wounds
B. RTA	3. Graze abrasions
C. Blade	1. Incised wound
D. Lathi	2. Tram track bruise



The above image shows a chop wound inflicted on scalp.



The above image shows a schematic illustrating heaping up of epidermis in a graze abrasion or brush burn



The above image shows production of a tramline bruise associated with a lathi strike

Solution to Question 3:

The correct match is A-3, B-2, C-1, D-4.

Adipocere formation starts in 4-5 days.

Relaxation of urethral sphincter by 2-3 days.

Marbling of veins starts in 24 hours.

Liquefaction of tissues by 5-10 days.

In postmortem marbling, blood in the vessels is haemolysed to sulfmeth-hemoglobin, which stains the vessel walls and the adjacent tissues, giving rise to a marbled appearance. The marbling of the skin begins by about 24 hours and becomes prominent in 36–48 hours after death. The image below shows postmortem marbling which is distinctly appreciable in the superficial veins of the abdomen, shoulders, chest, and inguinal region.



Urethral sphincter relaxation is of two types;- Primary and Secondary. Primary relaxation occurs before rigor mortis sets in. Secondary relaxation occurs at the time of decomposition i.e. around 2-3 days.

Adipocere formation, also called corpse wax, grave wax, or mortuary wax is a late sign of death. A wax-like organic substance is formed by the anaerobic hydrolysis or hydrogenation of fat in tissues by bacteria like *Clostridium welchii*. It has a greyish-white appearance with a distinctive offensive or sweetish smell. In a country like India, adipocere formation begins within 4-5 days and takes several weeks to complete.

Liquefaction or colliquative changes begin 5 days after death and become prominent in 10 days. Soft tissues change into semisolid masses which are black and get separated from bones and fall off.

Note: The first visible sign of putrefaction is greenish (due to sulfmeth-hemoglobin) discolouration of the anterior abdominal wall, usually in the right iliac fossa.

Solution to Question 4:

The given finding is known as Lichtenberg's figures or filigree burns which are pathognomonic for lightning injury. It is a superficial, thin, tortuous, fern-like pattern of erythema in the skin. It indicates the path taken by the electrical discharge and tends to follow skin creases and long axis of the body.

They are also known as arborescent burns or feathering as they have a resemblance to the branches of a tree. Lichtenberg's figures occur due to the rupture of small vessels. They are usually seen over the shoulder or flank within an hour of injury and disappear in 1-2 days.

Medico-legal aspect: Its recognition may be lifesaving in the unaccompanied comatose patient and is important because even delayed resuscitation of lightning victims can be very successful.

Lesions seen in electrical burns or electrocution are:

- Joule burns: When the skin has been in firm contact with an electrical conductor, this may split the layers of the epidermis or the epidermal-dermal junction and produce a raised blister, called Joule burns.
- Spark lesion: When the contact is less firm, an air gap (albeit narrow) exists between skin and conductor, the current jumps the gap as a spark. This causes the outer skin keratin to melt over a small area. On cooling, the keratin fuses into a hard brownish nodule, usually raised above the surrounding surface, the so-called spark lesion. The heat generated by the current may melt the calcium phosphate, which is radiologically seen as typical round dense foci known as bone pearls or wax drippings.
- Crocodile skin effect is caused by high-tension current burns. In high-voltage burns, such as those sustained from high-tension grid transmission cables, where the voltage is in the multi-kilovolt range, sparking may occur over many centimeters. This can cause multiple spark lesions giving rise to a crocodile-skin effect.

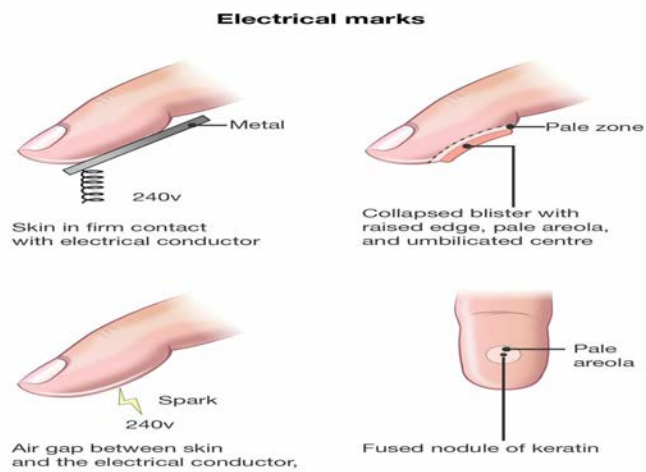
The image given below shows Joule burns



The image given below shows high voltage burns due to electrocution causing multiple spark lesions. Inset shows crocodile skin effect



Image below shows electrical marks



©MARROW

Marbling is a postmortem finding seen when blood in the vessels is haemolysed to sulfmeth-hemoglobin, which stains the vessel walls and the adjacent tissues, giving rise to a marbled appearance.



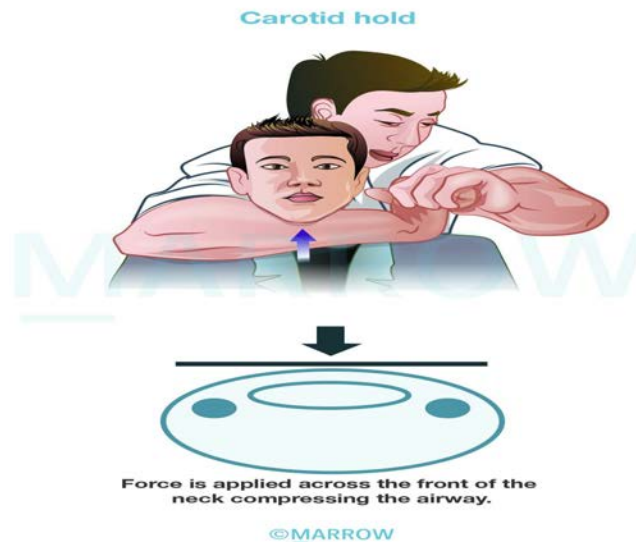
Solution to Question 5:

This type of physical restraint in which a person is placed in prone position with his wrists and ankles bound behind his back and secured by a rope is known as hog-tying or total appendage restraint procedure. This can cause breathing difficulty and the person can become unresponsive. Rarely, death may occur due to asphyxia, soon following the physical restraint.

Hog-tying
(total appendage restraint procedure)



Carotid hold also known as choke hold or bar arm involves placing the forearm straight across the front of the neck. The free hand grips the wrist and pulls it back causing obstruction of the airway and carotids resulting in immediate unconsciousness. Fracture of the hyoid or larynx can occur with greater force. Pressure on the carotid bodies can cause sudden cardiac arrest.



In carotid sleeper hold, also known as lateral vascular neck restraint, the arm is placed about the neck with the antecubital fossa centered at the midline of the neck. The free hand grips the wrist and the arm is pulled backward, compressing carotid arteries and jugular veins but not vertebral arteries or trachea. The person loses consciousness in 10 to 15 seconds and regains consciousness within 10 to 20 seconds upon releasing the pressure. Here, the neck structures are not damaged. Death may rarely occur due to catecholamine release or hypoxia which can trigger myocardial infarction or dysrhythmias in patients with ischemic heart disease.



Solution to Question 6:

Barberio's test detects the presence of spermine which is present in prostatic secretions and thus helps in the identification of seminal fluid.

In Barberio's test, a saturated aqueous or alcoholic solution of picric acid is added to the spermatic fluid. This produces yellow needle-shaped rhombic crystals of spermine picrate.

Option B: Acid phosphatase test is a quantitative test for the presence of acid phosphatase enzyme. High levels of acid phosphatase from prostatic secretions are present in seminal stains (The normal vaginal secretions have low levels of acid phosphatase). Positive reactions are found till 36 hours after which the enzyme levels fall and disappear within 72 hours. In the absence of demonstrable sperms or aspermia, this test can be considered conclusive.

Option C: Florence test detects choline which is secreted by seminal vesicles. The suspected seminal stain is extracted with 10% hydrochloric acid and a drop of it is placed over a slide. A coverslip is placed over it and a drop of Florence solution (potassium iodide, iodine, and water) is made to run under the coverslip. If semen is present, dark brown rhombic or needle-shaped crystals of choline iodide appear immediately. The test is not confirmatory for the presence of semen, however, a negative reaction rules out the presence of semen.

Option D: Fructose acts as an energy source for sperm. Its presence in seminal fluid indicates a patent efferent ductal system, as fructose will be absent in its obstruction.

The absolute proof of semen is finding at least one unbroken spermatozoon or electrophoretic LDH isoenzyme detection of sperms.

Solution to Question 7:

The correct match is A-4, B-1, C-2, D-3.

The respective ages of fusion are:

- The medial end of clavicle at 21-25 years
- Sacrum as a single bone at 22-25 years
- Crista scapulae at 50 years
- Lambdoid suture at 45 years

The secondary ossification center at the medial end of the clavicle appears at 18-19 years of age and fuses by 20-22 years of age. This occurs 2 years earlier in females. Thus the complete fusion of the medial end of clavicle can be used for establishment of age of 21 years in medicolegal cases.

The 5 sacral vertebrae are separated by cartilage till puberty. Fusion of epiphysis begins after puberty and ossification start from below upwards. Sacrum becomes a single bone by the age of 25 years.

Ossification of scapulae:

- 30-35 years: lippling starts on ventral margin of glenoid cavity
- 35-40 years: irregular lippling around the clavicular facet and inferior surface of acromion process.
- 45 years: localized bony atrophy
- 50 years: Cristae scapulae

Lambdoid suture starts closing endocranially at 25-35 years and completes fusion by 45-50 years. Endocranial fusion occurs before ectocranial fusion.

The sagittal suture is used for age determination even though age estimation by skull sutures is not considered to be very reliable. The last skull suture to fuse is the squamous temporal suture, which fuses after 80 years of age.

Age	Age determination
<5 years	Appearance of ossification centers
5-16 years	Analysis of hand and wrist
>16 years(male)	Complete fusion of distal radius or ulna
>16 years(female)	Complete fusion of distal ulna
>18 years(male)	Complete fusion of distal femur or proximal tibia
>18 years(female)	Complete fusion of iliac crest
>21 years	Complete fusion of medial end of clavicle

Solution to Question 8:

Larger frontal and parietal eminences, smooth glabella, and narrow mastoid are features of a female skull.

Differentiating features between male and female skull:

Note: The features given above develop after puberty and are modified by senility, so are applicable only between the 20th and the 55th year.

Trait	Male	Female
General size	Larger, longer (dolichocrania)	Smaller, rounder (brachyocrania)
Architecture	Rugged	Smooth
Glabella	Rough	Smooth
Supraorbital ridges	Prominent	Less prominent
Cheek bones	Heavier, laterally arched	Lighter, compressed
Zygomatic arch	More prominent	Less prominent
Mastoid process	Prominent, wide and long	Less prominent, narrow and short

Trait	Male	Female
Occipital region	Marked muscle markings	No marked muscle markings
Frontal eminences	Small	Large
Parietal eminences	Small	Large
Orbits	Square with round margins	Round with sharp margins
Mandible	Large, broad ramus	Small, small ramus
Chin shape	U-shaped	V-shaped

Solution to Question 9:

In cases of seawater drowning, hypernatremia and myocardial anoxia are seen. Seawater being hypertonic to plasma causes a shift of fluid from the blood to alveoli causing pulmonary edema. There is a net decrease in blood volume causing hemoconcentration and a fall in blood pressure.

This hemoconcentration in seawater drowning causes increased viscosity of blood and myocardial anoxia leading to heart failure. Hyperkalemia is not seen in seawater drowning because RBCs get shrunken in contrast to freshwater drowning.

Freshwater is hypotonic to plasma and is rapidly absorbed from the alveoli into the RBCs resulting in hemodilution and hyponatremia. The RBCs finally swell up and lyse resulting in the release of intracellular potassium causing hyperkalemia. Increased intravascular blood volume combined with hyperkalemia causes arrhythmia.

The table below summarises the cause of death in different types of drowning.

Type of drowning	Cause of death
Freshwater drowning Seawater drowning	Disruption of surfactant
Dry drowning	Laryngeal spasm/Vagal inhibition
Post immersion Syndrome / Secondary drowning	Cerebral anoxia Pneumonia Pulmonary edema Chemical pneumonitis Metabolic acidosis

Solution to Question 10:

Among the given options, a large acetabulum (only 5) is the characteristic of a male pelvis.

Triangular obturator foramen, U-shaped pubic angle, everted ischial tuberosity, large greater sciatic notch, deep preauricular sulcus are characteristics of a female pelvis.

The differences between male and female pelvis are summarized in the table below.

Characteristic	Male pelvis	Female pelvis
General Morphology	Massive, rougher, marked muscle attachments	Less massive, slender, smoother
Overall shape	Deep funnel	Flat bowl
Preauricular sulcus	Narrow, shallow	Broad, deep
Acetabulum	Large, wider, deeper	Small, narrower
Body of pubis	Narrow, triangular, thick	Broad, square
Obturator foramen	Large, oval with base upward	Small, triangular with apex forwards
Subpubic angle	V-shaped, narrow, (70°)	U-shaped, rounded, wide, (90°)
Greater sciatic notch	Smaller, narrower, deeper	Larger, wider, shallower
Ischial Tuberosity	Inverted	Everted, more widely separated
Pelvic Brim	Heart-Shaped	Circular or elliptical
Pelvic outlet	Smaller	Larger
Sacroiliac articulation	Large extends to 2 1/2 to 3 vertebrae	Small, oblique, extends to 2 to 2 1/2 vertebrae
Sacroiliac joint surface	Large and less sharply angulated	L-shaped and elevated anteriorly
Sacrum	Longer, narrower, with more evenly distributed curvature, promontory well marked. Body of first sacral vertebra larger.	Shorter, wider; upper half almost straight, curve forward in the lower half, promontory less marked. Body of first sacral vertebra small

Solution to Question 11:

The correctly matched option is 1-e, 2-c, 3-b, 4-a, 5-d.

The study of ridge patterns in the skin is Dactylography/ Galton-Henry system/ Dermatoglyphics. Fingerprints are the impressions of patterns produced by the papillary or epidermal ridges present on the fingertips.

Some changes in fingerprints include:

- Moderate epidermal ridge atrophy and loss of pattern in celiac disease
- Incomplete atrophy of the ridges in dermatitis
- Ridge alteration occurs in eczema, acanthosis nigricans, scleroderma, and dry or atrophic skin
- Permanent impairment of the fingerprint pattern occurs in leprosy, electric injury, and after exposure to radiation

- Distance between the ridges is changed but the pattern is not altered in infantile paralysis, rickets, and acromegaly

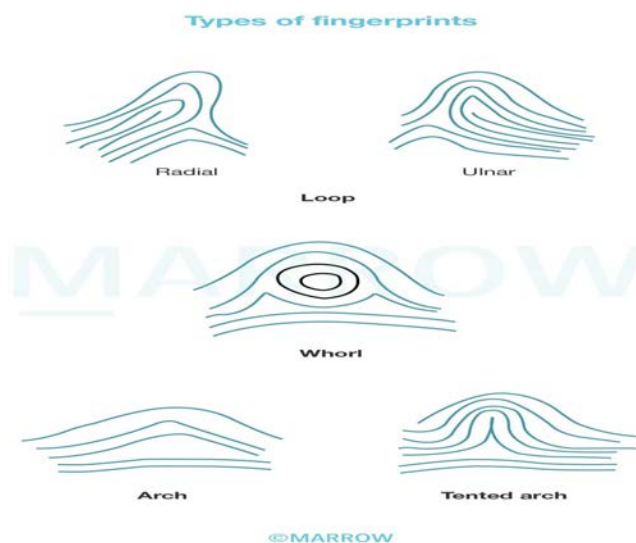
Classification of fingerprint ridges:

- Loops: radial or ulnar
- Whorls: concentric, spiral, double spiral, or almond-shaped
- Arches: plain, tented, exceptional
- Composite: They are a combination of 2 or more types

The fingerprint patterns are not inherited and cannot be used to prove paternity. The fingerprint pattern differs even between identical twins.

Edgeoscopy is the study of edges of friction ridges which normally show seven characteristic features. Ridgeology is the study of friction ridges which are composed of the edges of ridge units that vary in size, shape, and alignment.

1. Incomplete atrophy of ridges	e. Dermatitis
2. Loss of pattern with ridge atrophy	c. Celiac disease
3. Altered ridges	b. Scleroderma
4. Permanent loss of fingerprints	a. Radiation
5. Distance between ridges changed but pattern retained	d. Acromegaly



Solution to Question 12:

The description of the wound given in the question suggests a split laceration and the mechanism is the tissue being crushed between the underlying bone (i.e. skull) and the weapon.

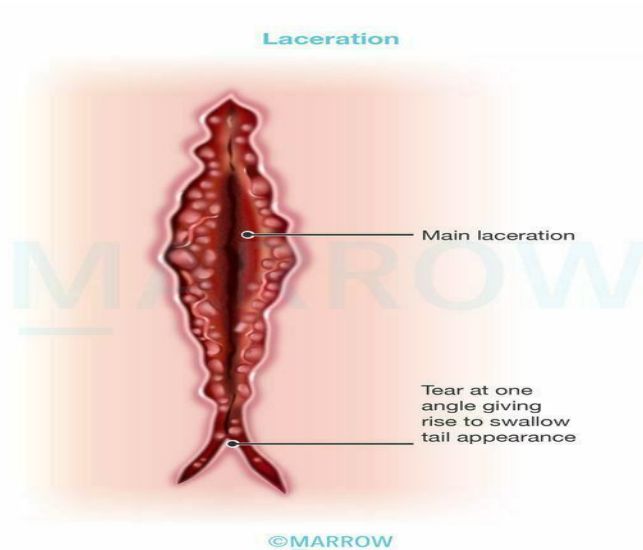
A laceration is a tear or a split in the skin, mucous membrane, muscle, or any internal organ produced by blunt force. The margins are irregular and may show minute tears and bruising. Tears that are seen at the ends, diverging away from the main laceration are called swallowtails. In skin lacerations, usually one of the margins overhangs the other. This helps in assessing the direction of the force.

Classification of a laceration:

- Split laceration (incised-like laceration): It occurs when the skin is crushed between two hard objects. For example, during a fall, the scalp gets crushed between the skull from inside and hard ground outside. This results in a wound, which usually mimics an incised one unless examined closely.
- Incised-looking lacerated wound is a wound with the linear splitting of the tissue caused by blunt force. It is seen where the skin overlies an area of the bone where there is scanty underlying subcutaneous tissue such as the scalp, forehead, shin, etc.
- Avulsed laceration: A lesion where the skin separates or peels off by tearing from the underlying tissues. An avulsion is a laceration produced by a shearing force delivered at an acute angle to tear off a portion of a traumatized surface or viscus from its attachments. It may overhang the wound. It is also called flaying. This is common in road traffic accidents. (Option A)
- Tears: Tearing of skin and subcutaneous tissue can occur from localized impact by or against irregular or semi-sharp objects. It may be due to blows by broken glass, or a fall on a rough projected object. (Option B)
- Stretch laceration: It occurs due to the over-stretching of the skin, beyond its elasticity. This is seen in wounds produced from running over by a motor vehicle, and when a fracture produces a sudden deformity of the bone. (Option C)
- Cut laceration: Injury by a heavy sharp-edged instrument. For example, cut from ax, chopper, etc

The image shows a split laceration which is incised looking laceration.





Solution to Question 13:

The given scenario is an example of burking.

Burking is a method of homicidal smothering and traumatic asphyxia where the assaulter kneels or sits on the chest (traumatic asphyxia) and close the nose and mouth with his hands (smothering). This was the method of killing performed by William Burke and William Hare.

Image shows burking



Other options:

Option B: Smothering is a form of asphyxia that is caused by closing the external respiratory orifices either by the hand or by other means or blocking up the cavities of the nose and mouth by a foreign substance, such as mud, paper, cloth, etc.

Below image shows smothering



Option C: Traumatic asphyxia results from respiratory arrest due to mechanical fixation of the chest, by a heavy substance or object. The normal movements of the chest wall are prevented. There is a gross compression of the chest and usually abdomen by a powerful force, due to which chest expansion and diaphragmatic lowering are prevented. For example stampede deaths, crushing by falls of the earth in coal mines, or earthquakes.

Option D: Overlaying or compression suffocation results due to compression of the chest, so as to prevent breathing. It occurs when an intoxicated parent or other person shares a bed with an infant. The thoracic movements are limited and respiratory exchange is either reduced or completely prevented which leads to death. It is always accidental.

Below image below shows overlaying.



Solution to Question 14:

The soddened injury with erythema, blisters, and a line of demarcation that extends down the limb is highly suggestive of a scald injury caused by moist heat.

Scald injury is an injury that results from the application of liquid above 60°C or from steam i.e moist heat. Redness appears immediately followed by blistering in a few minutes. Removal of the blistered skin leaves a raw, pink surface. Blisters are surrounded by a hyperaemic zone. Scalds appear soddened and bleached but the hair does not get singed. The skin does not get blackened or charred.

Scald injuries are limited to the area of contact and are more severe at the initial point of contact. It can occur through clothing, which worsens damage due to a longer period of contact with the hot liquid and a reduced cooling effect of evaporation.

It can be

- Immersion burns: Accidental or deliberate immersion in hot liquid, usually water.
- Splash or spill burns: Usually accidental.
- Steam burns: Exposure to superheated steam.

Clinical degree of scalding:

- Erythema (reddening)
- Vesication (blister formation)
- Necrosis of dermis when deeper layers of skin are involved.

The image below shows a scald burn due to moist heat.

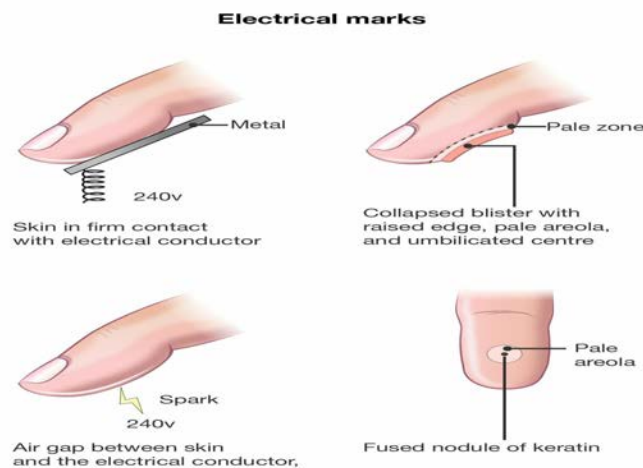


Other options:

Option A: Electric burns are injuries caused due to electric current, characterized by puckering of the skin at the edges of the burns. It is surrounded by an area of pallor. There is no red line at the point of entry and exit. The characteristic marks seen at the entry point are called Joule burns, or electrical burns/marks. They are chalky white, shallow, centrally collapsed blisters with a raised

border.

The image below shows electric marks.



©Marrow

Option B: Chemical burns by acid or alkalis appear as ulcerated patches without blisters. The hair is not singed and the red line of demarcation is absent. Sometimes, the burn shows distinct coloration and is usually uniform in character. Strong acids produce dark leathery burns on the skin. Strong alkalis cause the skin to slough and leave moist, slimy, greyish areas.

Option C: Dry heat injuries are caused by a flame, via direct conduction or radiation to the skin. These are characterized by singeing of the hair and blackening of the skin. Vesication may or may not be present.

Solution to Question 15:

Durham's rule and Curren's rule are used in cases as tests for insanity.

Durham's rule states that an accused person is not criminally responsible if, his unlawful act is the product of mental disease or mental defect.

Curren's rule states that an accused person is not criminally responsible if, at the time of committing the act, he did not have the capacity to regulate his conduct to the requirements of the law as a result of mental disease or defect.

In India, the law of Criminal responsibility is under Section 84 I.P.C. It states that "Nothing is an offense which is done by a person who, at the time of doing it, by reason of unsoundness of mind, is incapable of knowing the nature of the act, or what he is doing is either wrong or contrary to the law". This is slightly different from McNaughten's rule followed in Britain.

McNaughten's rule states that every man is to be presumed to be sane, and possesses a sufficient degree of reason to be responsible for his crimes until the contrary is proved.

Other rules to assess the criminal responsibility of insane persons:

- The Irresistible Impulse
- Federal rule or American Law Institute test (ALT)
- The Brawner rule.

Note: Ashley's rule is used to differentiate males and females based on the combined length of the manubrium and sternum. It is considered to be male sternum when the combined length is $\geq 149\text{mm}$, while if the combined length is $< 149\text{mm}$, then it is considered female.

Solution to Question 16:

The given clinical scenario of an industrial worker presenting with nausea, vomiting, dark bloody stools, conjunctivitis, burning in the throat and stomach is suggestive of acute arsenic poisoning.

There are 3 types of arsenic poisoning,

- Gastroenteric type is a common form and resembles bacterial food poisoning. There is a sweetish metallic taste. Features include:
 - GIT: Constriction in the throat and dysphagia, burning, and colicky pain in the esophagus, stomach, and bowel occur. Intense thirst and severe vomiting are constant symptoms. Involuntary and frequent dark-colored, stinking, and bloody stools that later become colorless, odorless, and watery resembling rice-water stools of cholera. A garlicky odor of breath and feces may be noted
- Renal: Oliguria, uremia, albuminuria, red cell casts.
- Cardiovascular system: Acute circulatory collapse with vasodilation, increased vascular permeability, ventricular tachycardia, and fibrillation.
- Central nervous system: Headache, vertigo, hyperthermia, convulsions, coma. Peripheral neuropathy is more sensory than motor.
- Death is usually due to circulatory failure.
- Fulminant type: Massive doses of arsenic when rapidly absorbed cause death in one to three hours. Death occurs due to shock and peripheral vascular failure. Arsenic also has a direct action

on the heart muscle. In this type gastrointestinal symptoms are absent.

- Narcotic type: The gastrointestinal symptoms are minimal. There is giddiness, formication, delirium. Arseniuretted hydrogen when inhaled as a gas, destroys hemoglobin and causes hemolysis, hemoglobinuria, and renal failure. Death is instantaneous.

Treatment:

- Repeated stomach washes with warm water and milk.
- Emetics should not be used
- Oral hydrated ferric oxide is given in small doses. It converts toxic arsenic to non-toxic ferric arsenite
- Dimercaptosuccinic acid (DMSA) is preferred, if not available British anti-Lewisite (BAL) is used.

Other options:

Option A: Potassium permanganate causes burning pain from the mouth to the stomach, intense thirst, dysphagia, stridor, and a persistent cough. The oral cavity is discolored, inflamed, and superficially corroded. Soon after poisoning the color is purple-brown, but later it appears coal-black.

Option C: Thallium poisoning is characterized by a triad of symptoms of acute gastrointestinal upset, painful combined polyneuropathy, and hair loss after poisoning. Other symptoms include muscle and joint pain, convulsions, optic neuritis, delirium. Loss of the outer third of eyebrows is significant. Maculo-papular butterfly-shaped skin eruption on the face is characteristic. There may be Mees lines on the nails.

Option D: Acute Lead poisoning causes dry throat, burning abdominal pain, peripheral circulatory collapse, insomnia, paraesthesias, depression, coma, and death. Cerebellar ataxia is common in children with acute lead poisoning. A screening test for lead poisoning is the presence of porphyrinuria, mainly due to coproporphyrin III. Peripheral smear shows basophilic stippling. A Burtonian line may be visible on the gums.

Solution to Question 17:

The given history of multiple small puncture wounds to the right side of the chest, below the nipple, and multiple small abrasions and bruising over the left lower abdomen are highly suggestive of blast injuries sustained due to flying missiles.

Blast injury is a complex type of physical trauma resulting from direct or indirect exposure to an explosion causing a sudden increase or decrease in atmospheric pressure.

T.K.Marshall's (Marshall's triad) described the characteristic injuries associated with bomb explosions. It includes:

- Abrasions
- Bruises/contusions
- Puncture lacerations

Blast injuries are classified as:

- Primary injuries: Effect of transmitted blast waves on gas-containing structures. The external injuries are absent and are characterized by internal injuries. The middle, ear, lungs, and gastrointestinal tract are the most susceptible to this.
- Secondary injuries: Injuries caused by shrapnel and other airborne debris propelled by the explosion. These injuries may affect any part of the body and sometimes result in penetrating trauma. Most casualties are caused by secondary injuries.
- Tertiary injuries: These are caused by blast wind that can throw victims against solid objects.
- Quaternary injuries: All other injuries caused by explosions are not included in the above three categories. eg, flash burns, crush injuries, respiratory effects due to toxic dust, gas, etc.

Option B: Blast of air or blast wave cause primary injuries which involve internal organs without external injury.

Option C: Explosion burns: When the bomb explodes, the temperature of the explosive gases can exceed 2000°C, and the heat radiated momentarily can cause flash burns. It is a quaternary type of blast injury.

Option D: Building collapse can cause crush injuries which is a quaternary blast injury.

Solution to Question 18:

Compression of the cervical sympathetic chain is the probable reason for an open right eye in this case.

Dilated pupils are seen when the ligature knot presses on the cervical sympathetic chain, the eye on the same side may remain open and its pupil dilated termed as *le facie sympathique*. It indicates antemortem hanging.

Dribbling of saliva is the surest sign of antemortem hanging. This occurs due to the increased salivation before death due to the stimulation of the salivary glands by the ligature or by congestive hypoxia. Stimulation of pterygopalatine ganglion also causes increased salivation.

Dribbling of saliva occurs from the angle of the mouth opposite to the side of the knot. When the knot is on the nape of the neck, it occurs across the middle of the lower lip.

The ligature mark is usually above the hyoid bone. It runs obliquely, backward, upwards, is non-continuous, and towards the point of suspension. The mark is noncontinuous because of a gap at the nape of the neck. When the knot is in an inverted V shape when it is in contact with the skin, because the ligature material extends downward on both sides. The skin is usually depressed/grooved, yellowish-brown, dry, hard, with small abrasions called rope burns at its edges.

Protrusion of tongue results from the pressure on floor of the mouth by ligature. It is usually swollen and blue. Injuries include bite marks with or without underlying small hemorrhages ('marginal' hemorrhages).

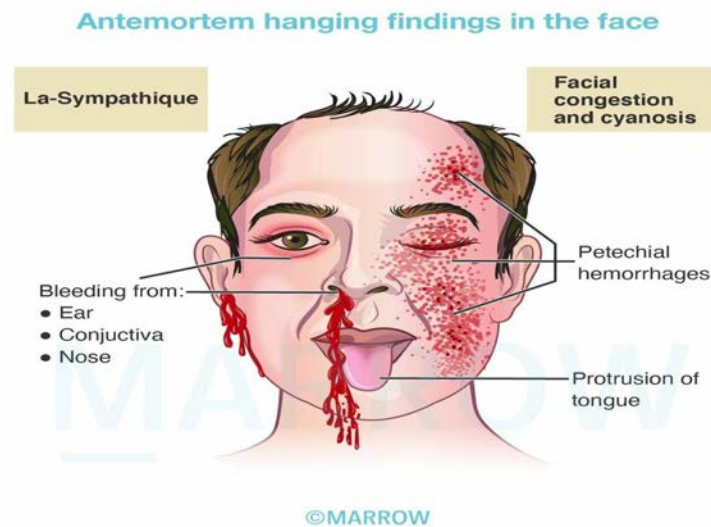
Tardieu's spots may be present on the forehead, over the eyelids, under the conjunctiva, and near the temple.

Other options:

Option A- In partial hanging, internal jugular veins are compressed, but vertebral arteries continue to supply blood to the head, resulting in congestion of the face and petechiae. When suspension is complete, face is pale and petechial haemorrhages are relatively uncommon due to the complete obstruction of the arterial system.

Option C- Vagus nerve compression leads to a sudden halt of the progression of the pure asphyxial process due to the superimposition of vagal inhibition. In these cases, the face remains pale.

Option D- Right internal carotid artery compression can cause transverse tears of carotid artery intima, known as the Amussat's sign. It is seen with antemortem hanging. These are usually seen in obese victims and in posteriorly placed knots.



Forensic Medicine NEET 2021

Question 1:

A guy was playing in a garden. Suddenly, he collapsed and was rushed to the hospital. His friend who played with him in the garden informed the doctors that he ate one of the fruits in the garden post which he seemed to have developed these symptoms. He also had irritability, restlessness, dry hot skin, and was unable to pass urine and stools. Identify the poison and its appropriate antidote.

- a) Datura, Pralidoxime
- b) Datura, Physostigmine
- c) Yellow oleander, Digoxin
- d) Yellow oleander, Physostigmine

Question 2:

Identify the lesion shown in the image below.



- a) Laceration
- b) Incised wound
- c) Laceration looking incised wound
- d) Incised looking laceration

Question 3:

A man was working in a field and he suddenly collapsed on a hot summer afternoon. On examination, there were no physical signs of dehydration. His serum electrolytes level was normal. The doctor found that the body temperature was 106 degrees Fahrenheit. Which of the following symptoms is least likely to be seen in this patient?

- a) Hot skin
- b) Hypotension
- c) Sweating
- d) Disorientation

Question 4:

A young teenage boy with a history of drug abuse had been instilled with a shot of intravenous cocaine at a party. He is rushed to the emergency room by his friends. Some of the symptoms were diaphoresis, headache, and acute coronary spasm. Which of the following symptoms noted would go against the likely diagnosis of cocaine poisoning?

- a) Myocardial ischemia
- b) Bradycardia
- c) Hyperthermia
- d) Agitation

Question 5:

The below phenomenon is due to



- a) Case of torture falanga
- b) Case of suspension/hanging more than 72 hours
- c) Case of immersion in water for 36 hours
- d) Case of colliquative liquefaction due to immersion in hot water

Question 6:

What is the active principle of the poison shown in the image?



- a) Ricin
- b) Bhilawanol

- c) Abrin
- d) Calotropin

Question 7:

Which of the following is true regarding the post mortem change shown in the image given below?



- a) Seen within 24 hours of death
- b) Due to sulphhemoglobin accumulation and formation
- c) Due to aseptic non bacterial cause
- d) It is associated with a case of lightning

Question 8:

A guy was sitting on the balcony which was overlooking a park. He always used to sit nude on his balcony and derived sexual gratification by observing the people that look at him. This is best described as?

- a) Voyeurism
- b) Fetishism
- c) Exhibitionism
- d) Masochism

Question 9:

A woman gave birth to twins. The father who believed that they did not belong to him requested DNA testing. After testing, it was found that one child did not belong to him. This case would be best described as?

- a) Superfetation
- b) Superfecundation
- c) Supposititious child
- d) Atavism

Question 10:

While recording evidence in the court of law, the lawyer asked the witness, "Did A kill B in your presence?" The witness said, "Yes." This type of question would be permitted in which procedural route in the court of law?

- a) Examination-in-chief
- b) Direct examination
- c) Cross examination
- d) Re-direct examination

Answer Key

Question No.	Correct Option
1	b
2	d
3	c
4	b
5	c
6	b
7	b
8	c
9	b
10	c

Detailed Explanations

Solution to Question 1:

The clinical vignette in the question is suggestive of Datura poisoning for which the appropriate antidote is physostigmine, a reversible acetylcholine esterase antagonist. It is an effective treatment for severe anticholinergic poisoning.

Datura poisoning:

The common names include thorn apple, Jimson weed, and Devil's trumpet. The active ingredients include the anticholinergic alkaloids atropine, scopolamine, and hyoscyamine. The seeds or other parts of the plant can be ingested or smoked.

Toxic manifestations of Datura:

- Mnemonic: "hot as a hare, blind as a bat, dry as a bone, red as a beet, and mad as a hatter"
- Also, remember the D's of Datura
- Dry hot skin (Temp rises up to 42 C, a scarlet rash may also be appreciated).
- Dilatation of pupils along with loss of accommodation
- Diplopia
- Dryness of the mouth and throat, unquenchable thirst
- Dysphagia
- Dilatation of cutaneous blood vessels
- Dysarthria (Difficulty in talking)
- Drunken gait
- Delirium
- Drowsiness
- Death due to respiratory or cardiac failure

Treatment

- Gastric lavage or decontamination with activated charcoal or tannic acid. Gastric emptying is often delayed, and the small, plentiful seeds can become trapped among the gastrointestinal folds after ingestion; thus gastric decontamination can be an important therapy.
- Physostigmine in severe toxicity given 1-2 mg IM/IV after every half hour as required. It is a physiological antidote. Pilocarpine may also be given.

Medico-legal importance:

In India, datura is used as a stupefying poison prior to robbery, kidnapping, and rape. It is also known as a rail-road poison, as it is commonly encountered during a journey. Robbers disguised as saints may offer 'prasad' mixed with datura seeds and rob the passengers.

Yellow oleander:

All parts of the plant are poisonous. The active principles are thevatin, and thevatoxin, which are cardiac glycosides. The nectar of the flower yields poisonous honey. Symptoms include

burning sensation in the mouth, tingling of the tongue, dryness of throat, vomiting, diarrhea, headache, dilated pupils, irregular action of heart followed by drowsiness, coma, collapse, and death.

The given image shows the flowers of *Cerebra thevatia*.



Solution to Question 2:

The image depicts a wound with linear splitting of the tissue, sharp margins, and crushed hair follicles, which is suggestive of an incised-looking lacerated wound.

Incised-looking lacerated wounds are seen at sites where the skin overlies an area of the bone where there is scanty underlying subcutaneous tissue. A blunt force causes linear splitting of the tissue which results in the laceration resembling an incised wound. There will be extensive tissue crushing but the margins will be clear and delineable.

Common areas are Scalp, Forehead, Eyebrows, Zygoma, Lower jaw, Perineum, Shin, and Iliac crest.

Laceration (Option-A)

A laceration is a tear or a split in the skin, mucous membrane, muscle, or any internal organ produced by blunt force. The full thickness of the skin is penetrated.

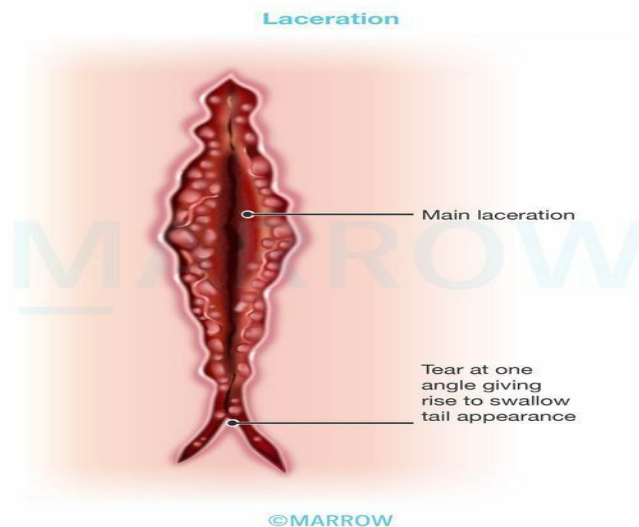
Features of laceration:

- Margins are irregular with pieces of tissue attached in between called tissue tags or bridges. Tears that are seen at the ends, diverging away from the main laceration are called swallow tails.
- Bruising is seen around the margins.
- Deeper tissues are unevenly divided with tissue tags.
- Hair bulbs are crushed.

- Depth of the wound contains abundant foreign material such as dust particles, paint material of the vehicle involved, etc
- In skin lacerations, usually one of the margins overhangs the other. This helps in assessing the direction of the force.
- A laceration, on healing, produces a permanent scar.

Classification of a laceration:

- Split laceration (incised - like laceration)
- This occurs when the skin is crushed between two hard objects
- For example, during a fall, the scalp gets crushed between the skull from inside and hard ground outside. This results in a wound, which usually mimics an incised one unless examined closely.
- Stretch laceration
- As the name implies, it occurs due to the over-stretching of the skin, beyond its elasticity.
- This typically occurs in the setting of a fracture, where the fracture fragment may stretch and pierce the skin to project out
- Avulsed laceration
- An avulsion is a lesion where the skin separates or peels off by tearing from the underlying tissues. It may overhang the wound.
- This is common in road traffic accidents.
- Tears - Tearing of skin and subcutaneous tissue can occur from localized impact
- Cuts - Injury leads to a deep and wide injury leading to a cut



Incised wound (Option-B)

An incised wound is a clean-cut wound made with a sharp-edged instrument, through the skin and subcutaneous tissues, including blood vessels. It is more in length than depth. The edges are well-defined and usually everted, free from contusions and abrasions. It is generally spindle-shaped. The depth is more at the commencement and progressively becomes shallower,

known as tailing. Tailing determines the direction of the force applied when creating the wound. The image below shows two spindle-shaped incisions.



Laceration-looking incised wounds (Option C)

In areas of the body where a loose layer of muscle fibres is attached to the skin, the edges of the incision may become inverted, adherent, and wrinkled leading to it simulating a lacerated wound. Such wounds are called lacerated-looking incised wounds. These are commonly seen in Scrotum and Axilla.

Solution to Question 3:

The given clinical vignette in a patient with no signs of dehydration and body temperature of 106°F is highly suggestive of heatstroke. Sweating is absent in heatstroke.

Heatstroke/ thermic fever/ heat hyperpyrexia is due to the failure of the heat regulation center of the hypothalamus. It is caused by failure of cutaneous circulation and sweating. The predisposing factors include high temperature, increased humidity, minor infections, old age, lack of acclimatization, alcoholism.

Heatstroke can be:

- Exertion-induced heat stroke is seen in people with prolonged exposure to or physical exertion in high temperatures, usually without adequate training.
- Sedentary heatstroke is a disease of elderly or infirm people whose cardiovascular systems are unable to adapt to the stress of a hot environment and release sufficient heat such that body temperature rises.

All the signs of shock are present. Patients usually present with a sudden collapse accompanied by loss of consciousness. Typical vital sign abnormalities include tachypnea, tachycardias, hypotension, and a widened pulse pressure. Other specific signs include rapid Kussmaul type of breathing, the absence of sweating with dry, hot skin, delirium, coma, convulsions,

rhabdomyolysis, multiorgan failure.

Although there is no single specific diagnostic test, clinical features of exposure to heat stress, CNS dysfunction, and core temperature >104.9 -degree Fahrenheit (>40.5 degrees Celsius) help establish the preliminary diagnosis. The body temperature must be measured rectally.

Treatment includes rapidly cooling the body with ice water till the rectal temperature falls below 102°F.

Autopsy findings include the presence of edema and reduction of Purkinje's cells in the cerebellum.

Solution to Question 4:

Cocaine intoxication presents with tachycardia and not bradycardia. Thus, the presence of bradycardia would go against the diagnosis of cocaine intoxication.

Cocaine is an alkaloid deliriant extracted from *Erythroxylum coca*.

Acute intoxication with cocaine leads to symptoms spanning 2 stages-

- Stage of excitement/euphoria - associated with hypervigilance, anxiety, hyperactivity along with hyper-adrenergic signs of mydriasis, tachycardia, elevated blood pressure, hyperthermia, and cardiac arrhythmias.
- Stage of depression - associated with loss of reflexes, muscle paralysis, circulatory and respiratory failure, and coma.

Chronic intoxication is characterized by toxic delirium and tactile hallucination of bugs (cocaine bugs) or vermin crawling under the skin (formication).

Common complications include cerebrovascular accidents and myocardial ischemia.

Treatment includes intravenous diazepam for controlling seizures. For ventricular arrhythmias, propranolol may be given intravenously. If the patient has psychosis, haloperidol may be given. Treatment of chronic cocaine abuse requires psychiatric support.

Cocaine withdrawal is a triphasic syndrome that presents after the sudden stoppage of chronic cocaine use. The phases are as follows -

- Crash phase (9 hours - 4 days): Anorexia, depression, excessive craving, hypersomnia, fatigue.
- Normal mood, anxiety, and anhedonia (4 - 7 days)
- Extinction phase (7 - 10 days): Absence of withdrawal symptoms but increased susceptibility to relapse.

Treatment for withdrawal includes bromocriptine and amantadine. They are useful in reducing cocaine craving.

Solution to Question 5:

The given image shows maceration of the skin of the soles suggestive of washerwoman's feet/ washerwoman's skin. It is seen in a case of immersion in water for 36 hours.

Washerwoman's skin occurs if the body of the person has been immersed in water for a few hours to 3 days. The first signs tend to be on areas with an appreciable keratin layer (such as the fingertips, palms, backs of the hands, and later the soles) where the surface becomes wrinkled, pale, and sodden. This is called washerwoman's skin. The first site affected is the fingerpads. It affects the hands earlier than the feet.

When a body falls into water in average temperate climates, the following is an approximate guide to timing:

The other signs of immersion are:

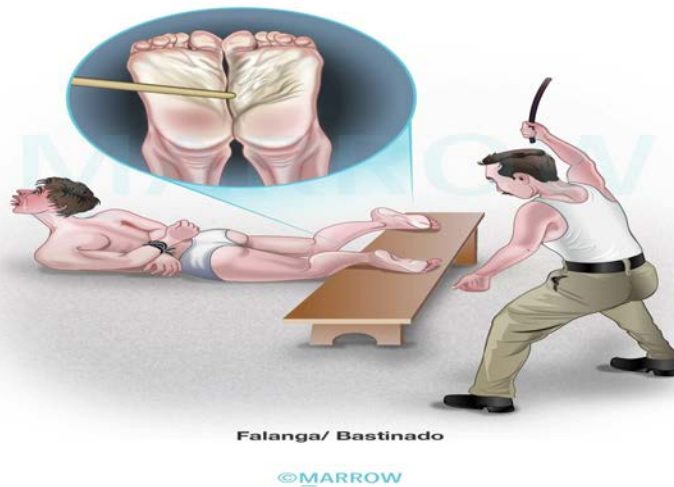
- Cutis anserina - goose-flesh or piloerection: This is usually found when the body is immersed in cold water.
- Post-mortem hypostasis - If the water is still, the postmortem hypostasis affects the heads and the limbs. This is because the chest and abdomen float higher because of the lungs and the abdominal gases.
- Foreign bodies in respiratory passages and stomach - Sand may be found deep in the respiratory passages and stomach, especially if the body has been rolled by the waves on a beach. This is not evidence of live aspiration.

Note: Signs of immersion just indicate that the body was immersed in water, they are not evidence of death from drowning.

Finding	Time since immersion
No wrinkling of finger pads	Less than a few hours
Wrinkled fingers, palms, and feet	Few hours to 3 days
Early decomposition (Head and abdomen)	4-10 days
The marbling of veins, peeling of the epidermis	2-4 weeks
Muscle loss with bone exposure	1-2 months

Falanga (Option-A) also known as Falaka or Bastinado, is a method of torture that involves repeatedly striking the soles of the feet with an object such as a truncheon, iron rod, cable, bamboo cane, wooden pole, or rifle butt.

Eponymous specific physical methods of torture – Beating



Glove-stockings-like staining (Option-B) is used to describe the purple-colored postmortem staining visualized in the dependent parts of the upper and lower limbs of a hanging victim. It is absent in postmortem hanging, where the pattern of hypostasis does not correspond with the position of suspension of the body

Liquefaction or colliquative changes begins by 5 days after death and becomes prominent in 10 days. Soft tissues change into semisolid masses which are black in color and get separated from bones and fall off.

Solution to Question 6:

The given image shows the seeds of *Semecarpus anacardium*, and its active principles are bhillawanol and semicarpol. Common names are marking nut, bhilwan, bibva, bhela, and oriental cashew.

The image shows black, heart-shaped seeds with hard rinds that identify them as seeds of *S. anacardium*. Within the fruit is a thick fleshy pericarp that yields the active principle, which is a brown oily resinous fluid called bhillawanol.

The juice when applied locally to the skin produces irritation, painful blisters, itching, and eczematous eruption. The seeds are used to produce an artificial bruise. Symptoms such as pain, fever, and painful passage of brownish urine may be seen.

A large dose taken orally can produce blisters in the mouth and throat, with gastroenteritis. It also produces dyspnea, cyanosis, tachycardia, coma, and death. The fatal period is 12 to 24 hours.

Medicolegal importance is the use of the juice by malingerers to fabricate wounds. Instilled in the eyes it can cause conjunctivitis or ophthalmia due to its local irritant effect.

Other options:

Option A: Ricin is the active principle of *Ricinus communis*. It is a toxalbumin or phytotoxin that inhibits protein synthesis by affecting the ribosomal RNA polymerase. The seeds are flat to oval-shaped and have dark brown markings (shown below). It is a chemical warfare agent.



Option C: Abrin is the active principle of *Abrus precatorius*. The seeds are egg-shaped, scarlet in color, and have a black spot at one end (shown below). On ingestion, it causes gastroenteritis with bloody diarrhea, rectal bleeding, tachycardia, and circulatory collapse. Injecting the active substance can mimic viperine snakebite. Rarely, it may lead to tetanic convulsions.

The seeds of *A. precatorius* are decocted and mixed with datura, opium, onion, spirit, and water to form a paste. From this, small sharp needles called suis are made. They are used for hunting animals and even for homicide purposes.



Option D: Calotropin is one of the active principles of *Calotropis*. The toxic parts of the plant are the stem, branches, leaves, and milky white latex (Madar juice). When applied locally, it can mimic bruises, and form pustules and vesicles. In the eyes, it can lead to conjunctivitis. On ingestion, it causes abdominal pain, vomiting, mydriasis, and tetanic convulsions and can eventually lead to collapse and death.

It is shown in the image below.



Solution to Question 7:

The postmortem change shown in the given image is marbling. It is seen when blood in the vessels is haemolysed to sulfmeth-hemoglobin, which stains the vessel walls and the adjacent tissues, giving rise to a marbled appearance.

Marbling:

The marbling of the skin begins at about 24 hours and becomes prominent in 36–48 hours after death. It is a visible postmortem change that is characterized by linear branching patterns on skin that correspond to the course of superficial veins. Intestinal bacteria infest the venous system and there is subsequent gas production and hemolysis which stains the vessel walls. This becomes prominent in the summer and is distinctly appreciable in the superficial veins of the abdomen, shoulders, chest, and inguinal region.

Lichtenberg's figures or filigree burns are pathognomonic for lightning injury. It is a superficial, thin, tortuous, fern-like pattern of erythema in the skin. It indicates the path taken by the electrical discharge and tends to follow skin creases and the long axis of the body. They are also known as arborescent burns or feathering as they have a resemblance to the branches of a tree. Lichtenberg's figures occur due to the rupture of small vessels. They are usually seen over the shoulder or flank within an hour of injury and disappear in 1-2 days.

Medico-legal aspect: Its recognition may be lifesaving in the unaccompanied comatose patient and is important because even delayed resuscitation of lightning victims can be very successful.

The image below shows the filigree burns seen in a case of lightning.



Solution to Question 8:

The given scenario is suggestive of exhibitionism.

It involves behaviors over a period of 6 months or more, involving the exposure of one's genitals to an unsuspecting stranger. The individual obtains sexual gratification from this. This is a punishable offense (Section 294 IPC/ Section 296 BNS) with imprisonment up to 3 months or a fine.

The important and common sexual perversions are summed in the pearl attached below. The various punishments in the legal provisions include:

IPC section	BNS Section	Section it deals with	Punishment
290	292	Punishment for public nuisance (Frotteurism)	Fine Up to Rs.1000
294	296	Obscene acts and songs (Exhibitionism)	Up to 3 months of imprisonment or fine (upto Rs. 1000) or both
297	301	Trespassing on burial places/ wounding the feeling of any person /indignity to any human corpse(Necrophilia, Necrophagia)	Up to 1 year of imprisonment or fine or both

Solution to Question 9:

The given clinical scenario where one twin is found to not belong to the father of the other twin is suggestive of heteroparental superfecundation.

Superfecundation is the fertilization of 2 different ova in the same menstrual cycle (as suggested by the same gestational age) by separate acts of coitus. It can occur even after artificial reproductive therapy. Hence, after embryo transfer, intercourse must be avoided. Heteropaternal superfecundation is used to refer to instances of two different males fathering fraternal twins. This leads to the possibility of twins also being half-siblings. This may give rise to the doubt of adultery and infidelity.

Option A: Superfetation is the fertilization of 2 different ova released in different menstrual cycles (i.e ovulation and fertilization during pregnancy). This can occur within 12 weeks of pregnancy after which the decidual space is obliterated and a detailed evaluation will reveal disparities in gestational ages.

Option C: Supposititious child is also known as a fictitious child. In such cases, a woman pretends to be pregnant and generally substitutes a male child for a female child born of her, or after an abortion. This is done for the purpose of claiming property.

Option D: Atavism is the reappearance of a characteristic in an individual after several generations of absence. For example, the child may not resemble his parents but resemble his ancestors. This is attributed to the recombination of genes.

Solution to Question 10:

In the given scenario, the kind of question asked by the lawyer to the witness is an example of a yes/no question or leading question. Leading questions are allowed during cross-examination according to Section 143 IEA (Indian Evidence Act).

Leading questions (Sec. 141 IEA) are defined as questions that suggest the answer that the person asking the question wishes or expects to receive. It is favourable for the case of the summoning counsel or in alignment with their claim. They are not allowed during the direct examination(Examination-in-Chief) conducted by the summoning counsel (Option A, B). The only exception to this is in the event that the Judge is satisfied that the witness is hostile, leading questions may be allowed during direct examination.

In cross-examination, the witness is examined by the defence lawyer or adverse party with the objective to elicit favourable facts to their case, to test the accuracy of the witness statements, unearthing new or old facts, and also, to discredit the witness. While leading questions are allowed in cross-examination, the Judge can disallow questions that are indecent or scandalous (Sec. 151 IEA) or questions asked with an intention to insult, annoy or offend the witness in any form (Sec. 152 IEA). There is no time limit to the duration of cross-examination.

The sequence of examination in the recording of evidence is as follows -

Sequence of Examination	Objective
Examination-in-Chief/Direct examination by the party that summoned the witness	To elicit the facts of the case from the witness. Leading questions are not allowed.

Sequence of Examination	Objective
Cross-examination by the defence lawyer	To elicit facts favourable to their case and test the reliability of the witness. Leading questions are allowed.
Re-examination by the party that summoned the witness	To resolve doubts that formed during cross-examination and re-emphasize the party's perspective

Forensic Medicine INI-CET 2022

Question 1:

Sagar was being tried in a court for the murder of Mohan in the park. Shikhar appeared in court and said he saw Sagar carrying a knife in his hand while in the park. What is this evidence known as?

- a) Direct
- b) Indirect
- c) Hostile
- d) Hearsay

Question 2:

Fingerprint first develops in how many weeks of intrauterine life?

- a) 12-16
- b) 20-24
- c) 28-32
- d) 32-36

Question 3:

Identify the given plant.



- a) *Datura alba*
- b) *Argemone mexicana*
- c) *Nerium odorum*
- d) *Papaver somniferum*

Question 4:

A male patient was presented with a history of consumption of an unknown substance. He had increased sweating, lacrimation, diarrhea, muscle weakness, and bradycardia. What's the management?

- a) Atropine
- b) Calcium gluconate
- c) Pralidoxime
- d) Obidoxime

Question 5:

Which of the following is not a feature of cocaine intoxication?

- a) Tingling and numbness of hands and feet
- b) Constricted pupils and bradycardia
- c) Hypertension
- d) Hyperthermia

Question 6:

Which of the following are seen in patients with chronic arsenic poisoning?

- a) 1,2,3 are correct
- b) 2,3 are correct
- c) 1,4 are correct
- d) Only 1 is correct

Question 7:

Post-mortem examination of a patient reveals the following. What is this finding known as?



- a) Hesitation cuts
- b) Grazed abrasion
- c) Laceration
- d) Patterned bruise

Question 8:

Poisoning with which of the following agents would cause hypertension?

- a) Organophosphorus
- b) Paracetamol

- c) Arsenic
- d) Cocaine

Question 9:

Treatment of choice for acute arsenic poisoning is :

- a) Ipecac
- b) Penicillamine
- c) Activated charcoal
- d) Dimercaprol

Question 10:

To detect vaginal cells on a person who is accused of rape, which test will you use?

- a) Florence
- b) Barberio
- c) Lugol's
- d) Precipitin

Question 11:

Which of these is not a cardiac poison?

- a) Aconite
- b) Cerbera thevetia
- c) Nicotiana tabacum
- d) Atropa belladonna

Question 12:

The image given below is a type of _____.



- a) Contusion
- b) Abrasion
- c) Laceration
- d) Stab wound

Question 13:

Match the following:

- a) A-1, B-3, C-2, D-4
- b) A-3, B-4, C-1, D-2
- c) A-3, B-2, C-1, D-4
- d) A-2, B-3, C-4, D-1

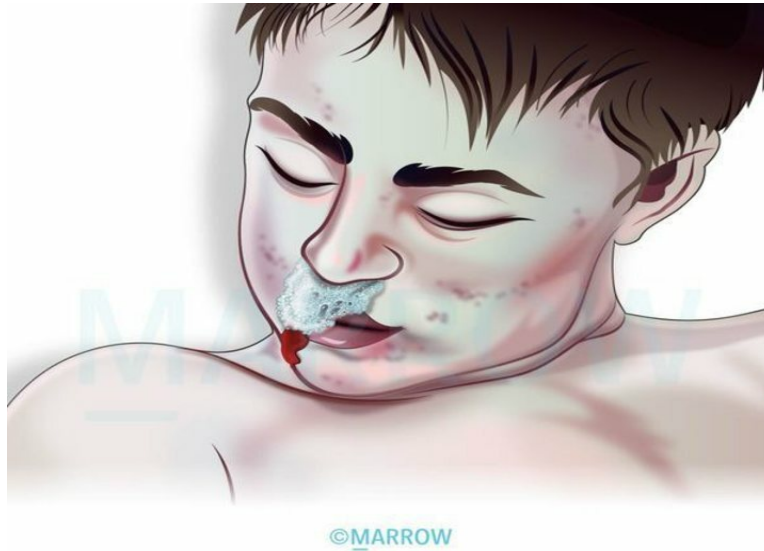
Question 14:

Match the following plant poisons with their active principles.

- a) 1-b,2-c,3-d,4-a.
- b) 1-a,2-c,3-b,4-d.
- c) 1-a,2-b,3-c,4-d.
- d) 1-c,2-b,3-a,4-d.

Question 15:

The dead body of a child was recovered from a river with froth around his mouth, which reappeared even after wiping it off. Which postmortem finding is not seen in this case?



- a) Voluminous and ballooned out lungs
- b) High concentration of arsenic from shaft of hair
- c) Paltauf's hemorrhage in sub pleura
- d) Cadaveric spasm with mud in closed fist

Question 16:

Spanish windlass is a type of which strangulation method?

- a) Bansdola
- b) Garroting
- c) Mugging
- d) Throttling

Question 17:

A person of eonism derives pleasure from _____.

- a) Wearing clothes of opposite sex
- b) Kissing and licking of anus by a sexual partner

- c) Underclothing of female
- d) Seeing a female undressing

Question 18:

A lady underwent cholecystectomy and now she presents with abdominal pain, one month later. An erect abdomen x-ray was taken, and the image is given below. The following offense comes under _____.



- a) Novus actus interveniens
- b) Res ipsa loquitur
- c) Res judicata
- d) Respondeat superior

Question 19:

Tattooing around the entry wound of a bullet is due to?

- a) Unburnt particles
- b) Smoke
- c) Flame
- d) Muzzle impression

Question 20:

Pick the mechanism of cold water drowning.

- a) Asphyxia
- b) Laryngeal spasm
- c) Vagal inhibition
- d) Cardiac arrest

Question 21:

Identify the pattern of abrasion shown in the image below.



- a) Ligature mark
- b) Graze abrasion
- c) Pressure abrasion
- d) Imprint abrasion

Question 22:

Gas shadow is present in the great vessels of a fetus in the case of intrauterine demise. What is this sign called?

- a) Roberts sign
- b) Spalding sign
- c) Ball sign
- d) Halo sign

Question 23:

A man was brought to the emergency department with a history of assault by a mob using heavy blunt objects and sharp objects. On examination, crushing of the hair bulb was seen. Identify the type of wound.



©MARROW

- a) Clean cut incised wound
- b) Lacerated wound
- c) Chop wound
- d) Avulsion injury

Question 24:

Identify the postmortem change given in the image.



- a) Marbling
- b) Tattooing
- c) Lividity
- d) Filigree burns

Question 25:

A male patient working in the matchbox industry for the past 20 years reports to the dentist with multiple sinuses and fistulae in and around his jaw with foul-smelling discharge. Which of the following chemicals are responsible for this?

- a) Bromine
- b) Antimony
- c) Fluorine
- d) Phosphorus

Question 26:

Which of the following rules of law are associated with insanity?

- a) 1, 2, 3 and 4
- b) 2, 3, 4 only
- c) 1, 2 only
- d) 3, 4 only

Question 27:

The steps in the management of snake bite include:

- a) 1, 2, 3, 4, and 5
- b) 1, 2, and 4
- c) 4 only
- d) 4 and 5

Question 28:

Hooch tragedy is related to:

- a) Phosphorus
- b) Mercury
- c) Ethanol
- d) Methanol

Answer Key

Question No.	Correct Option
1	b
2	a
3	b
4	a
5	b
6	a
7	a
8	d
9	d
10	c
11	d
12	a
13	c

14	b
15	b
16	b
17	a
18	b
19	a
20	c
21	c
22	a
23	b
24	a
25	d
26	c
27	d
28	d

Detailed Explanations

Solution to Question 1:

This evidence is known as indirect evidence. Indirect evidence is also known as circumstantial evidence. It is not the direct testimony of an eyewitness. It has an effect on other facts and subsidiary facts which are depended upon as consistent.

Other options:

Option A: Direct evidence is the evidence of a fact that is actually in issue. Eg: The witness saw A shooting B, prescription, consent form

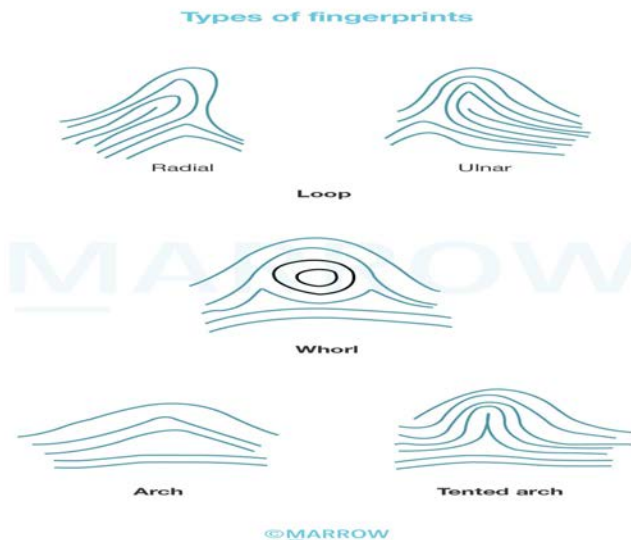
Option C: Hostile evidence is given by someone who is supposed to have some interest or motive for hiding part of the truth. He might give completely false evidence too.

Option D: Hearsay evidence is given by someone about something that he did not personally witness. Sometimes it might be obtained from a third party. It is presented in the court to claim that the facts contained in the statement are true.

Solution to Question 2:

Fingerprint first develops in 12-16 weeks of intrauterine life. It is complete by 24 weeks. In identical twins also the pattern is different.

The image given below shows the different patterns of fingerprints:



Changes might occur in fingerprints in certain conditions, some of them are given below:

Fingerprint changes	Condition
Incomplete atrophy of ridges	Dermatitis
Loss of pattern with ridge atrophy	Celiac disease
Altered ridges	Scleroderma
Permanent loss of fingerprints	Radiation
Distance between ridges changed but pattern is retained	Acromegaly

Solution to Question 3:

The given plant is *Argemone mexicana*.

The flowers are yellow in color. All parts of the plant are poisonous. Oil extracted from argemone seeds is commonly used as an adulterant in mustard oil. Sanguinarine and dihydrosanguinarine are the toxic alkaloids present in the oil and it causes epidemic dropsy.

Epidemic dropsy is characterized by vomiting, diarrhea, edema of the feet, and generalized anasarca. Some patients may develop hypersecretory glaucoma. It can lead to death due to cardiac failure.

Other options:

Option C: *Nerium odorum* - Flowers are usually white, pink, dark red, or pale yellow in colour. All parts of the plant including nectar, are poisonous. It contains several cardiac glycosides such as neriodorin, neriodorein, and karabin. The action of neriodorin mimics digitalis causing death

due to cardiac failure. It can resist heat and hence can be found even in burnt dead bodies.



Option D: *Papaver somniferum* - The flowers are white in color. Opium is the dry juice of this plant. Morphine is isolated from opium. The clinical features include euphoric feeling in the stage of excitement followed by headache, giddiness, and a tendency to sleep in the stage of stupor. The stage of narcosis is characterized by a patient in a deep coma, who cannot be aroused.



Solution to Question 4:

The clinical scenario is suggestive of organophosphate consumption and atropine is used for its management. Atropine is a muscarinic receptor antagonist and it competitively blocks the acetylcholine at muscarinic receptors.

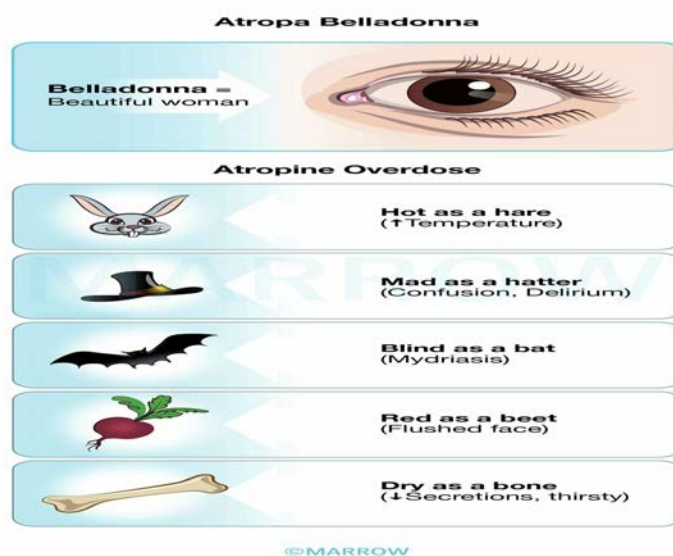
Organophosphates are powerful acetylcholinesterase inhibitors. It is absorbed by inhalation or through the skin. Mucous membranes and the gastrointestinal tract are also the sites for absorption.

The predominant initial signs are of muscarinic excess which includes: (DUMBBELSS)

- Diarrhoea
- Urination
- Miosis
- Bronchospasm
- Bradycardia
- Emesis
- Lacrimation
- Salivation
- Sweating

Nicotinic manifestations include twitching of eyelids, tongue and facial muscles along with tachycardia which then leads to paralysis. CNS stimulation results in headache, tremors, ataxia, coma and death.

Atropine in larger doses can lead to atropine toxicity. The features are depicted in the following image:



Other options:

Option B: Calcium gluconate is used as the antidote for oxalic acid poisoning.

Option C: Pralidoxime is also used in organophosphate poisoning which is used to counter the nicotinic effects of organophosphates.

Option D: Obidoxime is used in organophosphate poisoning if pralidoxime is not available.

Note: In OP poisoning, the frequency of occurrence is Muscarinic --> CNS --> Nicotinic signs and symptoms. The definitive treatment, however, is atropine.

Solution to Question 5:

Constricted pupils and bradycardia are not a feature of cocaine intoxication. Dilated pupils and rapid pulse are seen in cocaine intoxication.

Cocaine is obtained from the leaves of *Erythroxylum coca*. Cocaine can be administered by snorting, intravenous route, or smoking.

The manifestations of cocaine intoxication occur in two stages- the stage of excitement and the stage of depression.

In the stage of excitement, the features seen are dry mouth, dysphagia, tachycardia, tachypnea, hypertension, dilated pupils, and increased temperature. There may be a feeling of well-being, and the patient may be excited, restless, and talkative. This progresses into a calm, dull state. Neurological manifestations include exaggerated reflexes, tremors or convulsions, hallucinations, tingling, and numbness in the hands and feet.

The stage of depression is characterized by feeble respiration, excessive sweating, collapse, convulsions, and death.

The fatal dose of cocaine is 1g when taken orally. The fatal period ranges from a few minutes to hours.

Chronic cocaine users may have the feeling of insects crawling on their skin or the presence of grains of sand under the skin. This is known as cocaine bugs or Magnan's symptom. Ulceration of the nasal septum may be seen in chronic abusers who sniff cocaine.

Treatment includes gastric lavage with warm water containing potassium permanganate, charcoal, or tannic acid if taken orally. When it is through the nose or throat, a wash with water is done. In case of intravenous intoxication, a ligature is applied above the site of the injection.

The antidote for cocaine intoxication is inhalational amyl nitrite. Other measures include control of convulsions, airway and circulatory stabilization, administration of iv thiamine, and naloxone.

Solution to Question 6:

Among the given options, raindrop pigmentation, peripheral neuritis, and mees line are seen in patients with chronic arsenic poisoning. Luminous vomit is a feature of yellow phosphorous poisoning.

Arsenic is an odorless, tasteless, metallic poison. Gastrointestinal features of acute arsenic poisoning include nausea and vomiting, colicky abdominal pain, and profuse diarrhea with rice water stools.

In chronic arsenic poisoning, skin symptoms include erythematous lesions, flushing, and can also lead to hyperkeratotic skin and desquamation. Pigmentation is patchy and resembles raindrops on a dusty road. In the distal parts of the body, hyperkeratosis is present. Mees lines are

transverse white striae of the fingernails, called striate leukonychia. Renal manifestations include chronic nephritis and urine contains albumin, casts, etc.

Arsenic can be detected in the bones for prolonged periods. Hence it can be found even in skeletonized remains of the body. Dimercaprol is the treatment of choice for acute arsenic poisoning.



Solution to Question 7:

The finding is known as hesitation cuts. It is a feature of suicide. They are small multiple cuts which involve only the superficial skin. They are seen at the beginning of the incised wound. The throat, wrist and front of the chest are the most common sites. Unintentional cuts are seen on the fingers in which the razor blade are held.

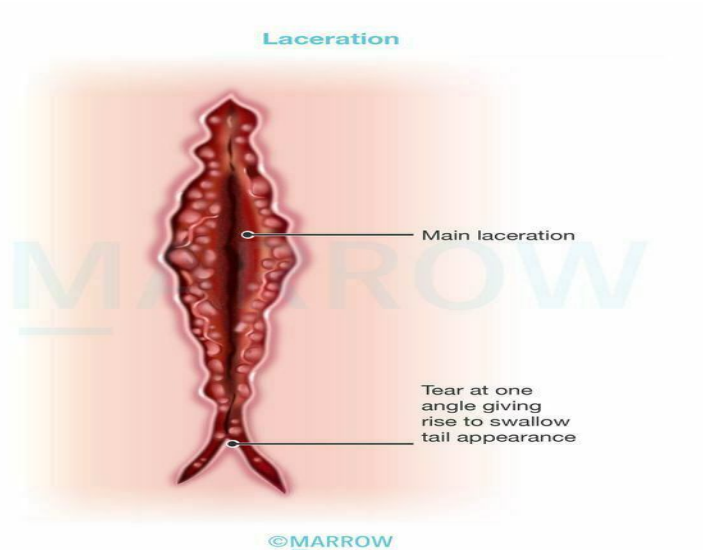
Other options:

Option B: Grazed abrasion occurs when there is movement between skin and some rough surface. The epidermis is destroyed and shows uneven longitudinal parallel lines in the direction of the force. Cleaning and dressing is the treatment.



Option C: In laceration tearing of the skin, mucous membrane, muscles or internal organs occurs due to blunt force application. The tissues are crushed and stretched beyond their elasticity limit.





Option D: In Patterned bruise, it mirrors the pattern of the object used. The tissue is crushed if heavier objects are used and will produce a solid contusion pattern.



Solution to Question 8:

Poisoning with cocaine cause hypertension. This is due to the inhibition of norepinephrine at the adrenergic terminals leading to increased sympathetic stimulation. This effect occurs at all concentrations of cocaine and causes hypertension and ventricular ectopics.

Cocaine is obtained from the plant *Erythroxylum coca*. The symptoms of cocaine poisoning are divided into two stages:

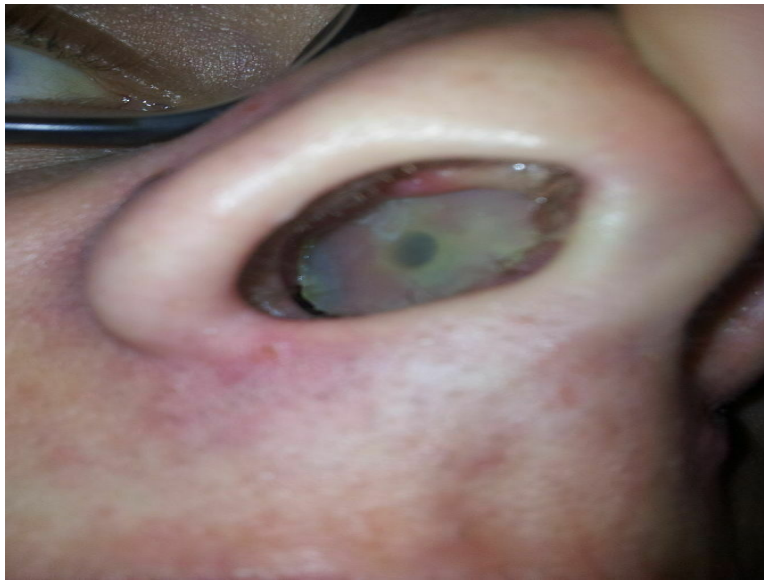
- Stage of excitement: This stage is characterized by features like dysphagia, dryness in the mouth, tachycardia, and dilated pupils. The reflexes are exaggerated and are associated with tingling and

numbness in the hands and feet.

- Stage of depression: This stage presents with feeble respiration, profuse sweating, convulsions, and death. In most cases, the cause of death is cardiopulmonary failure or vascular collapse.

Chronic users develop a feeling of insects crawling through the skin or the presence of grains of sand under the skin. This is pathognomic for cocaine poisoning and is known as cocaine bugs or Magnan's symptoms. Ulceration and sometimes perforation of the nasal septum are seen in chronic abusers due to the sniffing habit.

The image of the nasal septum of a chronic abuser is shown below:



The image of the plant is shown below:



Other options:

Option A: Organophosphorous poisoning produces autonomic instability. Both tachycardia and bradycardia can be seen, however, usually it is the parasympathetic effects which predominate.

Option B: Paracetamol poisoning causes hypotension, tachycardia, diaphoresis, etc.

Option C: Arsenic poisoning causes vasodilation, ventricular tachycardia, ventricular fibrillation, etc.

Solution to Question 9:

Dimercaprol is the treatment of choice for acute arsenic poisoning.

Arsenic is an odorless, tasteless, metallic poison. Gastrointestinal features of acute arsenic poisoning include nausea and vomiting, colicky abdominal pain, and profuse diarrhea with rice water stools.

In chronic arsenic poisoning, skin symptoms include erythematous lesions, and flushing and can also lead to hyperkeratotic skin and desquamation. Pigmentation is patchy and resembles raindrops on a dusty road. In the distal parts of the body, hyperkeratosis is present. Mees lines are transverse white striae of the fingernails, called striate leukonychia. Renal manifestations include chronic nephritis and urine contains albumin, casts, etc.

Arsenic can be detected in the bones for prolonged periods. Hence it can be found even in skeletonized remains of the body.

Other options:

Option A: Ipecac is an emetic used in ergot and hydrocarbon poisoning.

Option B: Penicillamine is used in lead poisoning as a chelating agent.

Option C: Activated charcoal acts as an adsorbing agent in many poisonings like aspirin, phenobarbital poisoning, etc.



Solution to Question 10:

Lugol's iodine test will be used to detect the vaginal cells present on the penis of a rape accused. It shows brown color on filter paper due to the presence of glycogen in epithelial cells of the vagina.

Other options:

Option A: Florence test is used to detect choline in that originates from seminal vesicles. It produces dark brown, rhombic, or needle-shaped crystals of choline iodide if semen is present.

Option B: Barberio test is used to detect spermatic fluid. It produces yellow rhombic crystals of spermine picrate when picric acid is added to spermatic fluid.

Option D: Precipitin test is used for species identification of semen stain. A positive test will confirm human origin.

Solution to Question 11:

Atropa belladonna is a deliriant poison that acts by inhibiting the muscarinic effects of acetylcholine. All the parts of the plants are toxic. The active principle is 1-hyoscyamine.

Clinical manifestations of *A. belladonna* poisoning:

- Fever, flushing, rash
- Neurological symptoms - delirium, slurred speech, headache, choreoathetosis, myoclonus, visual and auditory hallucinations, convulsions, and coma
- Eye symptoms - photophobia, blurred vision, dilated pupils poorly reactive to light
- Cardiovascular symptoms - arrhythmia, tachycardia, hypertension
- Gastrointestinal symptoms - severely dry mouth and throat, decreased bowel sounds, constipation, ileus
- Renal symptoms - urinary retention

Anticholinergic symptoms include tachycardia, hypertension, flushing, dry mouth, mydriasis, and urinary retention.



Other options:

Option A: Aconite is a cardiac poison derived mainly from the root and root tubers of the plant *Aconitum napellus* (shown below). Symptoms of intoxication are due to action on the voltage-gated sodium channels on the cell membrane of nerves and skeletal and smooth muscles. Symptoms include intestinal (nausea, vomiting, perioral tingling and numbness, abdominal pain, and diarrhea), cardiovascular (chest pain, palpitations, hypotension, bradycardia), and neurological (vertigo and restlessness) manifestations.



Option B: Thevetin is a cardiac poison derived from the plant *Cerbera thevetia* (shown below). Symptoms of intoxication are characterized by vomiting, diarrhea, headache, dizziness, mydriasis, and loss of muscular power. Cardiovascular signs include a weak and irregular pulse rate with hypotension. Possible complications are heart block and cardiorespiratory failure.



Option C: Nicotine is a plant alkaloid derived from the *Nicotiana tabacum* plant, shown in the image below. It is a cardiac poison.

All components of this plant are poisonous except its ripe seeds. It is a cardiac poison that initially stimulates the autonomic ganglia to produce symptoms of acute intoxication (tachycardia, hypotension, tachypnea, nausea, vomiting, abdominal pain, and headache). These symptoms later subside to give way to respiratory depression and collapse.



Solution to Question 12:

The type of injury shown in the image is a contusion.

A contusion is characterized by the infiltration of extravasated blood into the subcutaneous and/or subepithelial tissues. It results from the rupture of small blood vessels due to the application of blunt force.

- A pure bruise lies beneath an intact epidermis.
- Bruises are caused by damage to veins, venules, and small arteries.
- The usual bruise from a blunt impact is situated in the subcutaneous tissues, often in the fat layer. (Because of the greater volume of soft subcutaneous tissue in fat people, they tend to bruise more easily than thin ones).
- Delayed bruising - It is common knowledge among lay persons, as well as doctors, that a bruise may 'come out' - that is, become more prominent with the passage of hours or days. This is partly caused by continued bleeding from the ruptured vessels, but mainly by the percolation of free blood from its origin deeper in the tissues upwards towards the epidermis.

Other options:

Option B: Abrasion, also known as gravel rash, is the destruction of superficial layers of the epidermis. It is caused by friction against a rough surface or by compression. It is of four types:

- Scratches

- Grazes
- Pressure abrasions
- Impact abrasions

Option C: Laceration refers to a tear or split of the skin, mucous membranes, muscle, or internal organs. It is due to the application of blunt force to a broad area of the body. They are also called tears or ruptures. The laceration may indicate the cause of the injury and the shape of the blunt weapon. The age of the injury can also be determined.

Option D: A stab wound is formed when force is delivered along the long axis of a narrow or pointed object. The object can be a knife, sword, chisel, scissors, or needle. It is deeper than its length and width on the surface of the skin. The edges are clean-cut and inverted.

Solution to Question 13:

The correct option is Option C: A-3, B-2, C-1, D-4.

The time of emergence of secondary dentition or permanent teeth:

The time of emergence of primary dentition or temporary teeth:

The deciduous teeth all emerge into the mouth by 2.5-3 years.

Note: 6-13 years is the age of mixed dentition. The first molar emerges at 6-7 years. The second molar emerges at 12 years, and the third molar emerges beyond 18 years of age.

Tooth	Emergence
First molar	6-7 years
Central incisors	6-8 years
Lateral incisors	7-9 years
First premolar	9-11 years
Second premolar	10-12 years
Canine	11-12 years
Second molar	12-14 years
Third molar	17-25 years

Tooth	Emergence
Mandibular central incisor	6-8 months
Maxillary central incisor	7-9 months
Maxillary lateral incisor	7-9 months
Mandibular lateral incisor	10-12 months
First molar	12-14 months
Canine	17-18 months
Second molar	20-30 months

Solution to Question 14:

The correct match is 1-a,2-c,3-b,4-d.

The active ingredients of *Datura* include the anticholinergic alkaloids atropine, scopolamine, and hyoscyamine. The seeds or other parts of the plant can be ingested or smoked.

Manifestations of *Datura* toxicity:

- Dry hot skin (Temperature rises up to 42 C, a scarlet rash may also be appreciated).
- Dilatation of pupils along with loss of accommodation
- Diplopia
- Dryness of the mouth and throat, unquenchable thirst
- Dysphagia
- Dilatation of cutaneous blood vessels
- Dysarthria (Difficulty in talking)
- Drunken gait
- Delirium
- Drowsiness
- Death due to respiratory or cardiac failure

Morphine, codeine, and thebaine are natural opioids derived from poppy *Papaver somniferum*.

Abrin is a toxalbumin of the plant *Abrus precatorius*, and it acts by inhibiting protein synthesis.

Toxic principles of *Abrus precatorius*:

- Abrin (a toxalbumen)
- Abrine (an amino acid)
- Abrialin (a glucoside)

Abrin acts by inhibiting protein synthesis, thereby, resulting in cell death. The clinical features after oral ingestion of these seeds include abdominal pain, nausea, vomiting, bloody diarrhea, and miosis. Faintness, dyspnea, and convulsions can also occur.

The poison obtained from *Strychnos nux-vomica* seeds is strychnine. Strychnine acts at the ventral horn by blocking the postganglionic glycine receptor sites in the spinal cord and brainstem. Glycine is an inhibitory neurotransmitter that acts at the postsynaptic membranes of ventral horn motor neurons. Hence, the blockade of these receptors results in unopposed muscle contraction, resulting in the classical clinical features:

- Tonic-clonic seizures
- Opisthotonus (body arched in hyperextension)
- Risus sardonicus

Solution to Question 15:

The above case scenario with frothing is suggestive of antemortem drowning. High concentration of arsenic in the shaft of hair is seen in patients with chronic arsenic poisoning.

Given below are the features of antemortem drowning.

- Froth - The presence of copious amounts of fine, shaving-lather-like froth at the mouth and the nostrils is the most characteristic finding of antemortem drowning. It is caused by the vigorous agitation of the tracheal and bronchial mucous secretions, edema fluid, and alveolar surfactant during respiratory efforts.
- Cutis anserina - The contraction of the erector pilorum muscles causes puckering and a grainy appearance of the skin. This resembles goose skin; hence, it is called cutis anserina.
- Washerwoman's hand - Maceration of the skin on the palms and soles gives a wrinkled, sodden, and bleached appearance due to submersion.
- Cadaveric spasm - Gravel, weed, soil, grass, etc. is firmly held in the clenched hands due to cadaveric spasm. This is a vital sign of antemortem drowning. The material also indicates the place of submersion.
- Rigor mortis - This appears early in cases of a violent antemortem struggle.
- Face - It is bloated and discolored, with cyanosis.
- Eyes - Conjunctiva are suffused and pupils dilated.
- Tongue - It may be swollen and protruded.
- Antemortem injuries - These may be sustained by striking objects, such as rocks, during the fall into the water body. Examination for such injuries must be performed only after the body is dry.

Internal findings:

- Lungs - They appear voluminous and distended show ballooning. The cut section will exude copious amounts of frothy, bloodstained fluid.
- Larynx, trachea, and bronchioles - The presence of sand, silt, and aquatic vegetation in the trachea and lower respiratory tract are characteristic findings of antemortem drowning.
- Heart and blood vessels- Gettler test can be done to estimate the chloride content in the right and left sides of the heart to differentiate between freshwater and saltwater drowning.
- Diatom test- Diatoms are microscopic algae. Their presence in the organs, particularly in the bone marrow, is suggestive of antemortem drowning.
- Stomach and small intestines- These organs may contain water in some cases. Water present in the small intestine is positive evidence of death by drowning.
- Middle ear - The presence of water in the middle ear is highly suggestive of antemortem drowning.
- Brain - It is congested, and edematous, with loss of gray-white junction.

Solution to Question 16:

Spanish windlass is an improvised version of garrotting, and it involves tightening an iron collar (Spanish windlass) around the neck and forcing a spike against the neck into the spinal cord.

This was a form of judicial execution once employed in Spain. The rod-like object used to tighten the ligature is called a Spanish windlass; this may be a ruler, stick, or screwdriver.

Garrotting involves attacking the victim from behind using a ligature. The ligature is thrown over the victim's neck and then tightened by twisting with a lever.

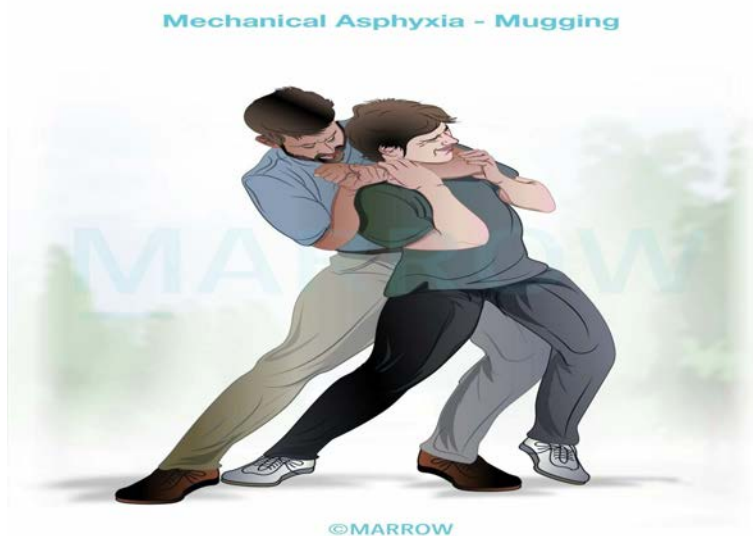
The image given below shows garrotting.



Option A: Squeezing to death using sticks is called bansdola. The neck or the chest is squeezed between two sticks, one placed in front, and one behind. The neck may also be squeezed with a

single stick in front if the victim is lying down.

Option C: Squeezing the neck with the arm and forearm is called mugging. The image below shows mugging.



Option D: Throttling or manual strangulation occurs due to compression of the neck by humans leading to asphyxia. Thumb impression and finger marks may be seen over the victim's neck.

Solution to Question 17:

A person of eonism (transvestism) derives pleasure from wearing clothes of the opposite sex. It is usually found in males. His dress, manner, occupational interests, and associations are all designed to increase his feeling of being a woman.

Other options:

Sexual pleasure from

- Kissing and licking of anus by a sexual partner - Anilingus
- Sexual interest in nonliving objects that are used to generate sexual excitement. -Fetichism
- Seeing a female undressing - Voyeurism or scoptophilia

Solution to Question 18:

The x-ray shows a pair of surgical scissors left behind in the abdomen after surgery. The offense comes under res ipsa loquitur. It is obvious evidence of the negligent act committed.

"Res ipsa loquitur" means "the thing or fact speaks for itself." The negligent act is so obvious that the patient need not prove the negligence. He has to merely state what according to him was the negligent act.

This doctrine applies to both civil and criminal negligence. All of the following conditions should be fulfilled for this doctrine:

- In the absence of negligence, there would not have been any injury usually
- The doctor had complete control over the injury-producing instrument or treatment.
- The patient was not a part of contributory negligence.

Other options:

Option A: Novus actus interveniens: A person is responsible not only for his actions but also for the logical consequences of those actions. This principle applies to cases of assault and accidental injury.

Option C: Res judicata (Section 337, BNSS) -If a question of negligence against a doctor has already been decided by a court in a dispute between the doctor and his patient, the patient will not be allowed to contest the same in another legal proceeding, based on the same set of facts. Multiple appeals cannot be made.

Option D: The principle of respondeat superior is used to explain vicarious liability, which explains that the employer is responsible for his employees' negligence if such an act occurs in the course of the employment and within its scope.

Solution to Question 19:

Tattooing around the entry wound of a bullet is due to the deposition of unburnt powder particles. These are difficult to remove. They are also called stippling or peppering. This is seen up to 50 cm of firing

When the distance between the firing and the victim is between 30-50 cm (in the case of pistols and revolvers), it is a near-shot wound. The victim is within the range of powder deposition, so there is some amount of tattooing present. But it is outside the range of flame, so there will not be any burning or singeing. There will not be any blackening, as it is outside the range of smoke as well.

The injuries related to firearms are:

- Burn injuries - occur due to flame released during the firing. It is seen in contact shot wounds or distances up to 5-8 cm. Singeing of hair is caused due to burns
- Blackening - occurs due to the deposit of smoke. It can be removed with a wet cloth. This can be seen up to around 15 cm of firing
- Tattooing - occurs due to the deposition of unburnt powder particles. These are difficult to remove. They are also called stippling or peppering. This is seen up to 50 cm of firing
- Greasing and collaring - occur due to the deposition of the lubricant of the bullet in the tissues, during its penetration. It can be found in contact, near-shot, or distant shot wounds.

In a firm contact shot, the imprint of the muzzle of the gun is seen on the skin around the wound. The victim is within the range of flame, hence singeing of the hair will be present. Tattooing begins only when the muzzle-to-target distance is >1cm, like in a loose contact shot.

When the distance between the victim and the firing is 5-8cm, it is called a close shot. The victim is within the range of the flame, hence, features related to flame, smoke, and powder deposition will be seen. Singeing, burning, blackening and tattooing will be present.

When the distance between the victim and the firing is ≥ 50 cm, it is a distant shot. The victim lies outside the range of flame and powder deposition. Hence, features related to flame, smoke, and powder deposition are absent. Burning, blackening, and tattooing are absent.

Solution to Question 20:

In cold water drowning or immersion syndrome or hydrocution, death occurs due to vagal inhibition, leading to cardiac arrest.

Hydrocution/ immersion syndrome refers to sudden death, following immersion in cold water. Death results from vagal inhibition of the heart, i.e. cardiac arrest occurring due to increased stimulation of the vagus nerve.

Vagus stimulation occurs due to the following mechanisms:

- Sudden entry of water into the nasopharynx or larynx
- Falling or diving into the water in a manner, so that it suddenly strikes the abdomen, especially the epigastric region
- Sudden in-rush of cold water into the ears

Types of drowning:

- Wet drowning/ typical drowning
- Atypical drowning/dry drowning
- Immersion syndrome/ hydrocution
- Post-immersion syndrome/ secondary drowning/ near-drowning
- Submersion of the unconscious/ shallow-water drowning

Solution to Question 21:

The given image shows an abrasion due to ligature mark, which is a pressure abrasion.

Abrasions are superficial wounds produced due to injury to the epidermis. They are usually caused by friction against a rough surface or by compression.

Types of abrasions:

- Linear abrasions (scratches): Caused by relative movement of a pointed object on the skin
- Examples: nail, thorn, the point of sword or knife
- Grazed abrasions (sliding /scrapping/grinding abrasion): Caused by relative movement of a rough surface on a wide area of skin
- brush burns- tangential rubbing over the surface, usually the ground

- friction burns are extensive excoriated areas without bleeding
- Examples: fall, a victim being dragged on the ground, road traffic accidents
- Imprint abrasions (patterned abrasions): Caused by the impact of a rigid hard rough surface over a localized area of the skin without any movement; usually show the pattern of the object causing them
- Examples: tire, grills, bicycle chain, bite marks
- Pressure abrasions (pressure abrasion): Caused by crushing of the superficial layers of the epidermis
- Examples: ligature marks around the neck in hanging

Solution to Question 22:

Gas shadow in the great vessels of a fetus in intrauterine death is called the Roberts sign. It is the earliest sign and is seen within 12 hours of death. The gas shadow may be present within the chambers of the heart or in large vessels of the fetus.

Signs of intrauterine death:

- Earliest sign - reddening of the skin with peeling in 12 hours
- Robert's sign - radiographic finding showing gas in the great vessels like the aorta in 12 hours
- Large blebs containing a red serous or serosanguineous fluid appear at around 24 hours
- Serous cavities may contain a turbid reddish fluid after 48 hours
- Spalding's sign (Option B) radiographic finding is seen as a loss of alignment and over-riding of bones of the cranial vault. This occurs due to shrinkage of the brain after death. It is seen 7 days after fetal death.
- A sweetish and disagreeable odor
- Sodden and coppery red/purplish skin (never greenish as with putrefaction) with blisters, and cuticle peeling. The skin detaches easily and leaves moist and greasy areas
- Distended abdomen and thick, red, and soft umbilical cord
- Abnormally mobile or flexible joints
- Soft, edematous, and loss of morphology of the organs or viscera (lungs and uterus remain unaffected for long period)
- The body is soft and pliable and flattens out when placed on an even surface

Option C: Ball sign is hyperflexion of the spine with crowding of ribs seen in intrauterine death.

Option D: Halo sign is a radiological sign of intrauterine fetal death showing an increase in soft tissue width between subcutaneous fat and the cranial vault.

Solution to Question 23:

The type of wound in which the crushing of the hair bulb is seen is a lacerated wound.

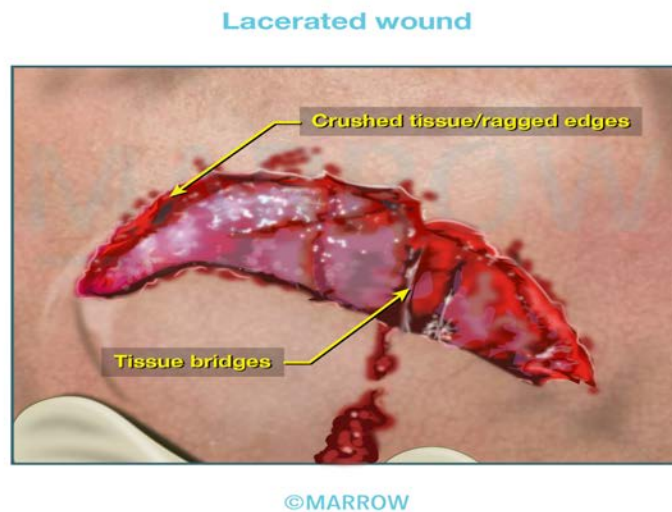
Cut lacerations may be produced by blunt weapons or heavy relatively sharp-edged weapons such as axe, hatchets, and choppers. Scalp lacerations can be found in association with bruising and abrasions. Fascial strands, hair bulbs, nerves, and vessels, that run in the depth of the wound, are irregularly torn and crushed.

Margins are irregular, ragged, and uneven. The ends are pointed or blunt. The edges of lacerations, especially over the skull are undermined. The depth of the wound varies according to the thickness of the soft parts at the site of the injury and the degree of force applied.

Medicolegal importance of laceration:

- The type of laceration indicates the cause of the injury and the shape of the weapon.
- Foreign bodies found in the wound may indicate the circumstances in which the crime was committed.
- The age of the injury can be determined.

The image below shows a lacerated wound.



Other options

Option A: An incised wound is clean-cut, regular, and well-defined. Bruising does not occur when caused by light-cutting weapons but bruising and some irregularity of the margins may be present when heavy and not-so-sharp-edged weapons like a shovel, Kassi, axe, tangi, etc. have been used.



Option C: Chop wounds are caused by a blow with the sharp cutting edge of a fairly heavy weapon like an axe, sword, or hatchet. The margins of the wound depending on the condition of the weapon, a sharp margin with slight bruising is seen with a sharp-edged weapon, and a ragged and bruised margin is seen with a blunt edge. Undermining occurs in the direction towards which the chop is made.



Option D: Avulsion or shearing laceration is a lesion where the skin separates or peels off by tearing from the underlying tissues. It is due to shearing force delivered at an acute angle. Commonly seen in road traffic accidents.

Solution to Question 24:

The postmortem change shown above is marbling, which is an external sign of putrefaction.

In marbling, the superficial veins of roots of the limb, thighs, sides of the abdomen, and shoulders are stained greenish-brown or purplish-red, in a linear branching pattern. It is usually prominent 36–48 hours after death occurs due to the formation of sulfhemoglobin subsequent to hemolysis of red blood cells.

In putrefaction, bacteria from the gut enter into and multiply in the surrounding tissues, eventually spreading through the blood vessels to other parts of the body. The bacterium *Clostridium welchii* is largely responsible for the putrefactive changes spreading through the body, due to its production of the enzyme lecithinase. The first external sign of putrefaction in a body lying outside in the air is a greenish discoloration of the skin over the region of the cecum.

Other options:

Option B: Tattooing around the entry wound of a bullet is due to the deposition of unburnt powder particles. These are difficult to remove. They are also called stippling or peppering. This is seen up to 50 cm of firing

Option C: Postmortem hypostasis/Postmortem staining/Postmortem lividity/Livor mortis/Sugillation refers to bluish-purple or reddish-purple discoloration of the superficial layers of the dermis. It is an early postmortem change that occurs due to the stagnation of blood in the vessels of the dependent parts due to the effect of gravity.

Option D: Filigree burns also known as Lichtenberg's flowers are superficial, thin, irregular, and tortuous markings on the skin indicative of a lightning strike. They resemble the branches of a tree and are usually found over the skin of the shoulders or the flanks.

Solution to Question 25:

The clinical scenario is suggestive of a phossy jaw, which occurs due to chronic phosphorus poisoning. Phossy jaw refers to osteomyelitis and necrosis of the jaw, with foul-smelling pus discharging multiple sinuses.

Frequent inhalation of phosphorus fumes causes necrosis of the lower jaw in the region of a decayed tooth. Patients initially present with toothache, followed by swelling of the jaw, loosening of the teeth, necrosis of gums, and sequestration of the bone in the mandible producing considerable facial disfigurement.

Chronic phosphorus poisoning is common among the workers of phosphorus industries (matchbox factories), who have had chronic exposure to phosphorus fumes for months to years. It is managed surgically and by prevention of further exposure to the poison.

Phosphorus is a non-metallic, inorganic irritant poison that affects cellular oxidation. It exists in two forms, white phosphorus, a highly toxic form, and red phosphorus, the non-toxic form.

Fulminant poisoning is seen when more than one gram is consumed, and death occurs within 12 hours due to peripheral vascular collapse. In acute poisoning, they present with pain in the throat and abdomen, with intense thirst, nausea, diarrhea, and severe abdominal pain. System toxicity occurs after 2–3 days and results in hematemesis, enlarged tender liver, jaundice, pruritus, and hemorrhages. The gastric and intestinal content have garlic odor and are luminous.

Gastric lavage with potassium permanganate oxidizes phosphorus into phosphoric acid and phosphates. Activated charcoal can be used to adsorb the poison.

Other options:

Option A: Bromine is a corrosive substance that causes necrosis of skin and tissues.

Option B: The symptoms of antimony poisoning, both acute and chronic are similar to that of arsenic. Inhalation of fumes causes a flu-like illness called metal fume fever. In cases of chronic poisoning with heavy metals such as lead, arsenic, mercury, antimony, copper, etc., the shaft of a long bone, strands of scalp hair, fingernails or toenails, and a wedge of the quadriceps muscle should be taken for sampling.

Option C: Prolonged ingestion of fluorides results in the mottling of teeth, an increase in the density of bones, lipping of bony margins of vertebrae, which sometimes produce pressure on the spinal cord leading to paraplegia.

Solution to Question 26:

Durham's rule and Curren's rule are laws associated with insanity.

Durham's rule states that an accused person is not criminally responsible if his unlawful act is the product of mental disease or mental defect.

Curren's rule states that an accused person is not criminally responsible if, at the time of committing the act, he did not have the capacity to regulate his conduct to the requirements of the law as a result of mental disease or defect.

In India, the law of Criminal responsibility is under BNS Section 22 (Section 84 IPC) It states that "Nothing is an offense which is done by a person who, at the time of doing it, by reason of unsoundness of mind, is incapable of knowing the nature of the act, or what he is doing is either wrong or contrary to the law". This is slightly different from McNaghten's rule followed in Britain.

McNaghten's rule states that every man is to be presumed to be sane, and possesses a sufficient degree of reason to be responsible for his crimes until the contrary is proved.

Other rules to assess the criminal responsibility of insane persons:

- The Irresistible Impulse
- Federal rule or American Law Institute test (ALT)
- The Brawner rule

Note: Ashley's rule is used to differentiate males and females based on the combined length of the manubrium and sternum. It is considered to be a male sternum when the combined length is ≥ 149 mm, while if the combined length is < 149 mm, then it is considered female.

Solution to Question 27:

The management of snakebites includes immobilization and administration of anti-snake venom (ASV) after hospitalization.

Immobilization is done to minimize the spread of the venom. Hospitalization, followed by ASV administration is the definitive management of snake bites.

Measures that can be taken in the case of snakebite:

- Apply firm pressure over the bitten area, which delays the absorption of venom
- Pressure immobilization is recommended for elapid and sea-snake bites, but not for viper bites as it may cause local necrosis
- Immediately apply a broad firm bandage (Sutherland wrap) on the bitten area and around the limb. As much of the limb should be bandaged as possible. It should be tight enough to occlude the superficial venous and lymphatic return, but not the arterial or deep venous flow. A pressure of 50 to 70 mm Hg is maintained.
- In bites on the trunk, head, or neck, apply firm pressure over the bitten area
- Immobilize the limb, as movement can accelerate the spread of venom
- Make the patient lie on one side in the recovery position so that the airway is clear, in case of vomiting or fainting.

Measures contraindicated in the case of snakebite:

- Suction and incision of the wound
- Electric shock treatment of wound site
- Application of tourniquets because they obstruct arterial flow and cause ischemia
- Ice water immersion of the bitten limb
- Do not attempt to kill or catch the snake as it may be dangerous.
- Administration of ASV locally at the wound site.
- Herbal/local remedies

Solution to Question 28:

Hooch tragedy is due to the consumption of cheap illicit liquor containing methanol, particularly by the lower socio-economic classes.

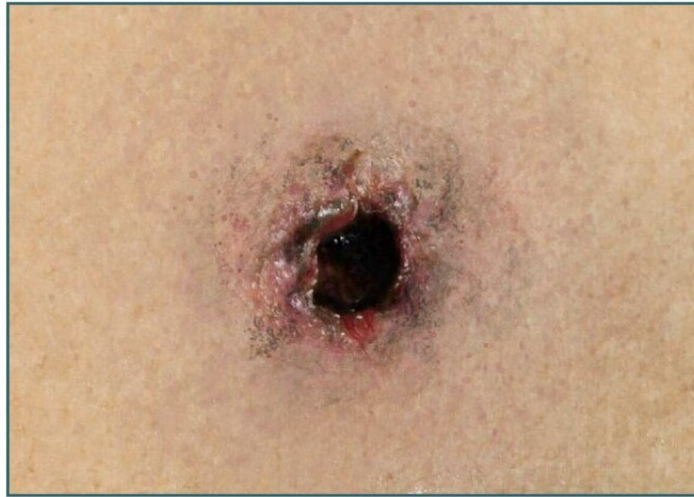
The central organ for metabolism is the liver. Methanol or methyl alcohol is first oxidized to formaldehyde with the help of alcohol dehydrogenase. It is further metabolized to formic acid via aldehyde dehydrogenase. These metabolites are responsible for the adverse effects of intoxication.

Features seen are ophthalmological complications such as blindness, and optic neuropathy. Brain edema, acute renal failure, and severe metabolic derangement are also seen.

Forensic Medicine NEET 2022

Question 1:

Identify the gun that caused the wound and the range of the shot.



- a) Shot gun, intermediate range
- b) Shot gun, close range
- c) Pistol, near shot
- d) Pistol, close shot

Question 2:

Identify the type of wound from the image.



- a) Defense wounds
- b) Antemortem wound
- c) Postmortem wound
- d) Hesitation cuts

Question 3:

A 56-year-old man presents with fatigue, pallor, and abdominal pain. He reports memory loss and reveals that he works at a battery recycling plant. On examination, a noticeable foot drop in both lower extremities is seen. Laboratory testing shows elevated blood lead levels. Which enzyme is affected in the given scenario?

- a) Aminolevulinic acid (ALA) dehydratase
- b) Uroporphyrinogen III
- c) Uroporphyrinogen I
- d) Pyruvate dehydrogenase

Question 4:

A 45-year-old female patient is told about the benefits and complications of hysterectomy and she agrees to the procedure. What kind of consent is this?

- a) Informed consent
- b) Implied consent
- c) Opt out

- d) Passive consent**

Question 5:

A patient came with complaints of hair loss. His wife mentions that she has noticed some behavioral changes. The doctor notices that there is a loss of eyebrows on the lateral side. He then comes to a conclusion by examining the nails. What is the type of poisoning in this case?

- a) Thallium**
- b) Arsenic**
- c) Mercury**
- d) Lead**

Question 6:

A dead body is brought for evaluation. On post-mortem examination, a ligature that was completely encircling the neck, horizontal, and below the level of the thyroid was seen. There was no dribbling of saliva. What is the cause of death?

- a) Throttling**
- b) Ligature strangulation**
- c) Gagging**
- d) Hanging**

Question 7:

A surgeon returns home from a party after many pegs of alcohol and was called to perform an emergency operation. During the operation, the assisting staff noticed the surgeon's hand shaking and instruments falling. He eventually nicks an artery and the patient collapses. Under which of the following terms will this incident be tried?

- a) Criminal negligence**
- b) Civil negligence not amounting to criminal negligence**
- c) Therapeutic misadventure**
- d) Dichotomy**

Question 8:

During the court proceedings, the defense lawyer asks a leading question, which the prosecutor appeals against. The judge grants the appeal. Leading questions are not allowed in all of the following except?

- a) Re-examination
- b) Cross examination
- c) Examination in chief
- d) Dying declaration

Answer Key

Question No.	Correct Option
1	c
2	a
3	a
4	a
5	a
6	b
7	a
8	b

Detailed Explanations

Solution to Question 1:

The given image showing a gunshot entry wound with powder burn around the wound is likely due to a near-shot injury by a pistol.

A near-shot injury is one when the range of flame is within 60 cm.

The features of a near-shot wound are a circular entry wound with the size same as that of the bullet. The edges are lacerated, inverted, and surrounded by a narrow zone of abrasion. There are no signs of burning or singeing. Powder tattooing around the entry wounds is another important feature.

Other options:

Option A: In an intermediate range which is up to 7 feet, the edge of the wound is abraded and the appearance is referred to as a rat hole. There is no burning and the tattooing is appreciable till 3-4 feet and then starts disappearing.

Option B: In shotgun injuries due to a close-range i.e between contact and 3 feet, the entry wound is circular or elliptical with features of the scorching, singing, and less dense tattooing.

Option D: In a close shot pistol injury, the range of flame is within 5-8 cm with a circular entry wound. Other features such as burning, singeing, blackening and tattooing of skin are seen in and around the entry wound.

Solution to Question 2:

The given image shows irregular and ragged cut wounds over the palmar surface of the hands and fingers, which is suggestive of defense wounds.

Defense wounds are the wounds that occur as a result of the victim's attempt to save himself from an injury.

It can be active or passive. Active defense wounds occur when the victim tries to grasp the weapon, while passive wounds result when the victim raises the hands, arms, or legs in an attempt to protect the eyes, head, or neck.

The morphology of the wound depends upon the type of weapon and the attack. In a sharp injury due to a single-edged weapon, a single cut may be seen on the palm or the bends of fingers. In the case of a double-edged sharp weapon, cuts are seen on both palm and fingers. The cuts are irregular in-depth and are ragged.

The defense wounds signify homicide, and in females, they often indicate sexual assault.

Other options:

Options B and C: Antemortem and postmortem wounds are of medicolegal importance while evaluating trauma. They can be differentiated by the following features:

Option D: Hesitation cuts are a feature of self-inflicted wounds or suicide. It typically appears as multiple, small, and superficial wounds on the skin, at the beginning of an incised wound. The common sites are the throat, wrist, and front of the chest.

Feature	Antemortem wounds	Postmortem wounds
Edges	Swollen edges with wound gaps	Closely approximated with no wound gaps
Hemorrhage	Arterial type of hemorrhage	Venous hemorrhage
Spurting	Spurting of blood seen with firmly coagulated blood	No spurts and no clotting
Vital reaction	Inflammation and repair seen	No signs of inflammation and repair

Solution to Question 3:

The given scenario is suggestive of chronic lead poisoning, and the enzyme affected is aminolevulinic acid dehydratase or ALA dehydratase.

Lead is a heavy metal, which mainly acts by depressing the enzymes in heme synthesis, blocking the enzyme aminolevulinic acid dehydratase and inhibiting ferrochelatase.

Chronic lead poisoning also referred to as plumbism or saturnism occurs following inhalation of lead dust or vapors that arise as a result of burning of paints, batteries, smouldering, glass blowing, and polishing.

Features of lead poisoning:

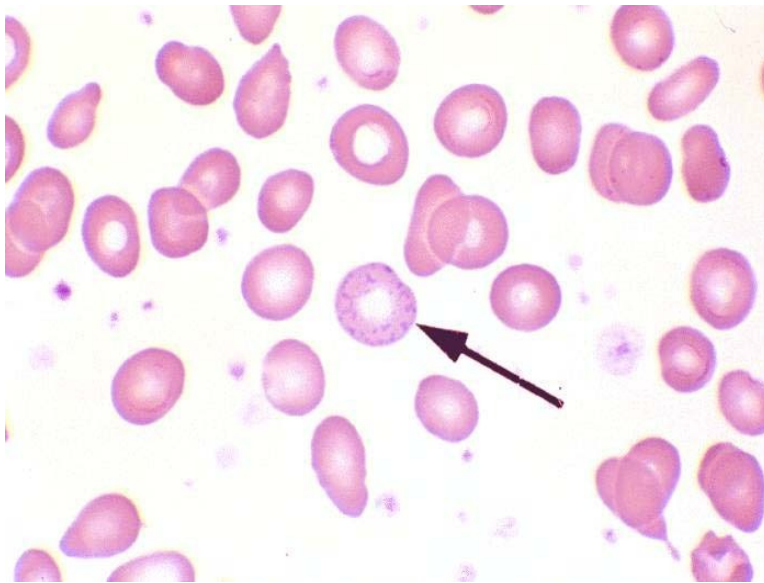
- Facial pallor, which is the earliest and most consistent sign
- Anemia
- Burtonian line, or the lead line, is seen on the gums as a blue line. It occurs as a result of lead sulphide formation, which is deposited as granules in the upper jaw at the junction of the teeth.
- Abdominal colic, constipation
- Lead palsy is a late manifestation, commonly seen in adults and is characterized by numbness, hyperesthesia, and muscle cramps, followed by weakness. Other features like foot drop, wrist drop, and fatigue may be seen.
- Lead encephalopathy is characterized by vomiting, convulsions, headache, insomnia, visual disturbances, and irritability.

Diagnosis is mainly by history and clinical findings. Increased urinary lead and porphyrin excretion are suggestive of chronic lead poisoning. Radiograph, erythrocyte porphyrin assays, and basophilic stippling in the peripheral smear are also useful.

Prophylaxis in the form of proper ventilation, personal hygiene, periodic medical examination, supplementation of calcium-rich diet and sulphuric acid in the water, and weekly saline purgative in factory works is useful.

Management includes British Anti-Lewisite, or BAL, for severe acute poisoning, with and without encephalopathy; ethylenediamine tetraacetic acid (EDTA) for moderate poisoning; and D-penicillamine in mild cases.

The below image shows basophilic stippling.



Burtonian line - Lead poisoning



Solution to Question 4:

This kind of consent is informed consent since the patient has been made aware of the benefits and complications and has agreed to the procedure.

Informed consent is one given by the patient after understanding the nature of the condition, proposed treatment, outcomes of the recommended treatment, alternative options to treatment, and the risks and benefits involved. The patient can accept or reject the procedure. This is also called the rule of full disclosure.

Consent refers to a voluntary agreement, compliance, or permission. It can be expressed or implied.

Implied consent (option B) is that which is not written or expressly stated. It is implied that the patient is giving consent to examination in a general sense. For example, when a patient visits the

hospital with an illness, it implies that he has given consent to a physical examination to find out the nature of his illness. Another example of implied consent is when a patient exposes his arm for injection. It is legally effective.

Expressed consent can be oral or written and refers to consent in which the terms are clearly stated in distinct and explicit language.

Opt out (Option C) refers to a patient's refusal of the procedure, test, or treatment. It is the physician's duty to disclose the risks or consequences of refusal adequately and appropriately to the patient.

Passive consent (Option D) is not commonly described. However, in regard to research, it refers to parents signing and returning a form if they do not want to allow their child to participate in research.

Solution to Question 5:

The given scenario of hair loss, loss of eyebrows on the lateral side along with behavioral and nail changes point to thallium poisoning.

Thallium poisoning occurs due to an overdose or idiosyncrasy. Thallium sulfate and thallium acetate are the two salts of thallium used in dyes, in the glass industry, and as rodenticides and pesticides.

Mild cases of poisoning have features of joint pain, loss of appetite, and drowsiness. Severe cases present with colic, vomiting, muscle pain, convulsions, psychosis, tremors, delirium, and coma are seen in severe cases. Hair loss is seen after 2 weeks and a characteristic feature is loss of the outer thirds of the eyebrows. Mees lines are seen on the nails. Maculopapular rashes in a butterfly pattern on the face are characteristic.

In chronic poisoning, alopecia, skin rashes, peripheral neuropathy, and mental confusion are present.

Treatment is mainly by stomach wash with potassium iodide or Prussian blue, activated charcoal. Hemodialysis is useful and forced diuresis can be given.

The image below shows Mees lines.



Solution to Question 6:

The most likely cause of death in a case where the ligature is completely encircling the neck, horizontal, and below the level of the thyroid, with no saliva dribbling, is ligature strangulation.

Strangulation is a type of asphyxia where there is neck constriction without suspension of the body. The pressure on the neck is applied either by bare hands, a ligature, or by any other instrument.

When strangulation is done by using a ligature like a constricting band or rope, it is called ligature strangulation. The constricting band is tightened by a force other than the victim's body weight. It is characterized by the ligature mark, which is transverse and completely encircles the neck below the thyroid cartilage. There is no dribbling of saliva. Bleeding may be present in the ears, nose, and mouth.

Ligature and manual strangulations are mostly homicidal. Marks of violence are usually seen on other parts of the body.

Other options:

Option A: Throttling or manual strangulation occurs due to compression of the neck by humans leading to asphyxia. Thumb impression and finger marks may be seen over the victim's neck.

Option C: Gagging is a type of asphyxia due to forcible closure of the mouth by a cloth or closing the mouth and nose and is tied around the head.

Option D: Hanging is characterized by an incomplete and oblique ligature mark on the neck. It is usually seen higher, between the chin and larynx. Saliva often dribbles from the mouth.

Solution to Question 7:

The incident mentioned in the given scenario will be tried under criminal negligence.

Criminal negligence is when a doctor lacks a gross skill or care while treating, which results in serious injury/death of the patient, or when an illegal act is performed by a doctor or when an assaulted person dies. Most cases of criminal negligence occur due to drunkenness or usage of drugs, which leads to impaired efficiency.

Some of the examples are amputation of the wrong finger; leaving an instrument, tube, or sponge in the abdomen; incompetent administration of general anesthetic; and administering the wrong substance in the eye, leading to visual loss.

The doctor is prosecuted by the police and charged in criminal court, when it is proved that no medical doctor in his ordinary senses would have done it. Contributory negligence cannot be used as a defense in criminal negligence.

Other options:

Option B: Civil negligence occurs as a result of lacking care and skill in the profession. Contributory negligence can be cited as a defense in civil negligence.

Option C: Therapeutic misadventure is a case that causes injury or death to the individual due to an unintentional act by a doctor or hospital. The physician is not liable for the injuries unless some negligence on his part contributed to the occurrence of the injury.

Option D: Dichotomy is a type of unethical act, wherein a physician offers or receives any gift or bonus for referring or procuring any patient for medical or surgical treatment.

Solution to Question 8:

Leading questions are not allowed in all of the given options except cross-examination, i.e leading questions are allowed during cross-examination.

As per section 141 of the Indian Evidence Act, a leading question is one which is suggestive of the answer that is desired or one which includes a fact and leads to the admitting of a conclusive answer, such as yes or no.

Cross-examination is one wherein the witness is questioned by the opposite party's lawyer, i.e the lawyer for the accused or the defense lawyer. It is undertaken with the objective to demonstrate favorable facts to their case, to test the accuracy of the witness statements, to unearth new or old facts, and to discredit the witness.

The sequence of examination while recording evidence is as follows:

- Examination-in-Chief/direct examination by the party that summoned the witness to elicit all the facts from the witness. Leading questions are not allowed (option C). The only exception is if the judge is satisfied that the witness is hostile.
- Cross-examination is done by the defence lawyer.
- Re-examination is done by the party that summoned the witness to correct or clarify or detail the statements made by the witness during cross-examination. Leading questions are not allowed (option A).

Other options:

Option D: A dying declaration is an oral or written statement made by a person who is dying. The doctor should certify that the person is conscious and with normal mental faculties before the declaration is recorded. It is recorded by the executive magistrate, or a doctor in the presence of two witnesses, if there is no time for the magistrate to come. Leading questions should not be asked.

Forensic Medicine NEET 2023

Question 1:

A child before playing consumed fruit from the garden. After some time he developed a high fever, confusion, photophobia, and unable to urinate. What are the likely causative agent and the appropriate antidote used in this case?

- a) Datura, Pralidoxime
- b) Datura, Physostigmine
- c) Yellow oleander, Pralidoxime
- d) Yellow oleander, Physostigmine

Question 2:

The method of autopsy carried out en masse to remove from tongue to prostate is_____.

- a) Virchow technique
- b) Rokitsansky technique
- c) Ghon technique
- d) Letulle technique

Question 3:

A 42-year-old HIV-positive patient died, and his body was brought for post-mortem. The method of autopsy to be done is_____.

- a) Ghon technique
- b) Letulle technique
- c) Virchow technique
- d) Rokitsansky technique

Question 4:

A 10-year-old male child came to the casualty with difficulty in walking and pain in the perianal region. On subjecting the specimen from the perianal region to a test, it produces

yellow needle-shaped rhombic crystals with picric acid. What is the test done?

- a) Barberio test
- b) Florence test
- c) Teichmann test
- d) Acid phosphatase test

Question 5:

A patient was brought to casualty with seizures. He had multiple episodes and was taking herbal medicines for the same. He was conscious in between the episodes. During the episodes, he had arching of his back. What is the likely cause?

- a) Strychnine
- b) Ricinus
- c) Nerium odorum
- d) Datura

Question 6:

BNS Section 90 (IPC 314) deals with

- a) Causing miscarriage with the consent of the mother
- b) Causing miscarriage without the consent of the mother
- c) Death of the mother by act done with intent to cause miscarriage
- d) Causing the death of the quick unborn child by an act amounting to culpable homicide

Question 7:

The temperature of a body of a deceased person is found to be 39 degrees Celsius. Which of the following is the most probable reason?

- a) Cyanide poisoning
- b) Septicemia
- c) Corrosive poisoning
- d) Intra-abdominal hemorrhage

Question 8:

A farmer was sleeping in the field, and he felt a sting on his leg. He saw something moving away quickly. He then got drowsy and was taken to the hospital. He developed pain around the site and continued to bleed profusely from the wound site. The wound became red with blisters. Which of the following is the most likely cause?

- a) Viper
- b) Cobra
- c) Wasp bite
- d) Scorpion

Answer Key

Question No.	Correct Option
1	b
2	d
3	d
4	a
5	a
6	c
7	b
8	a

Detailed Explanations

Solution to Question 1:

The clinical vignette in the question is suggestive of Datura poisoning, for which the appropriate antidote is physostigmine, a reversible acetylcholine esterase antagonist. It is an effective treatment for severe anticholinergic poisoning.

Datura poisoning:

The common names include thorn apple, Jimson weed, and Devil's trumpet. The active ingredients include the anticholinergic alkaloids atropine, scopolamine, and hyoscyamine. The seeds or other parts of the plant can be ingested or smoked.

Toxic manifestations of Datura:

- Mnemonic: "hot as a hare, blind as a bat, dry as a bone, red as a beet, and mad as a hatter"
- Also, remember the D's of Datura
- Dry hot skin (Temp rises up to 42 C, a scarlet rash may also be appreciated).
- Dilatation of pupils along with loss of accommodation
- Diplopia
- Dryness of the mouth and throat, unquenchable thirst
- Dysphagia
- Dilatation of cutaneous blood vessels
- Dysarthria (Difficulty in talking)
- Drunken gait
- Delirium
- Drowsiness
- Death due to respiratory or cardiac failure

Treatment

- Gastric lavage or decontamination with activated charcoal or tannic acid: Gastric emptying is often delayed, and the small, plentiful seeds can become trapped among the gastrointestinal folds after ingestion; thus, gastric decontamination can be an important therapy.
- Physostigmine in severe toxicity given 1-2 mg IM/IV after every half hour as required: It is a physiological antidote. Pilocarpine may also be given.

Medico-legal importance:

In India, Datura is used as a stupefying poison prior to robbery, kidnapping, and rape. It is also known as a rail-road poison, as it is commonly encountered during a journey. Robbers disguised as saints may offer 'prasad' mixed with Datura seeds and rob the passengers.

Yellow oleander:

All parts of the plant are poisonous. The active principles are thevatin and thevatoxin, which are cardiac glycosides. The nectar of the flower yields poisonous honey. Symptoms include a burning sensation in the mouth, tingling of the tongue, dryness of the throat, vomiting, diarrhea, headache, dilated pupils, irregular action of the heart followed by drowsiness, coma, collapse, and death.

Treatment includes

- Gastric lavage
- Single-dose activated charcoal
- To combat acidosis IV sodium lactate and 5% glucose can be given
- Atropine, adrenaline, or noradrenaline (if blood pressure is low)
- Digoxin-specific antibodies

The given image shows the flowers of *Cerebra thevatia*.



Solution to Question 2:

The en masse removal of organs is done in the Letulle technique.

It is the removal of the cervical, thoracic, abdominal, and pelvic organs as a single organ mass. They are subsequently dissected into organ blocks.

Solution to Question 3:

Rokitansky's technique of autopsy will be done on patients with highly transmissible diseases such as HIV, Hepatitis B, etc.

It involves in situ dissection in part, combined with en block removal of organs.

Solution to Question 4:

The test done in the scenario is the Barberio test.

Barberio is used to detect seminal fluid. When a saturated aqueous or alcoholic solution of picric acid is added to the spermatic fluid, yellow needle-shaped rhombic crystals of spermine picrate are produced.

Other options:

Option B: Florence test is used to detect semen. If semen is present, dark brown rhombic or needle-shaped crystals of choline iodide appear immediately

Option C: The Teichmann test (hemin crystal test) is used to detect bloodstains. If bloodstain is present, faint yellowish-red to brownish-black rhombic crystals of hemin or hematin chloride

appear. When hydrogen peroxide is added, the hematin crystals give bubbles of gas.

Option D: The acid phosphatase test is a quantitative test for the presence of acid phosphatase enzyme in the semen. High levels of acid phosphatase from prostatic secretions are present in seminal stains (Normal vaginal secretions have low levels of acid phosphatase). Positive reactions are found till 36 hours after which the enzyme levels fall and disappear within 72 hours. In the absence of demonstrable sperm or aspermia, this test can be considered conclusive.

Solution to Question 5:

The given clinical vignette suggests the diagnosis of strychnine poisoning.

Strychnine poisoning occurs if a person ingests crushed seeds of *Strychnos nux vomica*. If swallowed uncrushed, the seeds of *nux vomica* have no poisonous action. The fatal dose is 15-30 mg of strychnine or one crushed seed.

Strychnine is a competitive antagonist of glycine at the postganglionic receptors in the motor neurons of the anterior horn of the spinal cord. This leads to increased activity in the motor neurons.

Patients typically present with

- Convulsions are clonic initially but eventually become tonic.
- Risus sardonicus results from the contraction of the jaws and facial muscles in which the corners of the mouth are drawn back.
- The convulsions are most marked in anti-gravity muscles so that the body typically arches in hyperextension, the posture known as opisthotonus.
- If the spasm involves the abdominal muscles, the body bends forward (emprosthotonos), or to the side (pleurosthotonus).
- If death is preceded by a severe convulsion, as in strychnine poisoning, it causes an increase in the body temperature known as postmortem calorificity.

The first step in the management of strychnine poisoning is to control convulsions with benzodiazepines or barbiturates to control seizures, and the patients should be placed in a dark room to avoid triggering further convulsions. Activated charcoal can be used to adsorb unabsorbed poison after a thorough stomach wash. Acidification of urine can be done to increase the excretion of strychnine.

Other options:

Option B: Ricinus poisoning leads to symptoms such as the burning of the oral mucosa, salivation, nausea, vomiting, severe abdominal pain, and bloody diarrhea. It may also cause drowsiness, convulsions, delirium, and death.

Option C: Nerium poisoning presents with headache, dizziness, respiratory difficulty, and nausea. Ingestion causes difficulty in swallowing and articulation, abdominal pain, vomiting, profuse frothy salivation, and diarrhea. Blood pressure falls, fibrillation, AV block, coma, respiratory paralysis, and death can occur due to cardiac failure.

Option D: Datura poisoning presents with

- Dry hot skin (Temperature rises up to 42°C, a scarlet rash may also be appreciated)
- Dilatation of pupils along with loss of accommodation
- Diplopia
- Dryness of the mouth and throat, unquenchable thirst
- Dysphagia
- Dilatation of cutaneous blood vessels
- Dysarthria (Difficulty in talking)
- Drunken gait
- Delirium
- Drowsiness
- Death due to respiratory or cardiac failure

Solution to Question 6:

BNS Section 90 (IPC 314) deals with the death of the mother due to miscarriage.

Important sections of IPC relating to miscarriages:

Note: A quick child is a fetus that has developed to such a stage that it moves within the womb of the mother. It shall mean an unborn child whose heart is beating, who is experiencing electronically measurable brain waves, who is discernibly moving, and who is so far developed and matured as to be capable of surviving the trauma of birth with the aid of usual medical care and facilities available in this state.

BNS Section	IPC section	Deals with
88	312	Causing miscarriage with consent
89	313	Causing miscarriage without consent
90	314	Death of the mother by act done with intent to cause miscarriage
91	315	Preventing the child from being born alive or causing it to die after birth
92	316	Causing the death of the quick unborn child by an act amounting to culpable homicide

Solution to Question 7:

The condition in which the temperature of the body remains raised for about the first two hours after death is termed postmortem caloricity and the most probable reason for this is septicemia.

The condition occurs

- When the regulation of body temperature has been severely disturbed before death as in sunstroke or pontine hemorrhage.
- When heat production in muscles is increased due to convulsions as in tetanus and strychnine poisoning.
- When there is increased bacterial activity as in septicemia and cholera.

Solution to Question 8:

Based on the given clinical findings, the most likely cause is a bite of viper snake, resulting in a haemotoxic envenomation.

The Russell's viper has a flat and triangular head with a V-shaped mark, white in color, with the angle of the V pointing forwards. Along the back, it has three rows of diamond-shaped spots. Other points of identification include broad plates on its belly, the small scales on the head, and a shield beneath the tail divided into two rows. It is commonly found throughout India.

Patients with Russell's viper bite usually present with severe pain, and redness around the bite site. Swelling usually develops followed by blisters in and around the wound site. Persistent bleeding from the bite site is a constant feature. Patients may develop hemorrhagic syndrome due to bleeding from various sites like gums, nose, rectum, etc. In severe cases, shock may develop. Bleeding and clotting time are usually prolonged. Death usually occurs due to shock or hemorrhage.

Other options:

Option B: Cobra venom usually produces neurotoxic effects with minimal local effects.

Neurological manifestations of cobra venom include ptosis, paralysis of the face, palate, jaws, tongue, neck muscles, muscles of deglutition, and respiratory muscles.

Option C: Wasp stings cause immediate pain, a wheal and flare reaction, and local edema and swelling that subside in a few hours. Multiple stings can lead to vomiting, diarrhea, generalized edema, dyspnea, hypotension, and collapse.

Option D: Scorpion sting usually presents with localized redness and burning pain which is unable to localize due to its radiation along the dermatomes involved, localized swelling, and punctate hemorrhagic spots. Other features include headache, profuse perspiration, chest discomfort, paraesthesias, hypersalivation, cold extremities, and sometimes, priapism. Hypertension usually develops early, while pulmonary edema and myocarditis may develop later.

Forensic Medicine INI-CET 2023

Question 1:

Identify the following image.



- a) Blister due to barbiturate poisoning
- b) Blister due to antemortem burns
- c) Edema due to chemical burn
- d) Pemphigus vulgaris

Question 2:

A 5-year-old boy went to the temple in a village with his grandmother. He was crying inconsolably on the way back. He was taken to the emergency department within 3 hours. He had altered sensorium, cold clammy extremities, increased salivation, and excessive sweating. On examination, he was pale and had priapism with the following vitals - BP - 140/78 mmHg, HR - 152/min, RR - 36/min sPO₂ of 96%. Which of the following would you use in his management?

- a) Anti-snake venom
- b) Methylprednisolone
- c) Prazosin
- d) Adrenaline

Question 3:

In an examination for drunkenness, a person smells of alcohol and his conjunctiva appears red. During the examination of motor coordination, it is noted that he can walk in a straight line and can talk normally. Which of the following statements is true?

- a) He has consumed alcohol
- b) He has consumed alcohol and is under the influence of alcohol
- c) He has consumed alcohol but is not under its influence
- d) He has not consumed alcohol

Question 4:

The Blackfoot disease is an endemic peripheral vascular disease seen in toxicity of which of the following metal?

- a) Arsenic
- b) Antimony
- c) Barium
- d) Cadmium

Question 5:

A 12-year-old girl is brought by a lady police sub-inspector with an alleged history of sexual violence 2 days back. She was accompanied by her mother. The registered medical practitioner should do which of the following as the next step?

- a) Take a detailed history
- b) Preserve clothing
- c) Take consent for genital examination
- d) Documentation of injuries and examination findings

Question 6:

Which of the following is not true about postmortem changes?

- a) In drowning, postmortem hypostasis is seen only in the lower limbs

- b) Greenish discolouration in right iliac fossa is the earliest sign of decomposition
- c) Adipocere is seen in bodies recovered from water source
- d) Cadaveric spasm is seen only in specific group of muscles

Question 7:

A female was brought for autopsy with an alleged history of being found at a lake in the village. All of the following are suggestive of antemortem drowning except:

- a) Water in stomach
- b) Washerwoman's hands and feet
- c) White leathery foam around mouth and nostrils
- d) Positive diatom test

Question 8:

In death due to firearm injury, the entry wound shows only an abrasion collar. The range of the firearm is:

- a) Contact range
- b) Close range
- c) Near range
- d) Distant range

Question 9:

In the 18th century, the British East India Company smuggled which of the following drugs to China?

- a) Cocaine
- b) Magic mushroom
- c) Opium
- d) Marijuana

Question 10:

An industrial worker sustained burn injuries to the anterior region of his chest and abdomen, as well as burns to his right upper limb and a circumferential burn to his right thigh, as a result of a cylinder explosion. What is the percentage of total burns surface area?

- a) 27
- b) 36
- c) 45
- d) 53

Question 11:

In a Firearm injury, blackening surrounding the gunshot entry wound is due to?

- a) Flame effect
- b) Unburnt powder
- c) Smoke deposition
- d) Muzzle of the firearm

Question 12:

Match the following poisons to their site of action:

- a) A-3, B-1, C-4, D-2
- b) A-2, B-3, C-1, D-4
- c) A-4, B-2, C-1, D-3
- d) A-4, B-2, C-3, D-1

Question 13:

A 13-year-old female was brought by her mother to the emergency with abdominal pain for a few days. On clinical examination by the RMP, they found that she is 8 weeks pregnant. What should be your immediate next step?

- a) File an MLC and Inform Police
- b) Obtain consent form her mother for genital examination.
- c) Preserve the clothing
- d) Plan/perform MTP

Question 14:

A young man is brought to the casualty with pinpoint pupils. Which of the following is the most likely substance he would have consumed?

- a) Datura
- b) Opium
- c) Alcohol
- d) Cocaine

Question 15:

A 25 year old man was brought to the casualty in a semiconscious state following a road traffic accident. Which type of injury is unlikely to be seen in this patient?

- a) Graze abrasions
- b) Laceration Wound
- c) Multiple Contusions
- d) Chop Wound

Question 16:

An autopsy was performed on a person with a history of drowning; however, the report revealed that there was no water in his lungs, hence the case was labeled as dry drowning. What is the most likely cause of his death?

- a) Vagal inhibition
- b) Laryngospasm
- c) Pulmonary edema
- d) Electrolyte imbalance

Question 17:

Cadaveric spasm is not seen in?

- a) Drowning
- b) Traumatic asphyxia
- c) Firearm suicidal wounds

d) Electrocution

Question 18:

Which of the following is not considered a grievous hurt?

- a) Unilateral permanent hearing loss**
- b) Emasculation**
- c) Fracture of limb**
- d) 7 days admission in yellow zone of hospital**

Question 19:

All of the following can be used for the identification of a dead body, except?

- a) Blood**
- b) Nail**
- c) Muscle**
- d) Teeth**

Question 20:

Which chelating agent is not used in heavy metal poisoning?

- a) British anti-Lewisite**
- b) Ethylenediaminetetraacetic acid**
- c) Pralidoxime**
- d) Penicillamine**

Answer Key

Question No.	Correct Option
1	b
2	c
3	c

4	a
5	a
6	a
7	b
8	d
9	c
10	b
11	c
12	a
13	a
14	b
15	d
16	b
17	b
18	d
19	b
20	c

Detailed Explanations

Solution to Question 1:

The given image of burns with blisters with surrounding hyperemia points to a diagnosis of antemortem burns.

A zone of hyperemia (line of redness), which varies in width is present at the edge of the burnt area in antemortem burns. It is due to edema of tissues and capillary dilatation which merges with the edge of the burn. It involves the whole thickness of the actual skin. It is permanent and persists even after death. If the entire body is burnt, a line of redness will be absent.

The antemortem blister appears as a raised dome. It contains serous fluid with proteins and chlorides. The base shows reddening with swollen papillae. In antemortem burns, the skin adjacent to the burnt area shows an increase in enzyme reaction. The reaction is increased for sulfhydryl groups in all layers. There is also an increase in ATP, esterases, aminopeptidases, and acid and alkaline phosphatase. Marked cellular exudation and reactive tissue changes are called vital reactions. This is characteristic of antemortem burns.

Other options-

Option A: Blister due to barbiturate poisoning, is commonly seen in frictional areas such as interdigital cleft, genital regions, etc. It is due to the dermo necrotic effect of barbiturates.

Option C: Chemical burns by acid or alkalis appear as ulcerated patches without blisters. The hair is not singed and the red line of demarcation is absent. Sometimes, the burn shows distinct

coloration and is usually uniform in character. Strong acids produce dark leathery burns on the skin. Strong alkalis cause the skin to slough and leave moist, slimy, greyish areas.

Option D: Pemphigus vulgaris is an intraepidermal immunobullous disease. It occurs commonly between the 4th to 6th decades. They present with flaccid blisters. Mucosal involvement will be seen in the form of buccal and palatal erosions. Firm pressure over the lesion separates the epidermis from the dermis. Nikolsky's sign is positive.



Solution to Question 2:

The clinical stem points to a likely diagnosis of a scorpion bite, for which Prazosin can be used in the management. It acts as a physiological antidote counteracting the symptoms of the adrenergic surge.

The venom is a potent autonomic stimulator that causes the release of massive amounts of catecholamines from the adrenals. It also has a direct effect on the myocardium.

Scorpion sting usually presents with localized redness and burning pain which is unable to localize due to its radiation along the dermatomes involved, localized swelling, and punctate hemorrhagic spots. Other features include headache, profuse perspiration, chest discomfort, paraesthesias, hypersalivation, cold extremities, and sometimes, priapism. Hypertension usually develops early, while pulmonary edema and myocarditis may develop later.

Treatment:

- Paracetamol, NSAIDs, or opioids for pain.
- Benzodiazepines for sedation for agitation, oculomotor abnormalities, uncoordinated neuromuscular activity, and muscle spasms.
- Hypertension and pulmonary edema- Prazosin counteracts the adrenergic surge that can lead to these effects.
- Cardiogenic shock- Antivenom, prazosin, nitroglycerin, dobutamine.

- Cholinergic effects- Atropine.
- Supportive care, wound care.

Solution to Question 3:

The given scenario of a person who smells of alcohol but whose motor coordination is normal with no slurring of speech is suggestive that he has consumed alcohol but is not under its influence.

The effects of alcohol intoxication at various blood alcohol concentrations are listed below.

Smell of alcohol	Motor coordination	Inference
Present	Normal	Consumed alcohol, not under the influence
Present	Abnormal	Consumed alcohol, under the influence
Absent	Normal	Not consumed alcohol
Absent	Abnormal	Cerebellar disorders

Blood alcohol concentration (mg/dL)	Effects
30-50	Impairment of driving and similar skills
50-100	Reduced inhibitions, slurred speech, laughter
100-150	Incoordination, ataxia
150-200	Obvious drunkenness, nausea, staggering gait
200-300	Vomiting, stupor
300-350	Stupor, coma, risk of aspirating vomitus
350-450 and above	Danger of death from respiratory paralysis

Solution to Question 4:

The Blackfoot disease is due to chronic arsenic poisoning.

In China (Province of Taiwan) exposure to arsenic via drinking water has been shown to cause a severe disease of the blood vessels, which leads to gangrene, known as 'black foot disease'. The Blackfoot disease is due to platelet activation & hypercoagulability of blood in peripheral

arteries and is observed in endemic areas of arsenicism.

Clinicopathological findings in acute and chronic arsenic poisoning:

Option B: The symptoms of antimony poisoning, both acute and chronic are similar to that of arsenic. Inhalation of fumes causes a flu-like illness called metal fume fever. In cases of chronic poisoning with heavy metals such as lead, arsenic, mercury, antimony, copper, etc., the shaft of a long bone, strands of scalp hair, fingernails or toenails, and a wedge of the quadriceps muscle should be taken for sampling.

Option C: Barium poisoning causes nausea, vomiting, and diarrhea. It causes cramps and stiffness of the muscles, dilation of the pupil, paralysis of the tongue and larynx, and vertigo. There is increased blood pressure with slow and irregular pulse.

Option D: Cadmium poisoning causes proteinuria and painful bone lesions known as Ouch-Ouch disease.

System	Acute	Chronic
Skin	Delayed hair loss (alopecia)	Melanosis, Bowen's disease, facial edema, hyperkeratosis, cutaneous cancer, hyperpigmentation.
Nails	nil	Mees's lines
Neurologic	Hyperpyrexia, convulsions, tremors, coma	Encephalopathy, polyneuropathy, tremors, axonal degeneration
GI tract	Abdominal pain, dysphagia, vomiting, bloody or rice water diarrhea, mucosal erosions.	Nausea, vomiting, diarrhea, anorexia, weight loss
Liver	Fatty infiltration	Hepatomegaly, jaundice, cirrhosis
Kidney	Tubular and glomerular changes – oliguria and uremia	Nephritic finding
Haematologic	thrombocytopenia, impaired folate metabolism	Bone marrow hypoplasia, anemia, leukopenia, basophilic stippling, and karyorrhexis.
Cardiovascular	The heart will show S-T wave abnormalities, prolonged QT interval, ventricular fibrillation, atypical ventricular tachycardia with humic acid in the Southwest coast of Taiwan	Blackfoot disease due to platelet activation and hypercoagulability of blood in peripheral arteries is observed in endemic areas of arsenicism.
Reproductive system	Maternal ingestion of inorganic arsenic crosses the placental barrier and infant death shortly after birth.	-

Solution to Question 5:

The registered medical practitioner should take a detailed history as the next step.

The registered medical practitioner must take a detailed history and then take informed consent for examination. In case of refusal, an informed refusal is to be documented after explaining to them the consequences of choice. As part of the examination, they should collect specimens, preserve clothing, and document injuries and examination findings. Medical treatment and testing for pregnancy have to be done for all victims of sexual assault free of cost and police should be informed as per section 357-C of CrPc.

The Protection of Children from Sexual Offences (POCSO) Act, 2012 considers all individuals (both male and female) below the age of 18 years as children and protects them from sexual abuse. It defines various forms of sexual offenses which include:

- Sexual harassment: A person is said to commit sexual harassment upon a child when such person with sexual intent,
 - Carries out any act with sexual connotations
 - makes a child exhibit his/her body
 - shows any object to a child in any form/media for pornographic purposes.
- Sexual assault
 - Any act with sexual intent that involves physical contact without penetration.
- Aggravated sexual assault
 - Similar to sexual assault along with the criteria explained under the term "aggravated".
- Penetrative sexual assault
 - An act with sexual intent that involves physical penetration.
- Aggravated penetrative sexual assault
 - Similar to penetrative sexual assault along with the criteria explained under the term "aggravated".
- Pornography

The term aggravated is used for those assaults done:

- by a person in a position of trust or authority such as police officer, security forces, relative, doctor.
- in the course of communal violence.
- on mentally & physically disabled.
- gang sexual assault.
- using deadly weapons.
- causing grievous hurt

The Act prescribes stringent punishment graded as per the gravity of the offense, with a maximum term of rigorous imprisonment for life and fine. Section 21(1) of the POCSO Act, 2012 requires mandatory reporting of cases of child sexual abuse to law enforcement authorities and applies to

everyone including parents, doctors, and school personnel. Failure to report a suspicion of child abuse is an offense under the Act.

Solution to Question 6:

In the case of drowning, if the water is still, the postmortem hypostasis affects the head, chest and the limbs. This is because the chest and abdomen float higher because of the lungs and the abdominal gases. Drowning in fast-flowing water results in absent postmortem staining as the position of the body keeps on changing.

Postmortem hypostasis/Postmortem staining/Postmortem lividity/Livor mortis/Suggillation refers to bluish-purple or reddish-purple discoloration of the superficial layers of the dermis. It is an early postmortem change that occurs due to the stagnation of blood in the vessels of the dependent parts due to the effect of gravity.

Sites of postmortem hypostasis depending on the position of the body:

Postmortem changes are classified on the basis of time:

- Immediate signs of death (at the time of somatic or clinical death):
 - Insensibility and loss of voluntary power
 - Cessation of respiration
 - Cessation of circulation
- Early signs of death. It is seen in cellular death:
 - Pallor and loss of elasticity of the skin
 - Eye changes
 - Primary flaccidity of muscles
 - Algor mortis/Postmortem cooling
 - Postmortem lividity/Suggillation
 - Rigor mortis
- Late signs of death:
 - Putrefaction
 - Adipocere
 - Mummification

Other options-

Option B: The first external sign of putrefaction in a body lying outside in the air is a greenish discoloration of the skin over the region of the right iliac fossa. Sulphmethemoglobin is the agent imparting greenish color.

Option C: Adipocere is a modification of putrefaction. The transformation is brought about by the gradual hydrolysis and hydrogenation of pre-existing fats, including olein, into higher fatty acids. It is characterized by the formation of a waxy, soft, moist, and whitish substance with a

sweetish offensive or ammoniacal (early stages) odor. It is more commonly seen in bodies recovered from damp, warm environments or immersed in water.

Option D: Cadaveric spasm - This is a vital sign of antemortem drowning. Cadaveric spasm is limited to select muscles i.e., the muscles that were in a state of contraction at the time of death. The mechanism of cadaveric spasm is thought to be due to neurogenic phenomena. Gravel, weed, soil, grass, etc. is firmly held in the clenched hands due to cadaveric spasm. The material also indicates the place of submersion.

Position of the body	Sites of hypostasis
Supine	Back except for shoulder blades, buttocks, and calves (contact pallor)
Vertical as in hanging	Legs, feet, forearm, and hands (glove and stocking hypostasis)
Drowning	Face, the upper part of chest, hands, lower arms
Inverted like in drunken persons	Head and neck

Solution to Question 7:

Washerwoman's hands and feet are a non-specific sign of submersion. They occur due to the imbibition of water into the outer skin layers. They can be seen in antemortem as well as postmortem drowning victims.

The presence of water in the stomach, white leathery foam around the mouth and nostrils, and a positive diatom test are suggestive of antemortem drowning.

Given below are the features of antemortem drowning.

- Froth - The presence of copious amounts of fine, shaving-lather-like froth at the mouth and the nostrils is the most characteristic finding of antemortem drowning. It is caused by the vigorous agitation of the tracheal and bronchial mucous secretions, edema fluid, and alveolar surfactant during respiratory efforts.
- Cutis anserina - The contraction of the erector pilorum muscles causes puckering and grainy appearance of the skin. This resembles goose skin; hence, it is called cutis anserina.
- Washerwoman's hand - Maceration of the skin on the palms and soles gives a wrinkled, sodden, and bleached appearance due to submersion.
- Cadaveric spasm - Gravel, weed, soil, grass, etc. is firmly held in the clenched hands due to cadaveric spasm. This is a vital sign of antemortem drowning. The material also indicates the place of submersion.
- Rigor mortis - This appears early in cases of a violent antemortem struggle.
- Face - It is bloated and discolored, with cyanosis.

- Eyes - Conjunctiva are suffused and pupils dilated.
- Tongue - It may be swollen and protruded.
- Antemortem injuries - These may be sustained by striking objects, such as rocks, during the fall into the water body. Examination for such injuries must be performed only after the body is dry.

Internal findings:

- Lungs - They appear voluminous and distended and show ballooning. The cut section will exude copious amounts of frothy, bloodstained fluid.
- Larynx, trachea, and bronchioles - The presence of sand, silt, and aquatic vegetation in the trachea and lower respiratory tract are characteristic findings of antemortem drowning.
- Heart and blood vessels- Gettler test can be done to estimate the chloride content in the right and left sides of the heart to differentiate between freshwater and saltwater drowning.
- Diatom test- Diatoms are microscopic algae. Their presence in the organs, particularly in the bone marrow, is suggestive of antemortem drowning.
- Stomach and small intestines- These organs may contain water in some cases. Water present in the small intestine is positive evidence of death by drowning.
- Middle ear - The presence of water in the middle ear is highly suggestive of antemortem drowning.
- Brain - It is congested, and edematous, with loss of gray-white junction.

Solution to Question 8:

The entry wound shows only an abrasion collar when the range of the firearm is distant.

To determine the range of firing, the following parameters are used:

- The entry wound size and shape
- Blast effect – Muzzle end imprint and edge eversion
- Blackening, Tattooing, Singeing of hair, Abrasion collar, and Grease collar
- Hair should be preserved to determine the range of firing which can be examined for carbonization/desiccation in close range, also from lead deposition around the wound to assess the range of firing.

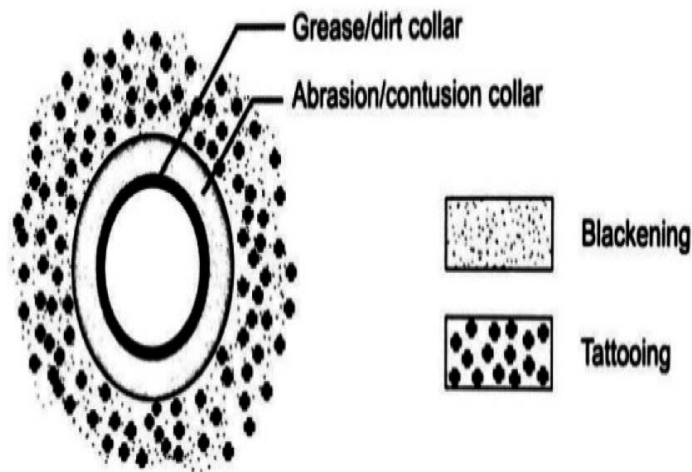
Abrasion collar: Gyroscopic movement of the bullet, the skin's elasticity, and the grease and dirt particles on the bullet's surface are the major causative factors of abrasion and grease collar. They are seen on the skin around the wound of entry at all ranges. Depending on the angle of entry made by the bullet, i.e.

- A perpendicular entry of the bullet develops the abrasion collar of rounded shape.
- An acute angular entry of the bullet develops the abrasion collar of an oblique or oval shape.
- Some contusion present in the abraded collar is known as contusion collar

Characteristics	Contact shot	Close Shot	Near Shot	Distant Shot
Range	In-touch with skin	2.5-7.5 cm	30-60 cm	>60 cm
Size	Largest size	Bullet size	Smaller	Smallest
Shape	Stellate	Circular	Circular	Circular
Blast effect				
Muzzle end imprint	+ve	-ve	-ve	-ve
Edge eversion	+ve	-ve	-ve	-ve
Other findings				
Blackening	-ve	+ve	-ve	-ve
Tattooing	-ve	+ve	+ve	-ve
Singeing	-ve	+ve	-ve	-ve
Abrasion collar	-ve	+ve	+ve	+ve
Grease collar	-ve	+ve	+ve	+ve

Grease collar: Mostly seen on the clothing and is seen as a black deposit, sharply outlined as if printed (better-seen on light color clothes and on fair-skinned individuals).

Medicolegal importance: Helpful in deriving information such as wound of entry and direction of firing



Note: A grease collar need not always be present. Hard metal jacketed bullets are associated less commonly with soiling. Often, the ring is not appreciable during gross inspection and may only be demonstrable on microscopic examination or other laboratory tests.

Solution to Question 9:

The British East India Company smuggled opium into China in exchange for tea, silk and porcelain pots.

Opium derived from the immature seedpods of the opium poppy (*Papaver somniferum*), was originally indigenous to the present-day territory of Turkey. The introduction of opium to China occurred during the late 6th or early 7th century CE, facilitated by Turkish and Arab traders.

During this time, tobacco smoking moved from North America to China, resulting in widespread opium smoking. Opium addiction and imports increased in the first century of the Qing era (1644–1911/12). To address the issue, the Yongzheng emperor (1722–35) banned opium trafficking and consumption in 1729. Despite hurdles, the trade continued, and in 1796, the Jiaqing emperor banned opium imports and cultivation. Opium sales survived despite these laws.

In the early 18th century, the Portuguese made a lot of money importing opium from India and selling it in China. British traders became China's main suppliers in 1773. The British East India Company monopolized Bengali opium farming. The company developed an inexpensive and productive opium poppies cultivation method in this location. The US and other Western nations traded Turkish and Indian opium.

The opium trade was pursued by Britain and other European nations as a means to address their persistent trade deficit with China. In Europe, there existed a substantial and notable demand for Chinese commodities such as tea, silks, and porcelain pots. Conversely, in China, there was a very limited market for Europe's manufactured goods and other trade items. As a result, Europeans were required to utilize gold or silver as payment for Chinese goods. The opium trade, which resulted in a consistent demand for opium among Chinese individuals addicted to the substance, effectively addressed the persistent trade imbalance by facilitating the importation of opium from Western countries.

Opium was not carried by the East India Company. Due to the Chinese restriction, the corporation outsourced it to “country traders”—private traders entitled to transport commodities from India to China. Smugglers on the Chinese shore bought opium from country traders. The traders gave the East India Company their sales of gold and silver. The corporation spent its gold and silver in China on profitable things to sell in England.

Opium imports into China rose from 200 chests in 1729 to 1,000 in 1767 and 10,000 between 1820 and 1830. Although chest weight varied by origin, it averaged 140 pounds (63.5 kg). By 1838, China imported 40,000 chests annually. China lost the balance of payments to Britain for the first time. Opium distribution networks had evolved across China, frequently with corrupt officials' help. High opium addiction levels affected imperial troops and officials.

Solution to Question 10:

The affected percentage of body surface area (BSA) is calculated using Wallace's rule of 9:

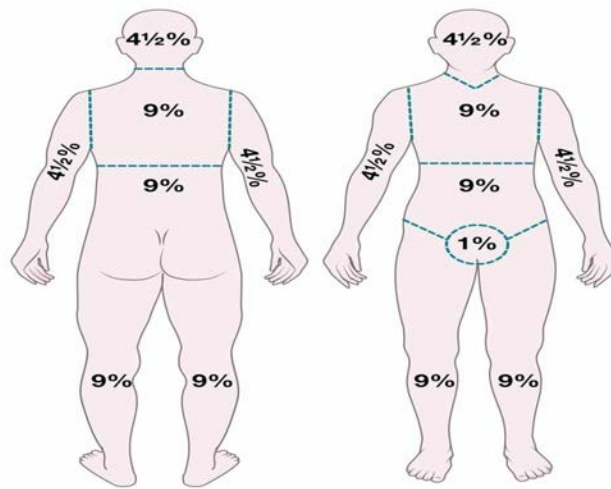
- Right upper limb - 9%
- Circumferential right thigh - 9% (Anterior thigh 4.5% and Posterior thigh 4.5% in Adults)

- Anterior chest - 9%
- Anterior abdomen - 9%

Thus, totaling to 36%.

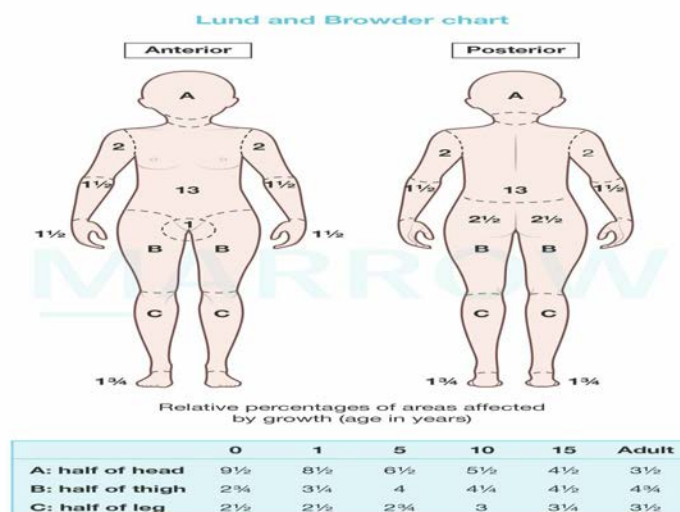
Note: According to Bailey's Short Practice of Surgery, 28th edition, the anterior thigh accounts for 4%, while according to Current Practice of Surgery, 14th edition, the anterior thigh accounts for 4.75 %. Either percentage, when totalled gives us the best possible answer to this question to be 36%.

The following image shows Wallace's rule of 9:



Lund and Browder's chart is an accurate method to measure burns if it is used correctly. In this, different proportions of body surface areas in children are taken into consideration. This chart compensates for the changes in the shape of the body with age and can accurately assess the burn area in children.

The below image shows the Lund and Browder chart.



Solution to Question 11:

Blackening occurs due to the deposits of smoke over the epidermis which can be removed with a wet cloth. This can be seen up to around 15 cm of firing.

Other injuries related to firearms are:

- Burn injuries - occur due to flame released during the firing. It is seen in contact shot wounds or distances up to 5-8 cm. Singeing of hair is caused due to burns.
- Tattooing - occurs due to the deposition of unburnt powder particles. These are difficult to remove. They are also called stippling or peppering. This is seen up to 50 cm of firing
- Greasing and collaring - occur due to the deposition of the lubricant of the bullet in the tissues, during its penetration. It can be found in contact, near-shot, or distant shot wounds.

In a firm contact shot, the imprint of the muzzle of the gun is seen on the skin around the wound. The victim is within the range of flame, hence singeing of the hair will be present. Tattooing begins only when the muzzle-to-target distance is $>1\text{cm}$, like in a loose contact shot.

When the distance between the victim and the firing is 5-8cm, it is called a close shot. The victim is within the range of the flame, hence, features related to flame, smoke, and powder deposition will be seen. Singeing, burning, blackening and tattooing will be present.

When the distance between the victim and the firing is $>50\text{ cm}$, it is a distant shot. The victim lies outside the range of flame and powder deposition. Hence, features related to flame, smoke, and powder deposition are absent. Burning, blackening, and tattooing are absent.

Solution to Question 12:

Datura poisoning

The common names include thorn apple, Jimson weed, and Devil's trumpet. The active ingredients include the anticholinergic alkaloids atropine, scopolamine, and hyoscyamine. The seeds or other parts of the plant can be ingested or smoked.

Toxic manifestations of Datura:

Mnemonic: "Hot as a hare, blind as a bat, dry as a bone, red as a beet, and mad as a hatter"

Also, remember the D's of Datura: Dry hot skin (Temp rises up to 42 C, a scarlet rash may also be appreciated).

- Dilatation of pupils along with loss of accommodation
- Diplopia
- Dryness of the mouth and throat, unquenchable thirst
- Dysphagia
- Dilatation of cutaneous blood vessels

- Dysarthria (Difficulty in talking)
- Drunken gait
- Delirium
- Drowsiness
- Death due to respiratory or cardiac failure

Medico-legal importance:

In India, Datura is used as a stupefying poison before robbery, kidnapping, and rape. It is also known as a rail-road poison, as it is commonly encountered during a journey. Robbers disguised as saints may offer 'prasad' mixed with Datura seeds and rob the passengers.

The following are images of the Datura plant and seeds:

Poison	Site of action
A. Datura	CNS
B. Nicotine	Cardiac muscle
C. Nux Vomica	Spinal Cord
D. Nitric acid	Skin





Nicotine is the principal constituent of tobacco responsible for its addictive behaviour. One of the newest agents introduced to aid in smoking cessation is varenicline.

Adverse effects of nicotine intoxication:

- Doses of 60 mg are toxic to the person and can cause respiratory paralysis.
- Nausea, vomiting, abdominal pain, diarrhoea
- Salivation
- Increased blood pressure, tachycardia, tremors and cardiac arrhythmias.

Other cardiac poisons include Oleander and Digitalis.

Strychnine poisoning occurs if a person ingests crushed seeds of *Strychnos nuxvomica*. If swallowed uncrushed, the seeds of *nuxvomica* have no poisonous action. The fatal dose is 15-30 mg of strychnine or one crushed seed.

Strychnine is a competitive antagonist of glycine at the postganglionic receptors in the motor neurons of the anterior horn of the spinal cord. This leads to increased activity in the motor neurons. Patients typically present with risus sardonicus, convulsions that are clonic initially but eventually become tonic and are most marked in anti-gravity muscles so that the body typically arches in hyperextension, the posture known as opisthotonus. If the spasm involves the abdominal muscles, the body bends forward (emprosthotonos), or to the side (pleurosthotonus)



Nitric acid is a corrosive inorganic acid. Due to the xanthoproteic reaction, the nitric acid combines with aromatic amino acids like tyrosine to form picric acid which is responsible for the yellowish discolouration of the skin.



Solution to Question 13:

The registered medical practitioner should take register a medico-legal case and immediately inform the police in this case. All three other options are steps to be taken eventually in this case but registering an MLC takes priority.

The Protection of Children from Sexual Offences (POCSO) Act, 2012 considers all individuals (both male and female) below the age of 18 years as children and protects them from sexual abuse. It defines various forms of sexual offenses which include:

- Sexual harassment: A person is said to commit sexual harassment upon a child when such person with sexual intent,
- Carries out any act with sexual connotations
- makes a child exhibit his/her body
- shows any object to a child in any form/media for pornographic purposes.
- Sexual assault
- Any act with sexual intent that involves physical contact without penetration.
- Aggravated sexual assault
- Similar to sexual assault along with the criteria explained under the term "aggravated".
- Penetrative sexual assault
- An act with sexual intent that involves physical penetration.
- Aggravated penetrative sexual assault
- Similar to penetrative sexual assault along with the criteria explained under the term "aggravated".
- Pornography

The term aggravated is used for those assaults done:

- by a person in a position of trust or authority such as police officer, security forces, relative, doctor.
- in the course of communal violence.
- on mentally & physically disabled.
- gang sexual assault.
- using deadly weapons.
- causing grievous hurt

The Act prescribes stringent punishment graded as per the gravity of the offense, with a maximum term of rigorous imprisonment for life and fine. Section 21(1) of the POCSO Act, 2012 requires mandatory reporting of cases of child sexual abuse to law enforcement authorities and applies to everyone including parents, doctors, and school personnel. Failure to report a suspicion of child abuse is an offense under the Act.

Solution to Question 14:

Pinpoint pupils can be observed in the overdose of opium.

Opioid overdose symptoms:

- Triad of coma, respiratory depression, and pinpoint pupils (miosis)
- Hypotension, bradycardia
- Shallow breathing

- Hypothermia

Naloxone is also used for reversing neonatal asphyxia due to opioid usage in labour. Naloxone is a pure opioid antagonist which reverses all effects except sedation.

Note: Naloxone is contraindicated in infants born to mothers with opiate addiction as it may precipitate acute neonatal withdrawal with seizures.

Causes of Miosis	Causes of Mydriasis
Carbolic acid poisoning	Datura poisoning
Morphine poisoning	Cyanides
Organophosphorous poisoning	Cocaine
Pontine haemorrhage	Ether
Barbiturate poisoning	Pethidine
	Nux vomica
	Viper venom
	Amphetamine

Solution to Question 15:

A chop wound would be unlikely to be found in this patient who has succumbed to a road traffic accident.

Injuries that are common following a blunt impact (here road and traffic accident) include:

- Bruises and Contusions
- Laceration
- Abrasion (patterned due to tyre marks)

Chop wounds or slash wounds are the wounds produced by chopping motion, made with heavy and sharp weapons like axes, cleave, etc. The dimensions of the wound correspond to the cross-section of the penetrating part of the weapon. The margins are sharp and show slight abrasion and bruising with the destruction of underlying organs. The presence of an incised wound on the skin with an underlying fracture or deep groove in the bone indicates a chop wound. The majority of these injuries are homicidal in nature and are typically inflicted upon the vulnerable regions of the body, such as the head, face, neck, shoulders, and extremities. Accidental injuries can be attributed to various sources such as power fans, band saws, or ship and aeroplane propellers. These incidents have the potential to cause significant lacerations to soft tissues or result in the amputation of body parts. Instances of self-inflicted lacerations with suicidal intent have seldom been observed.

Chop wound

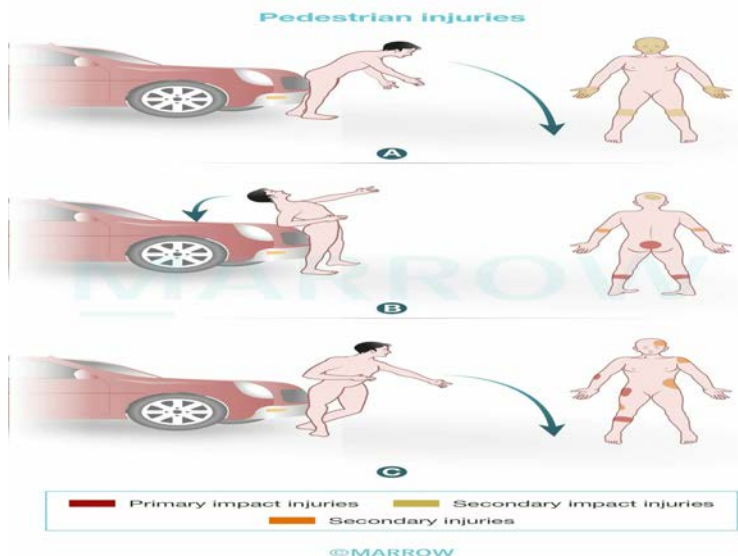


Individuals who sustain injuries as a result of road accidents can be categorized into three categories: pedestrians, cyclists (both pedal-powered and motorized), and occupants of cars, including both drivers and passengers. Among these three categories, pedestrians are frequently the ones who sustain injuries, however, the percentage of pedestrian casualties exhibits significant variation depending on the traffic patterns observed in different nations.

Injuries to pedestrians:

- Primary impact injuries are caused by the first impact of the vehicle on the victim.
- Secondary impact injuries are further injuries caused by the vehicle.
- Tertiary injuries are caused when the victim strikes other objects or falls on the ground after being hit by the vehicle.

Note: In case a pedestrian is hit by the vehicle only once (i.e only primary impact and no secondary impact by the vehicle) and then sustains injuries by falling on the road, it would be an example of secondary injury.



Injuries to occupants:

Approximately eighty per cent of vehicle impacts are against another vehicle or a stationary object and involve the front or front corners. Vehicle deceleration is rapid in this impact. Rarely, the car is hit from behind, causing 'acceleration'. The least common impacts are side and 'roll-overs'.

In a typical unrestrained deceleration condition, front-seat occupants surge forward and hit car structures in front of them.

An unrestrained impact may cause the following:

- The face and head following a collision with the windscreen glass, frame, or side pillars, cause skull and cervical spine fractures along with brain injury.
- Contact with the fascia or steering wheel can cause rib, sternal, heart, and liver damage.
- Hyperflexion may cause aorta tears at the descending part of the arch, where the vessel attaches to the vertebral column.
- The passenger's knees may hit the parcel shelf, causing fractures.
- The driver's legs, supported by stop and clutch pedals, may transfer impact force to the pelvis via the tibia and femur.
- The rebounding force from the contact may cause the head to violently swing backwards, damaging the cervical or thoracic spine.

Unrestrained rear occupants are projected forwards and impact the back of the front seats, including headrests if equipped, during severe deceleration. They may be hurled over the seats, striking and injuring the front-seat occupants, or they may be expelled through the windscreen, which has been damaged by them or the persons in front.

Injuries inflicted on motorcyclists following collision with other vehicles include patterned abrasions and various head, chest and spine injuries. A distinct injury can arise from the practice of "tail-gating" in which a motorcycle manoeuvres beneath the rear of a vehicle, resulting in significant cranial trauma or, in extreme cases, decapitation. The implementation of protective bars positioned at the sides and rear of trucks has effectively mitigated the occurrence of injuries resulting from bikes and vehicles passing beneath the vehicle.

Solution to Question 16:

In dry drowning death is due to sustained laryngeal spasm and subsequent airway closure. This occurs when there is sudden entry of water into the nasopharynx or larynx.

Dry drowning:

In this type of drowning, water does not enter the lungs, but death results from immediate sustained laryngeal spasm due to an inrush of water into the nasopharyngeal or larynx. Thick mucus, foam and froth may develop, producing a plug.

Immersion syndrome (Hydrocution/Submersion inhibition):

Death results from cardiac arrest due to vagal inhibition as a result of

- Cold water stimulates the nerve endings of the surface of the body.

- Water striking the epigastrium.
- Cold water enters ear drums, nasal passages, pharynx and the larynx which cause stimulation of the nerve endings of the mucosa.

Secondary drowning/Post immersion-syndrome/Near drowning:

It refers to a submersion victim who is resuscitated and survives for 24 hours. The patient may or may not be conscious. These patients may develop hypoxemia resulting in brain damage, electrolyte disturbances, pulmonary edema, sepsis, myocardial anoxia, etc., Death may occur from half to several hours after resuscitation in about 10-20% of people. Those who survive suffer from serious neurological damage.

Type of drowning	Cause of death
Freshwater drowning Seawater drowning	Disruption of surfactant
Dry drowning	Laryngeal spasm
Immersion syndrome	Vagal inhibition
Post immersion Syndrome/ Secondary drowning	Cerebral anoxia Pneumonia Pulmonary edema Chemical pneumonitis Metabolic acidosis Electrolyte imbalance

Solution to Question 17:

Cadaveric spasm cannot be observed in traumatic asphyxia as the occurrence of traumatic asphyxia is sudden and hence does not give the patient or victim time to contract his muscles during the time of death.

Cadaveric spasm has been observed in the following situations:

- Drowning death
- Brain Injury
- Fear or Excitement
- Electrocution

Perte's syndrome, also known as traumatic asphyxia, is a clinical syndrome characterized by craniocervical cyanosis, subconjunctival haemorrhage, numerous petechiae, and accompanying neurological symptoms. This syndrome manifests due to the abrupt or intense compression of the thorax or upper abdomen, or both. Frequently observed injuries encompass pulmonary contusion, hemothorax, and pneumothorax. Effective management of the related injuries is crucial for optimal treatment outcomes. The expeditious initiation of treatment with a focus on restoring adequate oxygenation and perfusion has the potential to yield a favourable prognosis. The recoveries of the patients were shown to be correlated with the severity of their injuries as well as the presence of concomitant injuries.

Cadaveric spasm is an antemortem change that indicates the cause and manner of death. This phenomenon is alternatively referred to as instantaneous rigor or cataleptic rigidity

The victims do not undergo postmortem flaccidity (unlike rigor mortis), thereby directly entering a state of instantaneous rigor. This results in muscle stiffening which indicates the victim's attitude at the time of death. Cadaveric spasm is limited to select muscles i.e., the muscles that were in a state of contraction at the time of death. The mechanism of cadaveric spasm is thought to be due to neurogenic phenomena.

Predisposing factors for this postmortem change include excitement, fear, fatigue, electrocution, nervous system injury, and convulsant poisons such as strychnine. It is a unique phenomenon that cannot be replicated by any other situation, and it is not possible to induce it through any means following the cessation of life. A significant amount of force is necessary to overcome stiffness. The phenomenon seamlessly transitions into typical rigor mortis and ceases to exist after rigor mortis subsides. The process of protein coagulation can be observed under microscopic analysis in burns, but not in cases of cadaveric spasm.

Solution to Question 18:

Among the options provided, a seven-day admission in the yellow zone of the hospital is not an example of grievous hurt, but any hurt which causes the sufferer to be in severe body pain for 15 days is considered to be an example of grievous hurt.

Grievous hurt (Sec 116 BNS) is inflicted voluntarily on another person and comprises any of the eight kinds (clauses) enumerated below:

Clause 1 — Emasculation.

Clause 2 — Permanent privation of the sight of either eye.

Clause 3 — Permanent privation of the hearing of either ear.

Clause 4 — Privation of any member or joint.

Clause 5 — Destruction or permanent impairment of powers of any member or joint.

Clause 6 — Permanent disfigurement of head or face.

Clause 7 — Fracture or dislocation of bone or tooth.

Clause 8 — Any hurt that endangers life or causes the sufferer to be in severe bodily pain or unable to follow his or her ordinary pursuits for 15 days.

Punishment for grievous hurt :

BNS Section 117(2) - previously, IPC 325: Punishment for voluntarily causing grievous hurt –imprisonment up to seven years and fine.

BNS Section 118(2) - previously, IPC 326: Punishment for voluntarily causing grievous hurt by dangerous weapons or means—life imprisonment or imprisonment of a minimum of one year, extendable up to ten years, and fine.

Related IPC's:

BNS Section 115(1) - previously, IPC 321: Voluntarily causing hurt.

BNS Section 117(1) - previously, IPC 322: Voluntarily causing grievous hurt.

BNS Section 115(2) - previously, IPC 323: Punishment for voluntarily causing hurt: Imprisonment up to one year, or with a fine up to one thousand rupees or both.

BNS Section 118(1) - previously, IPC 324: Voluntarily causing hurt by dangerous weapons or means (3 years imprisonment) or with fine or both.

Solution to Question 19:

Among the options, nail is the least preferred in the identification of a dead body.

Nails have been used in the identification of dead bodies, however, their use has been deemed uncommon. Skeletal muscles are resistant to degradation and are therefore more likely to persist during decomposition, thus making them useful for the identification of corpses. It is also important to note that few studies show the amount of identical loci decreases with time.

Suitable samples for DNA analysis encompass several types of nucleated biological material, including but not limited to white blood cells, hair root bulb cells, and spermatozoa. In the past, a substantial quantity of samples was necessary for doing this analysis. However, the advent of the polymerase chain reaction (PCR), a technique capable of amplifying minute quantities of DNA, has enabled the identification of DNA from even the most limited samples.

When obtaining samples, whether from a deceased individual, a living victim, or the accused, the size of the sample that can be collected is typically not a significant concern. The collection of samples from various locations of staining inside the scene, the deceased, the surviving victim, or the accused can provide distinct challenges. These areas may consist of minute traces of blood, semen staining, and similar substances, necessitating meticulous collection procedures to mitigate the risk of contamination.

Blood and hair with intact follicles are utilized in both live and deceased individuals, while samples of the spleen can be obtained during post-mortem examinations. Additionally, buccal smears can be collected from living individuals. In situations where sexual assault is a potential occurrence, or in instances of suspicious deaths, it is advisable to collect vaginal, anal, and penile swabs. In the event of a potential delay in transporting samples to the laboratory, it is recommended that the samples be subjected to freezing conditions at a temperature of -20°C or as specifically directed by the laboratory.

Solution to Question 20:

Pralidoxime is used in organophosphate poisoning and not as a chelator in heavy metal poisoning.

In organophosphate poisoning, oximes like pralidoxime (2-PAM) regenerates active acetylcholinesterase restoring neuromuscular transmission and relieves skeletal muscle end-plate block. It is usually used as an adjuvant to atropine which act as competitive inhibitors to counter the nicotinic and muscarinic effects of organophosphates.

Pralidoxime should not be administered without concurrent atropine, in order to prevent worsening of symptoms due to transient oxime-induced acetylcholinesterase inhibition.

Pralidoxime attaches to the anionic site of the cholinesterase enzyme, which remains unoccupied in the presence of organophosphates. It binds and forms oxime-phosphonate which diffuses away, leaving the reactivated cholinesterase. Pralidoxime is not an effective antidote to carbamates like carbaryl and may potentiate its toxicity, as the anionic site required for attachment of oximes is occupied.

Other options :

Option A : British anti-Lewisite (BAL) also known as Dimercaprol is used as an antidote for gold, arsenic (except arsine), or acute mercury poisoning (except non-alkyl mercury). It is also an adjunct to edetate calcium disodium in acute lead poisoning.

Option B : Ethylenediaminetetraacetic acid (EDTA) is used for acute lead poisoning. It is not suitable for the management of chronic lead poisoning. The calcium salt of EDTA can be used in the management of heavy metal poisonings where the metal in question has a higher affinity for EDTA than calcium in vivo. Because of their lower affinity to EDTA, mercury and arsenic cannot be effectively chelated.

Option D : Penicillamine is used as a chelator in the treatment of mild cases of lead poisoning. D-penicillamine (Cuprimine) can also be used to treat scleroderma, Wilson's disease, and rheumatoid arthritis. Side effects include nephrotoxicity, hematologic derangements, and a characteristic rash that involves the neck and axillae—elastosis perforans serpiginosa. It is associated with membranous glomerulonephritis, drug-induced lupus, and myasthenia gravis.

Forensic Medicine INI-CET 2024

Question 1:

What is the Declaration of Oslo related to?

- a) Human Torture
- b) Therapeutic abortion
- c) Human experimentation
- d) Hunger strikes

Question 2:

A middle-aged housewife consumed rodenticide powder and presented with K^+ levels of 2.5meq. On examination, irregular pulse with muscle paralysis is present. ECG shows ventricular ectopics. What is the suspected poison?

- a) Aluminium phosphide
- b) Zinc phosphide
- c) Barium carbonate
- d) Warafarin tablets

Question 3:

Which of the following antidotes are used to treat cyanide toxicity?

- a) 1 and 2
- b) 1, 2 and 3
- c) 1, 3 and 4
- d) 3 and 4

Question 4:

What is the correct order for opening cavities in case of a foetal autopsy?

- a) 1, 3, 2
- b) 3, 2, 1

c) 2, 1, 3

d) 1, 2, 3

Question 5:

Match the following:

a) 1-b, 2-c, 3-d, 4-e

b) 1-a, 2-b 3-d, 4-c

c) 1-d, 2-c, 3-a, 4-e

d) 1-c, 2-b, 3-d, 4-e

Question 6:

The following finding was found in an autopsy. Identity the weapon.



a) Bullet

b) Hammer

c) Iron rod

d) Scissors

Question 7:

Marshall's triad (punctate lacerations, bruises, and abrasions) is seen in which type of blast injury?

- a) Primary blast injury
- b) Secondary blast injury
- c) Tertiary blast injury
- d) Quaternary blast injury

Question 8:

What is not seen in heat exhaustion?

- a) Loss of thermoregulatory mechanism
- b) Altered sensorium
- c) Orthostatic Hypotension
- d) Core body temp below 105°F

Question 9:

Correctly matched antidote with toxicity

- a) Organophosphorus- Physostigmine
- b) Atropine - L-carnitine
- c) Benzodiazepene-Flumazenil
- d) Arsenic-Desferrioxamine

Question 10:

The compound given in the image below can be used in the treatment of which of the following poisonings?



- a) Aluminium Phosphide
- b) Mercury
- c) Phosphorus
- d) Arsenic

Question 11:

A student consumed an unknown substance and came with shock and pulmonary edema. The silver nitrate breath test is positive. What is the antidote?

- a) BAL
- b) Pralidoxime
- c) MgSO_4
- d) Atropine

Question 12:

Cytochrome C oxidase present in ETC can be blocked by all except?

- a) Carbon monoxide
- b) Cyanide
- c) Nitric Oxide
- d) Hydrogen sulphide

Answer Key

Question No.	Correct Option
1	b
2	c
3	b
4	a
5	a
6	a
7	b
8	a
9	c
10	c
11	c
12	c

Detailed Explanations

Solution to Question 1:

In 1970, the World Medical Association adopted a resolution on therapeutic abortion known as the Declaration of Oslo.

Other options:

Option A: Declaration of Tokyo: related to human torture and other degradation

Option C: Declaration of Helsinki: concerned with human experimentation.

Options D: Declaration of Malta: related to hunger strikes.

Solution to Question 2:

The above clinical scenario with a presentation of hypokalemia, cardiac arrhythmia and paralysis after consumption of rodenticide poison is caused by Barium carbonate.

Barium carbonate is used in rat poisons, a local corrosive irritant that, after absorption, mainly acts on muscles including the heart muscles.

The initial symptoms manifest as intense gastrointestinal distress, including nausea, vomiting, and diarrhoea. Subsequently, muscle cramps and stiffness, pupil dilation, tongue and larynx paralysis, and vertigo occur. Early impacts on the heart lead to elevated blood pressure, a slow and irregular pulse, and potential cardiac arrest in systole. Among the most notable changes are a loss of reflexes and paralysis.

Treatment involves the following steps:

- Gastric lavage to remove contents and administration of either sodium sulfate or magnesium sulfate solution to precipitate barium as insoluble sulfate.
- Intravenous administration of 10 ml of 10% sodium sulfate every 15 minutes to convert the soluble salt into insoluble sulfate.
- Purging with magnesium sulfate and repeated bowel lavage.
- Symptomatic management.

Other options:

Option A: Aluminium phosphide (ALP) is a solid fumigant pesticide, insecticide and rodenticide. ALP has a garlicky odour. Phosphine is a toxic agent. It acts by inhibiting the electron transport resulting from preferential inhibition of cytochrome oxidase. Severe toxicity produces ARDS, cardiac arrhythmias, congestive heart failure, pulmonary oedema, convulsions and coma. Cardiogenic shock is the most common cause of death, associated with higher mortality (35%-100%) but does not cause hypokalaemia.

Option B: Zinc phosphide poisoning can cause symptoms such as abdominal pain, nausea, vomiting, diarrhoea, and weakness. However, these symptoms are generally less severe compared to aluminium phosphide poisoning due to the highly toxic nature of phosphine gas released by aluminium phosphide.

Option D: Warfarin is an anticoagulant used in rat poison but is not directly associated with hypokalaemia.

Solution to Question 3:

Amyl nitrite, sodium nitrite and sodium thiosulfate are antidotes used to treat cyanide toxicity.

Cyanide acts by inhibiting the activity of enzyme systems like cytochrome oxidase and carbonic anhydrase. It inhibits the final step of oxidative phosphorylation, preventing the synthesis of ATP and its usage as an energy source. Cyanide reduces the blood's oxygen-carrying capacity and combines with the ferric iron atom of intracellular cytochrome oxidase, inhibiting oxygen intake for cellular respiration.

Treatment primarily aims at reversing the cyanide-cytochrome combination by providing nitrites which convert haemoglobin to methaemoglobin.

Cyanides combine with methaemoglobin to produce non-toxic cyanmethaemoglobin, which, in the presence of rhodanase and sulphate donors such as thiosulphate, transforms cyanide to thiocyanate, which is eliminated in the urine.

The Cyanide kit contains:

- 21 amyl nitrite pearls
- Two 10 ml vials of 3% sodium nitrite
- Two 50 ml vials of sodium thiosulphate.

Using a handkerchief, a 0.2 ml ampoule of amyl nitrite is broken and held over the patient's nose for 15 to 30 seconds of every minute, until sodium nitrite infusion is started.

IV administration of 0.3 g of sodium nitrite in 10 ml of sterile water is given slowly over a 5-minute interval. Sodium nitrite forms methaemoglobin (Hb-Fe²⁺), then competes with cytochrome oxidase for the cyanide ion, thus protecting cytochrome oxidase. Following this, 25 g of sodium thiosulphate in 50% solution is given i.v. over a period of 10 minutes. It converts cyanide to non-toxic thiocyanate, which is excreted in urine.

Other antidotes such as Hydroxocobalamin (Vit B12) and Dicobalt EDTA can also be used.

Solution to Question 4:

During the postmortem examination of a foetus, cranium is examined first intact along with the dura followed by the abdomen and thorax. The abdomen should be opened before the thorax to note the highest point of the diaphragm. Hence the answer is HEAD> ABDOMEN>THORAX.

During the removal of body cavities, preference is given depending upon the findings. Cavities that give trace evidence are opened first as compared to cavities that give maximum evidence to reduce the various artefacts.

Body cavities to be opened first:

- New Born: Abdomen> thorax
- Poisoning: Cranium
- Asphyxial deaths: Cranium > Thorax > Abdomen > Neck

The protocol for fetal autopsies differs slightly from that of adult autopsies. It includes examining the umbilicus, umbilical cord, placenta, scalp hairs, lanugo hairs, nails and their length, skin color and texture, scrotal sac wrinkling, as well as specific fetal measurements of the head, chest, and abdominal circumference.

Solution to Question 5:

The correct answer is option A: 1-b, 2-c, 3-d, 4-e

Paraphilias or perversions are sexual stimuli or acts that are deviations from normal sexual behaviors but are necessary for some people to experience arousal and orgasm.

1. Eonism (also known as transvestism) is a sexual perversion, where a person's whole personality is dominated by a desire to be identified as a member of the opposite sex. This is usually seen in males who obtain sexual pleasure by wearing female clothes. There is no hormonal/genital abnormality.

2. Algolagnia is sexual pleasure derived from inflicting pain (active algolagnia) or enduring pain (passive algolagnia).

Active algolagnia also called Sadism, is the sexual gratification obtained from acts of physical cruelty or infliction of pain upon one's partner. It is seen more commonly in men. To obtain sexual gratification the sadist may bite, beat, cut, burn, or otherwise mistreat or abuse his sexual partner.

3. Frotteurism refers to the sexual gratification achieved by touching or rubbing against a non-consenting person. This behavior often occurs in busy and crowded places. It is predominantly described in men.

4. Necrophilia is an obsession with obtaining sexual gratification from dead bodies.

Bestiality is an act of using a lower animal for sexual gratification. It is an unnatural sexual offence and is punishable in India.

1. Eonism	b. Sexual pleasure by wearing clothes of the opposite sex
2. Algolagnia	c. Sexual gratification in inflicting pain
3. Frotteurism	d. Sexual gratification by rubbing private parts against another person
4. Necrophilia	e. Sexual gratification by intercourse with dead bodies

Solution to Question 6:

The given image shows a well-defined punched-out hole in the inner table of the skull. Hence, it could most probably be an entry wound produced by a bullet.

Entry wound: The initial contact of the missile punches a clean hole through the outer table of the skull. Where the bullet emerges internally, the inner table is then unsupported and a cone-shaped plug is detached, forming a crater that is appreciably larger than the external hole.

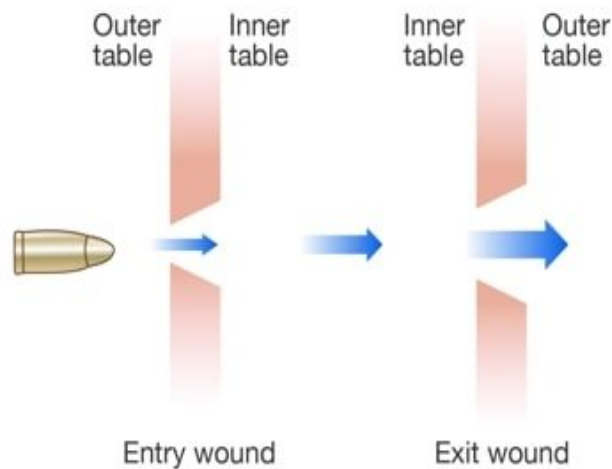
Bevelling: inner table

Punched out hole: outer table

Exit wound: If the bullet traverses the cranium and penetrates the opposite side, then the same pattern occurs, but this time with the small hole on the inner table and the crater on the outside.

Bevelling: outer table

Punched out hole: inner table



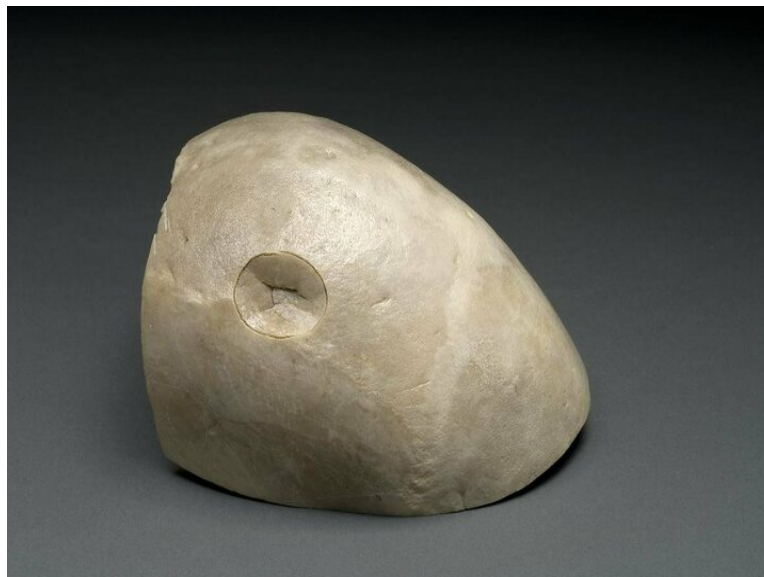
The sequence of bullet injuries is given by Puppe's rule.

Puppe's rule: It is possible to determine the sequence of injuries in skull fractures caused by blunt injuries to the head because the second fracture ends at the point where a break in cohesion is already present. The fracture lines created by every other shot extend only up to the fractured lines produced by the previous shot without continuing over them. Even though this rule is true for all blunt injuries to the skull, it is of particular importance in gunshot wounds.

Other options :

Option B: Hammer: When a hammer is used, the fracture is circular or in the shape of an arc of a circle. The diameter of the fracture is the same as the striking surface. There are generally no radiating linear fractures. A depressed fracture is caused by blows from a heavy weapon with a small striking surface. It is also called a signature fracture as the pattern of fracture resembles the weapon or object used.

The following is an image of a depressed fracture:



Option C: Iron Rod: An irregular missile such as an iron rod would produce several irregular lacerated wounds, and the exit wounds are larger than the entrance wounds.

The following is an image of the famous 3D model of Phineas Gage's skull, who survived a workplace injury of a tamping iron rod through his skull.



Solution to Question 7:

Marshall's triad in a bomb explosion is seen due to secondary blast injuries.

Marshall described the characteristic injuries associated with bomb explosions. It includes:

- Abrasions
- Bruises/ contusions
- Puncture lacerations

Blast injuries are classified as:

- Primary injuries- effect of transmitted blast waves on gas-containing structures. External injuries are absent and are characterized by internal injuries. The middle ear, lungs, and gastrointestinal tract are the most susceptible to this.
- Secondary injuries- injuries caused by shrapnel and other airborne debris propelled by the explosion. These injuries may affect any part of the body and sometimes result in penetrating trauma. Most casualties are caused by secondary injuries.
- Tertiary injuries- are caused by blast wind that can throw victims against solid objects.
- Quaternary injuries- all other injuries caused by explosions are not included in the above three categories. eg, flash burns, crush injuries, respiratory effects due to toxic dust, gas, etc.

Solution to Question 8:

Loss of the thermoregulatory mechanism is not seen in heat exhaustion. It is seen in heat stroke.

Heat exhaustion is caused by excessive exposure to heat, leading to nausea, vomiting, dizziness, altered sensorium(option B), thirst, fatigue, headache, sweating, and decreased urine output. Increases in sweating and dehydration can cause orthostatic hypotension(option C). Body temperature may rise but does not exceed 40.5°C (105°F). In heat exhaustion, the body maintains thermoregulation and no CNS dysfunction occurs. Prompt treatment is essential to prevent the progression of life-threatening heat stroke.

Solution to Question 9:

Benzodiazepine-Flumazenil is the correct match of toxicity with its antidote.

Flumazenil is a competitive antagonist at the benzodiazepine binding site of the GABA A receptor. It antagonizes both the agonist (benzodiazepines) and inverse-agonist (beta-carbolines) of this receptor.

The duration of clinical effects usually is only 30–60 min. It is absorbed rapidly after oral administration, less than a quarter of the drug reaches the systemic circulation. This is due to the extensive first-pass hepatic metabolism. Flumazenil is eliminated almost entirely by hepatic metabolism to inactive products with a $t_{1/2}$ of about 1 hour.

It is primarily indicated for the management of benzodiazepine overdose and the reversal of benzodiazepine anaesthesia. Flumazenil is only available for intravenous administration. It reliably reverses the sedation, respiratory depression, and amnesia caused by benzodiazepines.

Other options:

Option A: In organophosphate poisoning (insecticide) with both muscarinic and nicotinic manifestations, antidotes administered are atropine and pralidoxime. Atropine is effective against muscarinic receptors. Pralidoxime (2-PAM) is a cholinesterase-reactivating agent. Hence, both drugs must be co-administered to treat both muscarinic and nicotinic symptoms.

Option B: Treatment of atropine toxicity includes administration of a short-acting barbiturate or diazepam to control convulsions. Physostigmine is used as an antidote to atropine poisoning to treat delirium and coma. Other measures for symptomatic management are respiratory support and control of hyperthermia.

Option D: ■■■In Arsenic toxicity, the antidote preferred is freshly prepared hydrated ferric oxide, which converts toxic arsenic to non-toxic ferric arsenite. DMSA (Succimer), British anti-Lewisite (BAL), and penicillamine can also be used.

Solution to Question 10:

The compound given in the image is copper sulphate. It can be used in the treatment of phosphorus poisoning.

The initial stage of acute phosphorus poisoning through ingestion involves severe abdominal discomfort, vomiting, and diarrhoea. Excreta and breath have a garlicky odour. Luminescent faeces and vomit are characteristically seen (smoky stool syndrome). Direct contact with the skin

can result in second and third-degree burns. Shock, hematemesis due to liver failure, and renal insufficiency are potential causes of death.

Regular inhalation of phosphorus fumes, leading to chronic phosphorus poisoning, results in necrosis of the lower jaw around a decayed tooth. This condition is known as phossy jaw and is associated with the discharge of foul-smelling pus from multiple sinuses.

Gastric lavage with potassium permanganate oxidizes phosphorus into phosphoric acid and phosphates. Activated charcoal can be used to adsorb the poison.

Stomach wash with 0.2% copper sulphate solution or 0.2 g of copper sulphate every 5 minutes until vomiting occurs. It coats the particles of phosphorus with a film of copper phosphide, which is relatively harmless. As copper has caustic properties and can cause acute poisoning, adequate care should be taken. Copper sulphate was also used for topical application in phosphorus poisoning. It is not recommended now.

Other options:

Option A: Aluminium Phosphide has no antidote. Magnesium sulphate is given in ALP poisoning to correct hypomagnesemia and arrhythmias and reduce organ toxicity.

Option B and D: British anti-Lewisite (BAL), also known as Dimercaprol, is used as an antidote for Mercury poisoning (except non-alkyl mercury) and Arsenic poisoning.

Solution to Question 11:

The patient presents with clinical features suggestive of Aluminium phosphide (ALP) poisoning. Magnesium sulphate is given in ALP poisoning to correct hypomagnesemia, arrhythmias and reduce organ toxicity.

Aluminium phosphide (ALP) is a solid fumigant pesticide, insecticide and rodenticide. ALP has a garlicky odour. Phosphine is a toxic agent. It acts by inhibiting the electron transport resulting from preferential inhibition of cytochrome oxidase. Aluminium phosphite and Aluminium hypophosphite are non-toxic agents. Phosphine is a systemic poison and affects all organs of the body. The presence of HCl in the stomach accelerates this reaction.

Signs and Symptoms include irritation of mucous membranes and acute respiratory distress in mild toxicity. Moderate toxicity produces ataxia, numbness, tremors, muscular incoordination and paralysis. Severe toxicity produces ARDS, cardiac arrhythmias, congestive heart failure, pulmonary oedema, convulsions and coma. Cardiogenic shock is the most common cause of death and is associated with higher mortality(35%- 100%)

There is no specific antidote. Gastric lavage should not be done with warm water as on coming in contact with water, aluminium phosphide releases phosphine and increases the concentration of toxic agent phosphine, so it is done with potassium permanganate and activated charcoal. Magnesium sulphate is used to correct hypomagnesemia, arrhythmias and reduces organ toxicity. Supportive measures to treat acidosis, shock and hemodialysis can be done.

On post-mortem examination, a garlic-like odour is present at the mouth, nostrils and in the gastric contents. Silver nitrate paper test is done to detect phosphine which turns the paper black.

Silver nitrate paper test: This test is used to detect aluminium phosphide poisoning.

- Aluminium phosphide (ALP) is used as a pesticide, insecticide, and rodenticide.
- When ALP comes in contact with moisture, it releases an active component called phosphine.
- When the silver nitrate-impregnated filter paper, comes in contact with phosphine the paper turns black in colour.
- The test is done in two methods:
 - a. Gastric aspirate: 5 mL of gastric aspirate and 15 mL of water are put in a flask and the mouth of the flask is covered with silver nitrate-impregnated filter paper. The flask is heated at 50°C for 15 to 20 minutes. If the filter paper turns black, it indicates the presence of phosphine.
 - b. Breath: The patient is asked to breathe in and out through a face mask which is impregnated with silver nitrate. If the face mask turns black, it indicates the presence of phosphine. The breath test is positive only if the patient ingests more than 6 grams of aluminium phosphide.

Solution to Question 12:

Nitric Oxide cannot block cytochrome C oxidase present in ETC.

Nitric Oxide: While NO can modulate ETC activity and inhibit respiration temporarily, it does not universally block Complex IV like the other inhibitors. (Option C)

Cytochrome c oxidase (Complex IV) in the electron transport chain (ETC) is the terminal enzyme that facilitates the transfer of electrons to molecular oxygen, reducing it to water. Various inhibitors can block this process, leading to impaired oxidative phosphorylation and ATP synthesis:

Carbon monoxide: CO binds strongly to the heme iron of cytochrome c oxidase in its reduced (ferrous, Fe^{2+}) state. This prevents oxygen from binding and accepting electrons, halting the production of water and the proton gradient. Poisoning with CO can lead to severe hypoxia at the cellular level, even in the presence of normal oxygen levels. (Option A)

Cyanide: Cyanide binds irreversibly to the ferric (Fe^{3+}) iron in the heme group of cytochrome c oxidase. It completely blocks electron transfer to oxygen, making it a potent inhibitor. Cyanide poisoning results in rapid cellular asphyxiation and is life-threatening. (Option B)

Hydrogen Sulfide: Inhibits Complex IV at high concentrations by binding to its heme groups, similar to cyanide. Its inhibition is dose-dependent, but it remains a strong inhibitor at high levels. (Option D)

Forensic Medicine NEET 2024

Question 1:

During the post mortem examination following finding is seen. What is this indicative of



- a) Choking
- b) Throttling
- c) Hanging
- d) Smothering

Question 2:

An 18-year-old boy was brought by police, and accused of sexual assault, but the boy claims to be a minor (16 years old). Which radiographs will you order to solve this discrepancy?

- a) Elbow and hip joint
- b) Elbow and ankle joint
- c) Shoulder and skull joint
- d) Wrist and knee joint

Question 3:

A man sexually aroused when he wears his female partner's dress and heels. He denies getting attracted to male. What is this condition called?

- a) Gay sexual preference
- b) Transvestic fetishism
- c) Gender dysphoria
- d) Voyeurism

Question 4:

A woman gave birth to a child 8 months after the death of her husband. The in-laws claim that the child is not their son, however, the DNA testing confirms the paternity. The child is known as _____

- a) Illegitimate child
- b) Posthumous child
- c) Surrogate child
- d) Supposititious child

Question 5:

The railway police has brought a person with suspected poisoning. On examination, he was confused with slurred speech, hot skin, dry mouth, dilated pupils. What is the probable poisoning ?

- a) Cocaine
- b) Datura
- c) Cannabis
- d) Alcohol

Question 6:

A police officer brought a child to the hospital. There is redness and tenderness around perianal region. A rectal swab is taken and the image given below is the finding. What test is it indicative of?



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- a) Barberio test
- b) Florence test
- c) Teichmann test
- d) Takayama test

Question 7:

Rigor mortis is well established in 12-24 hours after death. What is the order of appearance of rigor mortis?

- a) Toe to head
- b) Head to toe
- c) Centre to periphery
- d) All the body parts simultaneously

Question 8:

A 5-year-old child has consumed overdose of iron syrup, around 60mg/kg. Which among the following is contraindicated?

- a) BAL
- b) Deferoxamine
- c) Calcium EDETATE
- d) DTPA

Question 9:

A sexual assault survivor had to testify in court and an in-camera trial was conducted. What is an "in-camera" trial?

- a) Open court proceedings
- b) Closed court proceedings
- c) Proceedings recorded on camera
- d) Proceeding held in a different location

Question 10:

A 36-year-old man died following a RTA. The doctor informed the police officer and shifted the body to the mortuary. Autopsy in this case will be conducted at the request of _____

- a) Forensic expert
- b) Investigating police officer
- c) Defense lawyer
- d) Prosecution lawyer

Question 11:

A person comes to emergency after consumption of a substance, the form and dose of which is not known to the relatives. The patient has tachypnea and hypotension. On metabolic assessment, high anion gap metabolic acidosis is noted with hypocalcemia. What is the diagnosis?

- a) Methyl alcohol
- b) Ethylene glycol
- c) Datura
- d) Digoxin

Question 12:

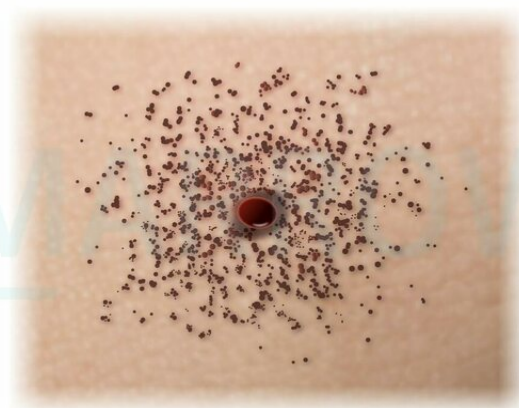
A patient presented to the OPD with findings as shown below. Identify the type of poisoning in this patient.



- a) Chronic malnutrition
- b) Copper
- c) Mercury
- d) Arsenic

Question 13:

The following findings were noted during the autopsy of a patient with a gunshot injury. What is the likely range of the gunshot?



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- a) Cannot be opined

- b) Distant range
- c) Contact range
- d) Close range

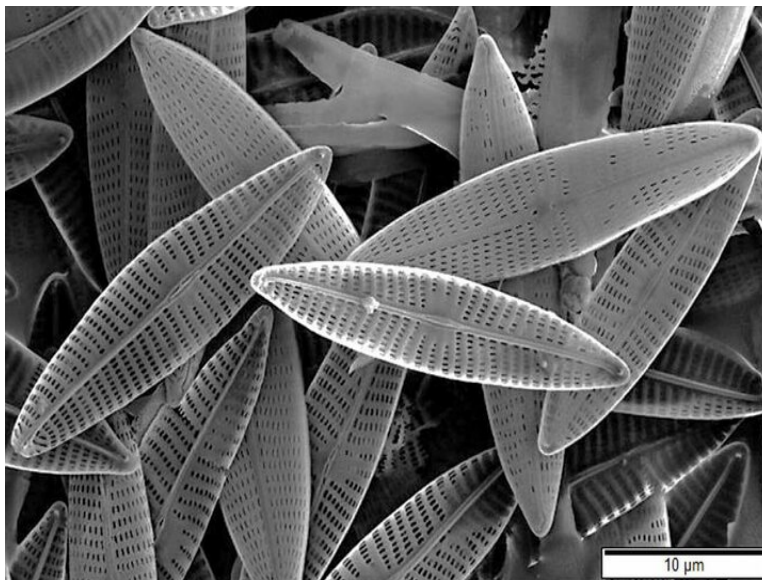
Question 14:

A large number of people who consumed cheap alcohol presented to the emergency department with abdominal pain, confusion, and decreased vision. Methyl alcohol level in blood was detected to be around 20 mg %. On assessment which of the following metabolites will be found in blood?

- a) Glycolic acid and Glyoxylic acid
- b) Formic acid and lactic acid
- c) Glycolic acid and lactic acid
- d) Oxalic acid and formic acid

Question 15:

In the case of drowning, following autopsy, an organism was isolated from the water present in the internal organs. Which of the following is true regarding the isolated organism?



- a) Present only in the bone marrow
- b) Dried outer wall contains silica and chlorophyll
- c) It is a microscopic unicellular bacteria
- d) It is not heat and acid resistant

Question 16:

A pregnant female comes to a gynaecologist, who sends her for USG. On USG, twin pregnancy was noted with about a one-month difference in the age of fetuses. Which of the following is true with regard to this patient?

- a) Superfecundation
- b) Superfetation
- c) Supposititious child
- d) Posthumous child

Question 17:

A 32-year-old woman G3P2L2 came to you in active labor. The labour process was arrested at around cervical dilatation of 8 cm. She was counselled regarding the need for an emergency cesarean section. The baby was delivered by cesarean section and was doing well but the mother developed intractable postpartum hemorrhage. Conservative measures were tried initially but failed. Emergency hysterectomy was done as a life-saving procedure. What are the medicolegal implications of this situation?

- a) Patient could sue the doctor for removing her uterus
- b) Hysterectomy was done as a life saving procedure, so the doctor is legally protected
- c) Consent can be taken from the patient on OT table
- d) Counselling of patient and attendants is not needed

Question 18:

A 30-year-old man who was found dead was brought in for autopsy. The doctor notices a greyish-white waxy layer over the body with preserved facial features. Which of the following statements is true concerning adipocere formation?

- a) It occurs in hot and dry temperatures
- b) It occurs in cold temperatures
- c) It is a type of body preservation with saponification of fats
- d) Occurs within few hours of death

Answer Key

Question No.	Correct Option
1	d
2	d
3	b
4	b
5	b
6	a
7	b
8	a
9	b
10	b
11	b
12	d
13	d
14	b
15	b
16	b
17	b
18	c

Detailed Explanations

Solution to Question 1:

Abrasions around the mouth, cheeks, lips or lesions within the lips may reliably indicate smothering.

The non-specific signs of death due to asphyxia:

- Petechial haemorrhages in the face, eyebrows, eyelids, forehead, ears, conjunctiva, sclera - When they occur in the visceral pleura, they are called Tardieu's spots
- Emphysematous, congested lungs with serosanguinous fluid exudates in alveoli.
- Cyanosis or dark-coloured blood due to accumulation of CO₂.
- Congestion of the larynx with slight oozing

Pressure marks on the face can occur as a part of normal postmortem changes. Circumoral and circumnasal pallor can be caused merely by passive pressure of the dependent head after death.

Note: The coexistence of these findings, in themselves, does not prove that death resulted from mechanical asphyxia. All these phenomena are non-specific and are in no way peculiar to this mode of death. They are of no value in proving that death resulted from mechanical asphyxia.

Solution to Question 2:

Wrist and knee joint radiographs will be ordered to solve this discrepancy as lower end of radius and upper end of tibia fuse at 18-19 years. If these joints are fused, It is certain that the boy is more than 18 years and his claim is false.

Bone/ joint	Max age of fusion
Wrist joint centers (lower end of ulna and radius)	19 years
Knee joint centres(lower end of femur, upper end of tibia and upper end of fibula)	20 years
Shoulder joint (head of humerus, greater tuberosity and lesser tuberosity)	19 years
Hip joint (head of femur, greater trochanter)	17 years
Elbow joint centers (lower end of humerus, upper end of radius, olecranon to ulna)	17 years
Ankle joint	17 years

Solution to Question 3:

The given scenario is suggestive of transvestic fetishism.

It is a paraphilia characterized by recurrent sexually arousing fantasies, sexual urges, or behavior involving the wearing of clothes of the opposite sex and appearing as a member of the opposite sex. It is also called transvestic fetishism. It is usually seen in males who obtain sexual pleasure by wearing female clothes.

Fetishistic transvestism is a paraphilia whereas transvestism is a gender identity disorder. There is no hormonal/genital abnormality in fetishistic transvestism.

Option A : Gay sexual preference is due to attraction to people of one's own gender.

Option C: Gender dysphoria is a term that describes a sense of unease that a person has due to a mismatch between their biological sex and their gender identity. This sense of unease or dissatisfaction can lead to depression and anxiety and can have a harmful impact on daily life.

Option D : Voyeurism/Peeping Tom/Scotophilia is sexual gratification by secretly watching others getting undressed, taking a bath, performing sexual intercourse, etc.

Solution to Question 4:

The child delivered after the death of the biological father, or a child delivered after the death of the biological mother, usually when delivered by a cesarean section is a posthumous child.

Legal issues involved with a posthumous child are legitimacy, inheritance of property, and the case of compensation for slander against the mother.

Other options :

Option A: Illegitimate child is a child born out of wedlock i.e. when the parents are unmarried.

Option C: A surrogate child is born out of a pregnancy in which a woman gives birth to a baby for a person who is not able to have children.

Option D: Supposititious child or fictitious child is when a woman pretends to go through pregnancy and delivery. Later, she produces a living child on her own to obtain money or claim the property.

Solution to Question 5:

The given clinical scenario is suggestive of Datura poisoning.

The two species are jimson weed (*Datura stramonium*) and angel's trumpet (*Datura candida*).

Seeds or other parts of the plant can be ingested or smoked. Poisoning leads to delirium, hallucinations, and seizures. Additionally, other classic anticholinergic effects, such as mydriasis, tachycardia, dry mouth and skin, blurred vision, urinary retention, and hyperthermia may present.

Gastric emptying is often delayed, and the small, plentiful seeds can become trapped among the gastrointestinal folds after ingestion. Thus gastric decontamination can be an important therapy.

Physostigmine, a reversible acetylcholine esterase antagonist, is an effective treatment for severe anticholinergic poisoning.

Solution to Question 6:

The test done in this scenario is the Barberio test.

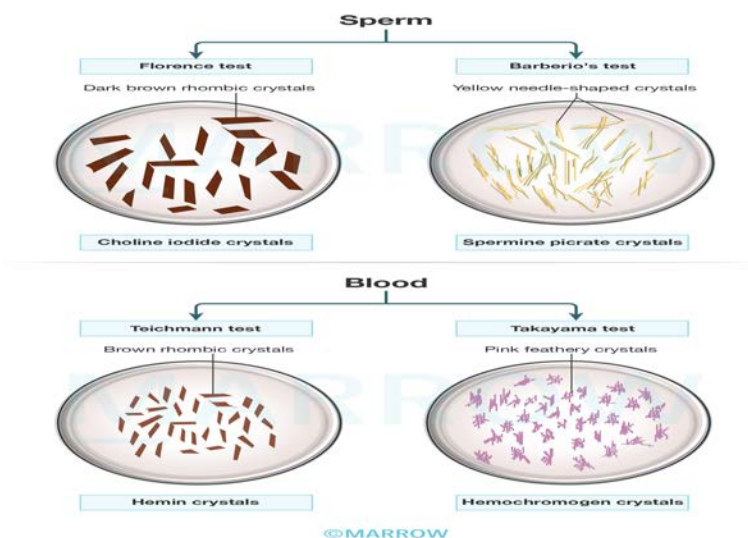
Barberio test is used to detect seminal fluid. When a saturated aqueous or alcoholic picric acid solution is added to the spermatic fluid, yellow needle-shaped rhombic crystals of spermine picrate are produced.

Other options:

Option B: The Florence test detects semen. If semen is present, dark brown rhombic or needle-shaped crystals of choline iodide appear immediately.

Option C: The Teichmann test (hemin crystal test) is used to detect bloodstains. If bloodstain is present, faint yellowish-red to brownish-black rhombic crystals of hemin or hematin chloride appear. When hydrogen peroxide is added, the hematin crystals give bubbles of gas.

Option D: Hemochromogen crystals are seen in the Takayama test. It is a microchemical test used to detect bloodstains. The stain used is Takayama reagent (Sodium hydroxide, pyridine, glucose). If bloodstain is present, pink feathery crystals of hemochromogen or reduced alkaline hematin appear in a few minutes. The test is negative if the crystals are not formed in 30 minutes. This test is more reliable and gives positive results with old stains also but not a definitive test to detect blood stains.



Solution to Question 7:

The order of appearance of rigor mortis is from head to toe.

Rigor mortis (cadaveric rigidity or death stiffening) refers to the stiffening and shortening of muscles following death. The basic sequence of the rigor mortis begins from the head and continues downwards the body. It disappears in the same order of appearance. Nysten's rule describes the sequential onset of rigor mortis in the various muscle groups.

The muscles are affected in the given order:

- Involuntary muscles - Heart (left chamber of the heart is affected first)
- Eyelids
- Neck and lower jaw
- Face
- Chest
- Upper extremities

- Abdomen
- Legs
- Fingers and toes

Solution to Question 8:

The clinical scenario points to iron toxicity, which is managed with deferoxamine (also known as desferrioxamine), calcium EDETATE and DTPA whereas Dimercaprol, also known as BAL (British anti-Lewisite), is used to treat arsenic, mercury poisoning, and Wilson's disease.

Large amounts of ferrous salts can be fatal in children. Initial symptoms include abdominal pain, diarrhea, vomiting, and gastrointestinal bleeding. In severe cases, cyanosis and hyperventilation due to acidosis can develop. This can sometimes be followed by an asymptomatic phase. Later, hypoglycemia and hepatic and renal failure can develop. Pyloric stenosis and gastric scarring can develop due to its corrosive nature. Ultimately, if left untreated, the child can die.

The gastric contents can be used for an iron color test to estimate the iron concentration in plasma. As iron tablets are radiopaque, an X-ray can be used to assess the number of pills taken. Deferoxamine (Option B) is selectively used as a chelating agent to manage cases of acute iron poisoning, hemochromatosis, and patients with iron overload from repetitive blood transfusions. Desferrioxamine can be administered intramuscularly, intravenously, or subcutaneously but not orally. Side effects include pruritus, wheals, rash, and anaphylaxis.

Deferiprone and deferasirox are other oral iron chelators used to treat iron overload in thalassemia patients.

Other options:

Option C : Ethylenediaminetetraacetic acid (EDTA) is used for acute lead poisoning. It is not suitable for the management of chronic lead poisoning. The calcium salt of EDTA can be used in the management of heavy metal poisonings where the metal in question has a higher affinity for EDTA than calcium in vivo. Because of their lower affinity to EDTA, mercury and arsenic cannot be effectively chelated.

Option D : Diethylenetriamine pentaacetate (DTPA) is a chelating agents which works by binding and holding on to radioactive materials (plutonium, americium, and curium) and poisons like iron. Calcium DTPA is well tolerated and effective as desferrioxamine in mobilising iron than Zinc DTPA.

Solution to Question 9:

'In-camera trial' is a closed court proceeding in which the case is heard in private i.e. in court where general public is excluded.

BNSS sections related to court proceeding are tabulated below:

BNSS	CrPC	Deals with
63-71	61-69	Summons
181	162	Oral statements made to the police and recorded by the police should not be signed
195	175	Power to summon persons by a police officer
196	176	Magistrate inquest
350	312	Expenses of complaints and witnesses
366366(2)	327	Criminal court is generally deemed to be an open court. In-camera trial for rape
389	350	Summary procedure for punishment for non-attendance by a witness in obedience to the summons
395	357	Order to pay compensation

Solution to Question 10:

In the case of the 36-year-old man who died following an RTA, the autopsy will be conducted at the request of the investigating police officer.

According to BNSS Section 194 (Section 174 CrPC), a police inquest is the standard procedure for investigating unnatural deaths in India. This process is typically carried out by a police officer, not below the rank of Senior Head Constable, who is responsible for conducting a preliminary inquiry into the circumstances surrounding the death.

During the police inquest, the investigating officer visits the scene of the incident, examines the body, and documents any visible injuries or signs of trauma. The officer's report will detail these findings and assess whether the death appears to be suicidal, homicidal, accidental, or suspicious. If the situation warrants, the body will be sent for a postmortem examination to be conducted by a qualified medical professional to determine the exact cause of death.

In certain cases, a magistrate inquest [BNSS 196 (Section 176 CrPC)] may be conducted instead of, or in addition to, a police inquest. This is particularly true in cases involving deaths under special circumstances, such as lock-up deaths, deaths while in police custody, custodial rape, death in a psychiatric hospital, dowry death or deaths resulting from police actions. The magistrate inquest is carried out by a District Magistrate, Subdivisional Magistrate, or another authorized magistrate, and involves a more formal judicial inquiry into the cause of death.

Solution to Question 11:

The patient's presentation of tachypnea, hypotension, high anion gap metabolic acidosis, and hypocalcemia, is suggestive of ethylene glycol poisoning.

Ethylene glycol is a clear, colourless, and odourless liquid commonly used as an antifreeze. It is known for its bitter-sweet taste and is absorbed mainly through ingestion rather than skin contact. Upon ingestion, ethylene glycol is metabolized in the liver to several toxic metabolites, including glycolic acid and oxalic acid, which contribute to its harmful effects.

Initially, ethylene glycol poisoning can cause symptoms such as vomiting, lethargy, and ataxia, progressing to severe effects including tachycardia, tachypnea, and circulatory collapse within 12-24 hours. As the condition progresses, typically within one to three days, patients may experience hypocalcemia, oliguria, tubular necrosis, and acute kidney injury (AKI). A notable feature of ethylene glycol poisoning is the formation of calcium oxalate crystals in the urine, which can lead to renal failure.

The postmortem findings often include cerebral edema, chemical meningoencephalitis, and significant damage to the liver and kidneys. Oxalate crystals may be observed in the brain, spinal cord, and kidneys, further confirming the diagnosis.

For treatment, immediate measures include gastric lavage, activated charcoal, and administration of ethanol or 4-methyl pyrazole (fomepizole), which inhibit the enzyme alcohol dehydrogenase to prevent further metabolism of ethylene glycol into toxic compounds. Additional treatments such as intravenous sodium bicarbonate and calcium gluconate are used to address metabolic acidosis and hypocalcemia, respectively, while hemodialysis is indicated in severe cases with significant renal impairment or high serum ethylene glycol concentrations.

Other options:

Option A: Methyl alcohol (Methanol) and Ethylene glycol poisoning can present very similarly. However, the key distinguishing feature of methanol poisoning is the presence of eye symptoms (as opposed to kidney damage in ethylene glycol) - like blurring of vision/loss of vision due to retinal damage by formate metabolites.

Methanol--Alcohol Dehydrogenase (ADH) $\Rightarrow$ Formaldehyde $\Rightarrow$ (Retinal damage)

Ethylene glycol--Alcohol Dehydrogenase (ADH) $\Rightarrow$ Glyoxyaldehyde $\Rightarrow$ (Kidney damage)

Option C: Datura is used as a stupefying poison before robbery, kidnapping, and rape. The active ingredients include the anticholinergic alkaloids atropine, scopolamine, and hyoscyamine. The seeds or other parts of the plant can be ingested or smoked. Toxic manifestations include dilatation of pupils along with loss of accommodation, diplopia, dryness of the mouth and throat, unquenchable thirst, dysphagia, dilatation of cutaneous blood vessels, dysarthria, drunken gait, delirium, drowsiness and death due to respiratory or cardiac failure.

Option D: Digoxin toxicity can occur in the elderly and patients with renal impairment and presents with nausea, vomiting, diarrhoea, confusion, blurred vision with a characteristic yellowing of vision (compared to the "starry night" painting), and irregular heart rhythms. Overdose and resultant arrhythmias are managed with anti-digoxin Fab fragments (Digibind or Digifab).

Solution to Question 12:

The image showing the raindrop pattern of skin pigmentation and the Aldrich-Mees lines on the fingernails of the patient are suggestive of chronic arsenic poisoning.

Arsenic in its metallic form is not poisonous, but when it unites with oxygen, it forms poisonous vapours of arsenic trioxide. These interfere with cellular respiration, damaging mitochondrial enzymes. It is present in fruit sprays, copper and gold mineral deposits, weed killers, insecticides, rat poisons, wallpapers, and artificial flowers.

There are several types of arsenic poisoning:

- Fulminant - When massive doses are rapidly absorbed, death occurs in 1–3 hours due to shock and peripheral vascular failure.
- Gastrointestinal - This is the most common form of acute poisoning and resembles bacterial food poisoning. There is a constriction in the throat and difficulty in swallowing. The stools are dark-colored, foul smelling, and bloody initially and later take a colorless odorless watery form similar to the rice-water stools of cholera.
- Chronic poisoning - It occurs due to repeated daily exposure. The main features include finely mottled, raindrop pigmentation that appears on the skin of flexures, temples, eyelids, and neck. In prolonged contact, hyperkeratosis of the palms and soles occurs along with the development of a transverse white line in the fingernails called the Aldrich-Mees lines.

Tests to detect arsenic poisoning:

- 24-hour urinary excretion of arsenic
- Detection of arsenic by atomic absorption spectroscopy
- Marsh and Reinsch's tests are now considered obsolete.

The preferred treatment option is freshly prepared hydrated ferric oxide, which converts toxic arsenic to non-toxic ferric arsenite. DMSA (Succimer), British anti-Lewisite (BAL), and penicillamine can also be used.



Other options:

Option A: Chronic malnutrition leads to distinct skin and nail changes depending on the specific condition. In kwashiorkor, there is flaky paint dermatosis with reddish-brown scaling, thin and sparse hair featuring a "flag sign," and soft nails. In marasmus, patients exhibit dry, wrinkled skin, slow-growing brittle hair, and nails, often accompanied by growth retardation.

Option B: Acute copper poisoning is due to inhalation of a large amount of copper dust or fumes, which leads to upper respiratory tract irritation, conjunctivitis, and sinus irritation. Contact with the skin may lead to irritant contact dermatitis and a greenish-blue discolouration of the skin.

In chronic poisoning, symptoms mostly resemble those of Wilson's disease, mainly including gradual anaemia, peripheral neuritis, and muscle atrophy.

Option C: Acute mercury poisoning presents with corrosive bronchitis with fever, chills, and dyspnea, which may progress to pulmonary edema and fibrosis. The symptom profile in children is similar to that of Kawasaki disease (mucocutaneous lymph node syndrome). Chronic exposure to elemental mercury leads to a classic triad of gingivitis, tremors, and neuropsychiatric changes.

Solution to Question 13:

The likely range of the gunshot injury shown above is close range, characterized by a circular wound with blackening and tattooing.

To determine the range of firing, the following parameters are used:

- The entry wound size and shape
- Blast effect – Muzzle end imprint and edge eversion
- Blackening, Tattooing, Singeing of hair, Abrasion collar, and Grease collar
- Hair should be preserved to determine the range of firing which can be examined for carbonization/desiccation in close range, also from lead deposition around the wound to assess the range of firing.

Blackening occurs due to the deposit of smoke and can be removed with a wet cloth. This can be seen up to around 15 cm of firing. Whereas, tattooing occurs due to the deposition of unburnt powder particles and are difficult to remove. They are also called stippling or peppering. This is seen up to 50 cm of firing.

Abrasion collar: Gyroscopic movement of the bullet, the skin's elasticity, and the grease and dirt particles on the bullet's surface are the major causative factors of abrasion and grease collar. They are seen on the skin around the wound of entry at all ranges. Depending on the angle of entry made by the bullet, i.e.

- A perpendicular entry of the bullet develops the abrasion collar of a rounded shape.
- An acute angular entry of the bullet develops the abrasion collar of an oblique or oval shape.
- Some contusion present in the abraded collar is known as contusion collar

Grease collar: Mostly seen on the clothing and is seen as a black deposit, sharply outlined as if printed (better seen on light colour clothes and fair-skinned individuals). Its medicolegal importance is that it helps derive information such as entry wound and direction of firing.

Solution to Question 14:

The given case scenario is suggestive of methanol poisoning. In these cases, formic acid and lactic acid are typically the metabolites found in the blood.

Methyl alcohol, or methanol, is commonly present in cheap or illicit alcoholic beverages due to its lower cost compared to ethyl alcohol. These inexpensive alcohols may be improperly distilled or adulterated, leading to methanol contamination. When ingested, methanol is absorbed into the bloodstream and metabolized by the liver, where it is first converted into formaldehyde, a highly toxic intermediate. Formaldehyde is then further oxidized into formic acid, which is responsible for many of the severe symptoms associated with methanol poisoning.

The accumulation of formic acid leads to significant metabolic acidosis, which manifests as confusion, abdominal pain, and visual disturbances. The visual symptoms can be particularly severe, including blurred vision or even blindness, due to the toxic effects of formic acid on the optic nerve. In addition to formic acid, methanol's disruption of normal oxidative enzyme activity results in the buildup of lactic acid, contributing further to acidosis. Brain edema, acute renal failure, and severe metabolic derangement are also seen.

The principles of treatment for this condition predominantly involve using fomepizole or ethanol to inhibit the metabolic conversion of these relatively non-toxic alcohols to their toxic metabolites.

Solution to Question 15:

Diatoms enter pulmonary circulation on antemortem drowning. Diatoms are a class of microscopic unicellular algae (option C) found suspended in most natural waters. Their cell walls are impregnated with silica, chlorophyll, diatomic, and a brown pigment.

The drowning fluid and the particles in it, e.g., diatoms and plankton, pass from the ruptured alveolar wall into lymph channels and pulmonary veins and thus enter the left heart. Only a live body with circulation could transport diatoms from the lungs to the brain, bone marrow, liver and other viscera, and skeletal muscle. They are also found in bile and urine. Bone marrow is highly suitable and reliable.

Isolation of diatoms involves acid digestion of tissue - bone marrow, lung, blood, and kidneys with subsequent centrifugation and the final residue is aspirated and examined for diatoms, which are birefringent, using dark ground illumination. A water sample is collected at the time of body retrieval in a clean container and the similarity of different species of diatom is compared. Diatoms are both heat and acid-resistant (option D).

- The presence of diatoms in the lung, brain, liver, kidneys, and bone marrow (option A) of the femur (best site for analysis) or humerus is suggestive proof of antemortem drowning.
- Since diatoms resist putrefaction, the diatom test may have some value in the examination of decomposed bodies.
- The test is negative in postmortem submersion and in dry drowning.

Diatoms are ubiquitous in the environment and may enter circulation via the gastrointestinal tract (as contaminants of foods) or via the respiratory tract (which are present in the air). Still, the diatom test is considered the gold standard for the diagnosis of typical drowning.

Solution to Question 16:

The above phenomenon is known as superfetation. It occurs when there is fertilization of 2 different ova released in different menstrual cycles. The consequence is two children are born at the same time and one of them might be mature while another might be immature or two fully developed fetuses are born at intervals ranging from one to three months apart.

Ovulation usually stops with the onset of pregnancy, but in a few individuals, ovulation may be possible till the first trimester, after which the decidual cavity is obliterated. The fertilization of the newly released ovum in a woman who is already pregnant may take place following coitus.

Other options:

Option A: Superfecundation is the fertilization of 2 different ova in the same menstrual cycle by separate acts of coitus. Heteropaternal superfecundation is used to refer to instances of two different males fathering fraternal twins. This leads to the possibility of twins also being half-siblings. This may raise doubts regarding adultery and infidelity.

Option C: A supposititious child is also known as a fictitious child. In such cases, a woman pretends to be pregnant and substitutes a living child as her own, or may substitute a male child for her female child, or abortion. This is done to claim property. This can be identified by verifying the signs of pregnancy and delivery in the woman and by DNA fingerprinting.

Option D: A posthumous child is a child born alive after the death of his or her father who was conceived before the death of the father.

Solution to Question 17:

According to BNS Section 30 (IPC 92), a doctor is permitted to perform life-saving procedures on a patient with life-threatening injuries. For example, in this case where conservative measures have failed to control intractable postpartum haemorrhage, an emergency hysterectomy can be performed without obtaining consent in order to save the patient's life.

The guiding principle is whether the delay in treatment in order to obtain consent would result in harm to the patient. This is the doctrine of anticipation. Under some circumstances like emergencies, informed consent may be neither possible nor necessary.

BNS Section 30 (IPC 92) states that nothing is an offence by reason of any harm which it may cause to a person for whose benefit it is done in good faith, even without that person's consent, if the circumstances are such that it is impossible for that person to signify consent, or if that person is incapable of giving consent, and has no guardian or other person in lawful charge of him from whom it is possible to obtain consent in time for the thing to be done with benefit.

Solution to Question 18:

The given scenario indicates adipocere formation, which is a late postmortem change where body fats undergo saponification through anaerobic hydrolysis and hydrogenation, resulting in a waxy substance. Therefore, the correct answer is that it is a type of body preservation with saponification of fats.

Adipocere formation or corpse wax, is a late sign of death, that occurs days to months after death. A wax-like organic substance is formed by the anaerobic hydrolysis or hydrogenation of fat in tissues. It has a greyish-white appearance with a distinctive offensive or sweetish smell.

Adipocere formation varies widely, taking as little as three weeks in temperate regions during summer, but typically requires several months for significant changes. In rare cases, the entire body may be affected, with complete conversion of an adult limb taking three to six months.

Medicolegal importance is as follows:

- It is the surest sign of death
- It gives a rough estimate of the time since death
- The cause of death can be identified.
- In the face, the features are well preserved and therefore used in identification.

Postmortem changes are classified based on time:

- Immediate signs of death (at the time of somatic or clinical death):
 - Insensibility and loss of voluntary power
 - Cessation of respiration
 - Cessation of circulation
- Early signs of death (cellular death that is observed in the first 12 to 24 hours of death):
 - Pallor and loss of elasticity of the skin
 - Eye changes
 - Primary flaccidity of muscles
 - Algor mortis/Postmortem cooling
 - Postmortem lividity/Suggillation
 - Rigor mortis
- Late signs of death:
 - Putrefaction
 - Adipocere
 - Mummification